Common free-flight selection and contact-phase sensitivity in a compliant Painlevé oscillator

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Abstract

We study the stiff limit of a compliant Painlevé oscillator with relaxing normal memory, smooth friction and square-root belt-speed detuning. From prescribed physical initial data to the first crossing of a fixed nearby angle section, we classify and realize all limiting ordered paths, including their infinite contact–flight cascades. For fixed mechanical parameters, the outgoing family is common to the prescribed compact parameter and preparation classes. Every free-flight target is selected by one compliance sequence uniformly across those classes, with each incoming datum retaining its own incoming arc. First-angle contact selection obeys an entrance-constant law. Unequal constants exclude common interior contact selection, even at different contact phases. Along a sequence selecting an interior contact target, smaller and larger entrance constants instead select admission and release, respectively. Explicit threshold sequences realize this ordering, which we establish for belt coefficients and for locally constructed friction-profile curves. The mechanism is a separation of scales: repeated return contraction leaves an entrance-phase difference of order $1/\log(1/\varepsilon)$, while an actual contact stop requires a touchdown window of order ε. A complete-state comparison from the physical preparation preserves the accuracy needed to distinguish them. This obstruction concerns first-angle stopping: a fixed local observation patch crossed strictly at a prescribed interior point instead permits common selection of that prefix.

1 Introduction

A rigid frictional system can lose forward uniqueness at a Painlevé singularity. Compliance restores an ordinary evolution, but its vanishing limit need not select a single outgoing motion. We study a compliant, smooth-friction unfolding of the critical oscillator of Nordmark, Várkonyi and Champneys [1, Section 6.1]. Two masses are joined by a rigid link, with the lower mass contacting a moving belt. Successive contacts and flights form a cascade whose number of returns tends to infinity as compliance vanishes.

Our question concerns two different forms of independence from the auxiliary contact parameters. Do they give the same family of limiting paths? Can one sequence of compliances

Running head: Path selection in a Painleve oscillator.

select the same member of that family for all parameter values? We answer both questions for paths stopped at the first crossing of a fixed nearby link angle. The system has relaxing normal memory and friction smoothed at slip scale ε, with belt speed $U_b = U_* - k\sqrt{\varepsilon}$ and fixed $k > 0$. Section 2 specifies the compact mechanical, profile and physical-preparation classes used throughout.

Earlier work and the present problem. Champneys and Várkonyi [2, Sections 4, 5.1 and 5.3] place contact regularization, dynamic jam and reverse chatter within the forward-continuation problem for frictional rigid bodies. Compliance-dependent continuation already appears in reverse chatter: Nordmark, Dankowicz and Champneys [3, Sections 5.2 and 6.1] give an asymptotic matching argument for selecting different forward motions along different compliance sequences, with numerical terminal curves for a conservative normal law and Coulomb friction. The relaxing normal law used here was introduced by Nordmark, Várkonyi and Champneys [1, Equations (20)–(21)] in their study of dynamic jam. The concluding discussion of their 2017 preprint identifies motion beyond the first lift-off or the onset of impact without collision as a further problem, including repeated contacts. We determine that subsequent local motion under the smooth friction and belt scaling fixed below, including every stopped path and the common-sequence question.

Rigorous passage results in related contact problems resolve several of the local difficulties. Hogan and Kristiansen [4, Theorems 3.1–3.2] prove compression, slip arrest and lift-off for a single impact without collision, and a threshold separating it from immediate lift-off, at a fixed negative Painlevé coefficient. Kristiansen and Hogan [5, Theorem 6] prove persistence of a strong canard at the degenerate point, using a second blow-up to resolve the ellipticity left by the first. Their erratum [6] requires sufficiently small initial tangential velocity for rigid-model slip arrest before the singular point; the canard theorem is unchanged. The additional tasks here are preparation of the independent normal memory into a repeatable return domain, comparison through a growing number of returns, and actual realization of every stopped contact prefix.

Regularization can also select one outgoing orbit in other nonsmooth systems. For a truncated visible–invisible two-fold normal form, Kristiansen and Hogan [12, Theorem 1] prove uniform contraction of a transverse incoming family towards one outgoing orbit under arctangent regularization, with a nonresonance condition and separation from the weak canard; their Section 5 discusses other regularization functions. In three-dimensional contact mechanics, Cheesman, Hogan and Kristiansen [13, Sections 4–5] use normal compliance and a separate smoothing of spatial Coulomb friction to analyse impact without collision. They resolve local sticking dynamics and identify a Painlevé-I equation in the transition from slipping to inconsistency.

For homogeneous singular ODEs, Drivas and Mailybaev [10, Theorem 5 and Corollary 2] study expelling regularizations approaching an exponentially attracting renormalized cycle with positive mean radial drift. Regularizations escaping into the basin of the same cycle share a phase family of limiting continuations. Their geometric sequence $\nu_n = \exp\{-T\langle F_r\rangle n + \chi\}$ selects phase $\zeta = c - \chi$, with regularization-dependent c. This gives

a related distinction between a common limiting family and a common selecting sequence. In the contact problem here, an entrance-phase difference disappears at a free target but remains larger than the fixed-angle contact stopping window. Selection therefore depends on the target type, and the order of the entrance constants determines the endpoint limits.

The scalar phase of a power-like return has a classical precedent. Goncharuk and Kudryashov [7, Lemmas 5–6] prove rectification and refined double-logarithmic parameter asymptotics, following root estimates of Ilyashenko, Kudryashov and Schurov [8, Sections 4.5–4.6] and the smooth saddle-loop work of Dumortier and Roussarie [9]. We construct the phase together with the complete mechanical graph and control its accuracy along actual returns. Bressan and Mazzola [11, Theorem 2] prove time–state graph convergence for specified approximations of impulsive feedbacks. Here the contact curves and their ordering are derived from the compliant force law.

Main conclusions. Theorem A classifies and realizes every stopped-path limit from the prescribed initial data. Each path consists of its incoming rigid arc, an infinite ordered contact–flight cascade leaving the singular point, and a terminal contact or flight prefix. At fixed mechanical parameters and observation section, the outgoing family is common to the prescribed auxiliary classes; different incoming data retain their own incoming arcs. Theorem B gives the stronger common-sequence conclusion for every free-flight target, uniformly over those compact classes.

Theorem C states the entrance-constant law for contact targets within the same physical model and classes. Two fixed unequal constants exclude common interior contact selection, even at different contact phases. Moreover, on any sequence selecting an interior contact target for one datum, smaller and larger entrance constants force admission and release for the other data. Proposition 7.1 gives explicit threshold sequences and uniform endpoint selection on compact subclasses separated from the threshold. Equal constants are necessary for common interior selection; these estimates do not make them sufficient. The belt application is Corollary 3.4; Corollary 7.7 gives the corresponding application along an admissible friction-profile curve constructed locally through each baseline profile.

Mechanism. Recurring compression consumes tangential slip; arrest changes the normal response, and the inherited compression drives release. The subsequent cubic free-flight balance gives a return scale of order $\delta^{3/4}$. On the invariant graph of the complete normalized limiting return, the exact base map and its phase satisfy

$$F_\lambda(\delta) \sim K_\lambda \delta^{3/4}, \qquad \mathcal{S}_\lambda(F_\lambda(\delta)) = \tfrac{3}{4}\mathcal{S}_\lambda(\delta).$$

The prescribed physical preparation determines a bounded entrance correction. Repeated return contraction reduces the difference of two such corrections to order $1/\log(1/\varepsilon)$ at a fixed positive section. An actual contact stop at $p = -as$ requires the much narrower window $\delta = s + O_s(\varepsilon)$. Thus

$$\varepsilon \ll \frac{1}{\log(1/\varepsilon)} \ll 1.$$

The same entrance difference vanishes in the free-target limit but still separates the contact stopping windows and orders their endpoint limits. The comparison uses the exact phase at the fixed section, not just its leading logarithmic approximation.

Proof route. Section 4 derives the complete prepared entrance and the repeatable local return from the physical initial data. Section 5 constructs the exact graph and compares the actual full states through the required event indices. Section 6 pairs whole returns and scans a fixed actual contact index to classify and realize complete paths. Section 7 compares the residual phase with the contact window and proves the belt and profile applications; Section 8 treats longer paths, first-angle truncation and strict local crossings. The local estimates and cancellations remain in the appendices.

2 The physical model and prescribed initial data

2.1 An independent belt parameter

The oscillator consists of two point masses joined by a rigid link. The upper mass moves vertically against a spring and damper; the lower mass can contact a belt moving to the left. In the nondimensional coordinates of [1, Section 6.1], both masses and the link length are one. The link angle ϕ is measured from the downward vertical, and the upper mass is at height $1+\psi$ above the undeformed belt. A torsional spring acts on ϕ. The observables y and u below are the lower mass's height and its slip relative to the belt. Let $q=(\phi,\psi)^T$, $\omega=\dot\phi$, and define

$$y=\psi+1-\cos\phi, \qquad v=\dot y, \qquad u=\omega\cos\phi+U_b, \qquad U_b=3/5+\eta. \tag{2.1}$$

The belt detuning η is independent of the mechanical parameters. We use the critical mechanical family of [1, Section 6]:

$$M(\phi)\ddot q=f_{\beta,\kappa}(q,\dot q)+Q_N(\phi)\lambda_N+Q_T(\phi)\lambda_T, \tag{2.2}$$

$$M=\begin{pmatrix}1 & \sin\phi\\ \sin\phi & 2\end{pmatrix}, \quad Q_N=\begin{pmatrix}\sin\phi\\ 1\end{pmatrix}, \quad Q_T=\begin{pmatrix}\cos\phi\\ 0\end{pmatrix},$$

$$f_{\beta,\kappa}=\begin{pmatrix}-(\phi-\phi_0)-\sin\phi\\ -\kappa\psi-c_\psi\dot\psi-2-\cos\phi\,\omega^2\end{pmatrix},$$

$$\mu=25/12, \quad c_\psi=25(18-7\beta)/108, \quad \phi_0=\phi^*+49/20-7\beta/12-9\kappa/50,$$

$$(\cos\phi^*,\sin\phi^*)=(3/5,4/5). \tag{2.3}$$

The prescribed completion is

$$\varepsilon\dot z=-z-\varepsilon^2\lambda_N, \qquad \lambda_N=\varepsilon^{-2}(z-y)_+, \qquad \lambda_T=-\mu\Phi_\theta(u/\varepsilon)\lambda_N. \tag{2.4}$$

Thus contact is determined by $g=y-z$, not by y alone. In flight $z(t)=z_re^{-(t-t_r)/\varepsilon}$, and $\dot g=v+z/\varepsilon$ at release. Every release below is the first zero of the active normal force on the

stated continuous contact passage, and every touchdown is the first zero of the subsequent positive free gap. Initial capture may contain other contacts.

Write $X = (\phi, y, \omega, v)$ and eliminate $\psi = y - 1 + \cos\phi$, $\dot\psi = v - \sin\phi\,\omega$. The free mechanical vector field and saturated contact field per unit normal force are

$$F_{\text{free}} = (\omega, v, (M^{-1}f)_1, b), \qquad Q_{\text{sat}} = (0, 0, (M^{-1}(Q_N - \mu Q_T))_1, p), \tag{2.5}$$

where

$$b = Q_N^T M^{-1} f + \cos\phi\,\omega^2, \qquad a_T = Q_T^T M^{-1} f - \sin\phi\,\omega^2, \qquad p = C - \mu B. \tag{2.6}$$

The exact contact equations for the observables are

$$\dot v = b + (C - \mu B \Phi_\theta(u/\varepsilon))\lambda_N, \qquad \dot u = a_T + (B - \mu A \Phi_\theta(u/\varepsilon))\lambda_N. \tag{2.7}$$

Direct inversion of M gives

$$A = \frac{2\cos^2\phi}{1+\cos^2\phi}, \qquad B = \frac{\sin\phi\cos\phi}{1+\cos^2\phi}, \qquad C = \frac{1}{1+\cos^2\phi}, \qquad D = C - B^2/A = \frac{1}{2}. \tag{2.8}$$

In particular the coefficient after eliminating tangential force is positive even when the saturated coefficient p is zero or negative. The chart is restricted throughout so that $\cos\phi$ has a common positive lower bound. Since $p = C(\phi) - \mu B(\phi)$, the rate $p_\phi(G) = a$ below permits a further common restriction with $p_\phi > 0$. Each observation section $p = -as$ then fixes one angle ϕ_s on this branch.

2.2 Critical rates and the working box

At $G = (\phi^*, 0, -1, 0)$ in the mechanical coordinates, $p = b = 0$ and the tuned slip is zero. The detuned slip at G is η. Substitution in (2.5)–(2.8) gives

$$\begin{gathered} A_G = 9/17, \quad B_G = 6/17, \quad C_G = 25/34, \quad h := \mu A_G - B_G = 3/4, \qquad a := 175/408, \\ F_{\text{free}} p(G) = -a, \quad F_{\text{free}} b(G) = -c_\lambda, \quad a_T(G) = d_\lambda, \quad Q_{\text{sat}} b(G) = a\beta, \end{gathered} \tag{2.9}$$

$$c_\lambda = \frac{126765 + 61425\beta - 30625\beta^2 - (9450\beta + 8262)\kappa}{55080}, \qquad d_\lambda = \frac{495 - 175\beta - 54\kappa}{180},$$

$$\ell := \lambda_* = \frac{c_\lambda}{a(\beta - 1)}, \qquad g_* := d_\lambda - h\ell = \frac{55825\beta + 17712\kappa - 213390}{31500(\beta - 1)}. \tag{2.10}$$

Here $\lambda = (\beta, \kappa)$; we also write $c = c_\lambda$ and $\gamma = -g_* > 0$. These formulas can be checked by differentiating the explicit fields in (2.5). At G the added generalized acceleration is $(-5/4, 1)$ in $(\ddot\phi, \ddot\psi)$, so its $(\dot\omega, \dot v)$ components are $(-5/4, 0)$, and

$$p_\phi(G) = a, \qquad \det D_{(\phi,\omega)}(p, b)(G) = -6125\beta/41616. \tag{2.11}$$

Thus (p, b, y, v) are analytic coordinates on a common neighborhood; write $\Xi(p, b, y, v)$ for their inverse mechanical chart.

Fix a sufficiently small closed box Λ about $(7/3, 0)$ in the half-plane $\kappa \geq 0$, with

$$2 < \beta_- \leq \beta \leq \beta_+ < 5/2, \qquad c, d_\lambda, \ell, -g_* > 0, \qquad |\det D_{(\phi,\omega)}(p, b)(G)| > 0, \tag{2.12}$$

all with common strict margins. Uniform statements use this compact box; its interior is relative to $\kappa \geq 0$. The upper bound $\beta_+ < 5/2$ records the scope of the present proof. The coefficient β is also the exponent in the incoming weak trace, as the scalar equation in Section A shows. We use the prescribed compact range (2.12) to construct the weak curve and obtain the common forcing bound (A.8) in the upstream normal-energy and weak-transport estimates. The strict sign and chart margins fix the event orientation and the common coordinate neighborhood.

2.3 Profiles and physical preparation

Let Θ be compact, and let $\theta \mapsto \Phi_\theta$ be continuous in the metric

$$d_{\bar\alpha}(\Phi, \Psi) = \max_{0 \leq j \leq 4} \sup_{q \in \mathbb{R}} e^{\bar\alpha |q|} |\partial_q^j (\Phi - \Psi)(q)|, \qquad 0 < \alpha < \bar\alpha < 2. \tag{2.13}$$

Its image is compact in the affine space consisting of tanh plus functions with finite weighted norm. Fix a compact $I_s = [s_-, s_+] \Subset (-1, 1)$ containing $B/(\mu A)$ on the working mechanical chart. The following bounds hold with common constants $0 < M_{\mathrm{sl}} < Q_\Phi$, $m_\Phi, \varkappa_{\mathrm{sl}}, C_\Phi > 0$:

(F1) Φ_θ is strictly increasing, $\Phi_\theta(0) = 0$, has values in $(-1, 1)$ and limits ± 1 at $\pm\infty$.

(F2) $\Phi_\theta([-Q_\Phi, Q_\Phi]) \supset I_s$, $\Phi_\theta' \geq m_\Phi$ there, and its first four derivatives are uniformly bounded there.

(F3) $\Phi_\theta(M_{\mathrm{sl}}) \geq s_+ + \varkappa_{\mathrm{sl}}$ and $\Phi_\theta(-M_{\mathrm{sl}}) \leq s_- - \varkappa_{\mathrm{sl}}$.

(F4) For $q \geq M_{\mathrm{sl}}$,

$$1 - \Phi_\theta(q) + \sum_{j=1}^{4} q^j |\Phi_\theta^{(j)}(q)| \leq C_\Phi e^{-\alpha q},$$

with the reflected bound on the negative tail.

The slip threshold M_{sl} is fixed with the profile class and mechanical chart. Nonodd profiles are allowed. Each profile is fixed before taking the stiff limit. The family $\tanh(\theta q)$, $1 \leq \theta \leq 5/4$, is an example: all derivatives through order four are bounded by $C(1 + |q|^4)e^{-2|q|}$ on the tails, as are the label derivatives up to an extra polynomial factor. This proves both (F4) and continuity in (2.13); monotonicity gives the remaining conditions after the strip is chosen large enough.

Choose a small fixed $p_0 > 0$ and a continuous compact family of saturated rigid data (ϕ_ν, ω_ν) on $p = p_0$. Their positive-slip rigid orbits reach G and have traces

$$b_{\mathrm{rig},\nu}(p) = \bar b_\lambda(p) + C_1(\lambda, \nu) p^\beta + \mathcal{R}_C(p), \qquad |\partial_p^j \mathcal{R}_C(p)| \leq C p^{\beta+1-j}, \quad 0 \leq j \leq 3, \tag{2.14}$$

where $\bar{b}_\lambda(p) = -\ell p + O(p^2)$ is the analytic weak curve constructed in Section A. On each closed rigid subarc before G, the force $-b/p$ and tuned slip have common positive lower bounds. The rigid trace is unchanged by detuning: U_b enters only the slip observable and the profile.

For a fixed fast triple $\mathbf{n} = (Y_0, V_0, Z_0)$ in any prescribed compact set, the complete physical initial state is

$$\mathfrak{X}_\varepsilon(0; \lambda, \theta, \nu, \mathbf{n}) = (\phi_\nu, \varepsilon^2 Y_0, \omega_\nu, \varepsilon V_0, \varepsilon^2 Z_0) \tag{2.15}$$

in coordinates (ϕ, y, ω, v, z). The slow datum and fast triple stay fixed as ε varies. The initial state need not be in contact. For $\mathbf{n} = 0$ it is the same full physical state for every compliance and profile. No later release or touchdown state is independently prepared. The class is nonempty: the analytic weak curve supplies a rigid datum at p_0, and a compact interval of nearby weak coefficients supplies more. The scalar construction in Section A gives (2.14). The positive limits of $-b/p$ and u_0/p then give the closed-arc margins after the chart is reduced.

Standing convention. Fix the compact profile and physical-preparation classes above and $k \in I = [k_-, k_+] \Subset (0, \infty)$; the nominal mechanical box may then be reduced jointly for these same classes. All error estimates and event margins are uniform on the resulting compact label classes. Unless explicitly stated otherwise, mechanical parameters and each sufficiently small positive observation section are fixed before $\varepsilon \downarrow 0$. The common graph, return and section cutoffs are chosen once before these targets. Uniform selection is asserted only where stated; each incoming datum keeps its own prescribed rigid arc.

The complete label is $\mathfrak{l} = (\lambda, k, \theta, \nu, \mathbf{n})$. We use $\mathfrak{X} = (\phi, y, \omega, v, z)$ for the full physical state and $X = (\phi, y, \omega, v)$ for its mechanical part. In a five-state norm, G includes its zero memory coordinate. Put $r = k\sqrt{\varepsilon} = -\eta$. The actual belt slip is $u_r = u = \omega \cos\phi + U_* - r$; the critical-speed slip is $u_0 = \omega \cos\phi + U_* = u_r + r$. The coefficient k and r remain constant along each trajectory; the physical belt speed follows $U_b = U_* - k\sqrt{\varepsilon}$ as compliance varies. The slow datum and scaled fast triple $\mathbf{n}$ remain fixed; the physical initial state follows (2.15).

3 Complete paths and their selection

We stop the original trajectory at its first crossing of one fixed downstream section. The limiting object retains the incoming rigid arc, every return in an accumulating cascade and the final contact or flight prefix. Its contact pieces carry an order, although their physical durations vanish.

3.1 The curves and their topology

For continuous oriented curves $f, g : [0, 1] \to \mathbb{R}^5$, put

$$d_{\mathrm{F}}(f, g) = \inf_{h \in \mathrm{Homeo}_+([0,1])} \sup_{q \in [0,1]} |f(h(q)) - g(q)|. \tag{3.1}$$

We identify curves at zero distance. Pauses and constant pieces are therefore allowed. Equivalently, one can use two nondecreasing continuous surjections in the comparison: approximate a surjection a by $(1-\eta)a+\eta\,\mathrm{id}$ and use uniform continuity, then compose the resulting homeomorphisms. For compact path images,

$$d_{\mathrm{H}}(A,B)=\max\{\sup_{x\in A}\inf_{y\in B}|x-y|,\sup_{y\in B}\inf_{x\in A}|x-y|\}\le d_{\mathrm{F}}(f,g),\quad A=f([0,1]),\ B=g([0,1]). \tag{3.2}$$

Write $I_{\lambda,\nu}$ for the prescribed saturated rigid incoming arc from $(\phi_\nu,0,\omega_\nu,0,0)$ to G, oriented in physical time. Appendix D and Section 5 construct an exact limiting graph return $F=F_\lambda$ and its ordered one-return curve $\mathcal{C}_{\lambda,d}:[0,3]\to\mathbb{R}^5$. The three parameter intervals describe saturated compression, inherited normal release and the following first free flight: $\alpha=0,1,2,3$ mark touchdown, slip arrest, release and the next touchdown, respectively. These are finite-scale curves, retaining their exact $\sqrt{d}$ dependence. They satisfy

$$\mathcal{C}_{\lambda,d}(3)=\mathcal{C}_{\lambda,F(d)}(0),\qquad \sup_{0\le\alpha\le3}|\mathcal{C}_{\lambda,d}(\alpha)-G|\le Cd^{3/4}. \tag{3.3}$$

On contact $p(\mathcal{C}_{\lambda,d}(\alpha))=-ad$; on the free part it strictly decreases from $-ad$ to $-aF(d)$. For fixed sufficiently small $s>0$, set $d_*=F^{-1}(s)$. There is consequently a unique continuous free-section parameter $j_s:[d_*,s]\to[2,3]$ with

$$p(\mathcal{C}_{\lambda,d}(j_s(d)))=-as,\qquad j_s(d_*)=3,\quad j_s(s)=2. \tag{3.4}$$

The possible terminal data form the compact set

$$\mathcal{A}_s=\{(d,j_s(d)):d_*\le d\le s\}\cup(\{s\}\times[0,2]). \tag{3.5}$$

Its graph branch records free-flight stopping; the vertical segment records contact stopping.

For $d\in[d_*,s]$, let $b_m=F^{-m}(d)$, $m\ge0$. These bases tend to zero. Define $L_d(0)=G$ and trace $\mathcal{C}_{\lambda,b_m}$ linearly on $[2^{-m},2^{-(m-1)}]$ for $m\ge1$. The joins and bound in (3.3) give a continuous curve. The complete curves are

$$\mathcal{W}_{\lambda,\nu,s}(d,\alpha)=I_{\lambda,\nu}\star L_d\star\mathcal{C}_{\lambda,d}|_{[0,\alpha]},\qquad (d,\alpha)\in\mathcal{A}_s. \tag{3.6}$$

Here $\star$ preserves order and joins equal endpoints. One may assign one third of $[0,1]$ to each factor; a zero final prefix is constant. Its compact image is

$$W_{\lambda,\nu,s}(d,\alpha)=I_{\lambda,\nu}([0,1])\cup\{G\}\cup\bigcup_{m\ge1}\mathcal{C}_{\lambda,b_m}([0,3])\cup\mathcal{C}_{\lambda,d}([0,\alpha]). \tag{3.7}$$

The parameter α records the order along each completed contact and free return. At admission the two terminal pairs represent one ordered path: $\mathcal{W}_{\lambda,\nu,s}(d_*,3)=\mathcal{W}_{\lambda,\nu,s}(s,0)$. Indeed, $F(d_*)=s$ identifies the final full return of the former with the last return already contained in L_s. Their parameter pairs remain distinct. Figure 1 shows the order and terminal domain.

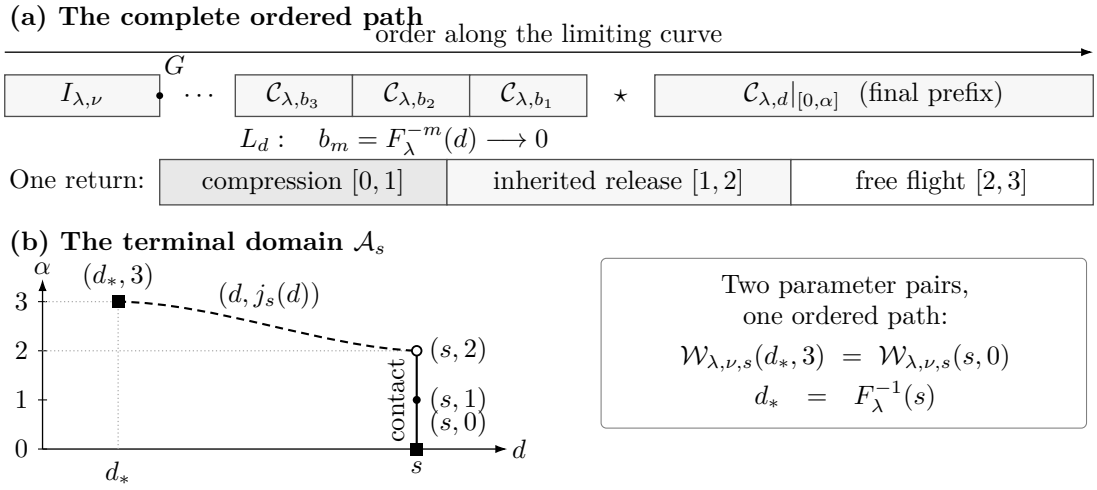


Fig. 1: Complete stopped paths and their terminal data. (a) The incoming arc is followed by returns accumulating at G and a final prefix; the displayed b_3, b_2, b_1 are the last three full-return bases. Strip widths encode order, not duration. (b) The free graph and contact segment form $\mathcal{A}_s$; the graph shape is schematic. Arrest is $(s, 1)$. Matching squares represent the single endpoint identification of complete ordered paths, while the parameter pairs remain distinct. Identifying the two interval endpoints gives the quotient circle of Corollary F.2.

3.2 Classification, common selection and contact sensitivity

Let t_n be the successive actual touchdowns after the critical release and put $\delta_n = -p(t_n)/a$ whenever $p(t_n) < 0$. The second and subsequent touchdowns have this sign. Let t_s denote the actual first crossing of $p = -as$. Existence of the required events is part of the following theorem. For sufficiently small compliance, put

$$\begin{gathered} N_s = \min\{n \geq 3 : \delta_n \geq s\}, \qquad d_\varepsilon = \delta_{N_s-1}, \qquad \alpha_\varepsilon = h_\varepsilon^{-1}(t_s), \\ f_\varepsilon(q) = \mathfrak{X}_\varepsilon(t_s q), \qquad H_\varepsilon = f_\varepsilon([0,1]). \end{gathered} \tag{3.8}$$

The increasing pairing $h_\varepsilon : [0,3] \to [t_{N_s-1}, t_{N_s}]$ is constructed in Lemma 6.1. Its compression parameter is the fraction of actual normal impulse up to the first $u_r = M_{\mathrm{sl}}\varepsilon$ event; the release parameter is the elapsed fraction from that event to the first force zero. The last interval uses elapsed free time. Thus α_ε is read from the actual return containing the first section time.

Theorem A (Classification and actual realization)**.** *Under the hypotheses of Section 2, there is a common $s_0 > 0$ such that, for every fixed $0 < s \leq s_0$ and all sufficiently small ε, the original solution reaches its first section time t_s transversely, and $N_s \to \infty$ as $\varepsilon \downarrow 0$. For each fixed complete label $\mathfrak{l} = (\lambda, k, \theta, \nu, \mathbf{n})$, the family f_ε is relatively compact for* (3.1)*, and*

$$\mathrm{Clust}_{\varepsilon\downarrow 0}(d_\varepsilon, \alpha_\varepsilon) = \mathcal{A}_s.$$

Its ordered-path accumulation set is exactly

$$\{\mathcal{W}_{\lambda,\nu,s}(d,\alpha) : (d,\alpha) \in \mathcal{A}_s\}.$$

Every listed curve is attained by an actual sequence of compliances with this label fixed. More precisely, for every sequence $\varepsilon_i \downarrow 0$ with $(d_{\varepsilon_i}, \alpha_{\varepsilon_i}) \to (d,\alpha)$, one has $(d,\alpha) \in \mathcal{A}_s$ and

$$d_{\mathrm{F}}(f_{\varepsilon_i}, \mathcal{W}_{\lambda,\nu,s}(d,\alpha)) \to 0, \qquad d_{\mathrm{H}}(H_{\varepsilon_i}, W_{\lambda,\nu,s}(d,\alpha)) \to 0. \tag{3.9}$$

The Hausdorff accumulation set is exactly the corresponding image family in (3.7)*.*

For fixed λ, ν, s, this accumulation family is independent of the profile, belt coefficient and fast preparation within the prescribed classes. Its contact segment records every possible stopping point of the terminal contact. A specified free target has the stronger selection property below.

Theorem B (Common selection of a free target)**.** *With the compact classes and s_0 of Theorem A, fix λ, $0 < s \leq s_0$ and $d \in (d_*, s)$. The normalized graph phase $\mathcal{S}_\lambda$ gives the explicit sequence*

$$\varepsilon_m(d) = \exp\{-\tfrac{32}{9}(3/4)^{-m}\mathcal{S}_\lambda(d)\}. \tag{3.10}$$

Along it,

$$\sup_{(k,\theta,\nu,\mathbf{n})} \{|(d_{\varepsilon_m}, \alpha_{\varepsilon_m}) - (d, j_s(d))| + d_{\mathrm{F}}(f_{\varepsilon_m}, \mathcal{W}_{\lambda,\nu,s}(d, j_s(d)))\} \to 0, \tag{3.11}$$

where the supremum is over the prescribed compact auxiliary class. The corresponding Hausdorff image convergence is uniform as well. The two free endpoints $(d_*, 3)$ and $(s, 2)$ have the same uniform convergence along the explicit sequences in Proposition 7.1 below.

The sequence fixes the target while the auxiliary labels vary; each incoming datum retains its own arc $I_{\lambda,\nu}$. The exact phase $\mathcal{S}_\lambda$ contains the finite-scale return information needed at the fixed section s. In particular, the explicit sequence uses this constructed phase; replacing it by $\log(K_\lambda^4/d)$ at fixed d need not select the same target.

Theorem C (Contact selection by the entrance constant)**.** *Under the hypotheses of Section 2, fix λ and $0 < s \le s_0$ sufficiently small before taking the compliance limit. The original physical preparation determines a continuous bounded entrance constant $\chi_\lambda(\mathfrak{a})$ for the prescribed auxiliary labels $\mathfrak{a} = (k, \theta, \nu, \mathbf{n})$; its construction is given in Proposition 4.1 and* (4.11)*.*

(i) *For every two fixed auxiliary labels $\mathfrak{a}_1, \mathfrak{a}_2$ with unequal entrance constants, there is $\varepsilon_0 > 0$, depending on this pair and the fixed data, such that for $0 < \varepsilon < \varepsilon_0$ their first section times cannot both satisfy*

$$t_{j_i}^{\mathfrak{a}_i} < t_s^{\mathfrak{a}_i} \le t_{L,j_i}^{\mathfrak{a}_i}, \qquad i = 1, 2,$$

where $t_{L,j_i}^{\mathfrak{a}_i}$ is the first release after touchdown j_i. Consequently no common compliance sequence selects contact-interior terminal pairs (s, α_i) for both labels, where $0 < \alpha_i < 2$, or the corresponding ordered paths $\mathcal{W}_{\lambda,\nu_i,s}(s, \alpha_i)$, even when $\alpha_1 \neq \alpha_2$.

(ii) *For one fixed auxiliary label $\mathfrak{a}_0$ and $0 < \alpha_0 < 2$, suppose that a sequence $\varepsilon_m \downarrow 0$ satisfies*

$$f_{\varepsilon_m}^{\mathfrak{a}_0} \longrightarrow \mathcal{W}_{\lambda,\nu_0,s}(s, \alpha_0) \qquad \text{in } d_{\mathrm{F}}.$$

Along this same given sequence, every other fixed label $\mathfrak{a}$ with $\chi_\lambda(\mathfrak{a}) < \chi_\lambda(\mathfrak{a}_0)$ has terminal-pair limit $(d_*, 3)$ and ordered-path limit $\mathcal{W}_{\lambda,\nu,s}(s, 0)$; if $\chi_\lambda(\mathfrak{a}) > \chi_\lambda(\mathfrak{a}_0)$, the limits are $(s, 2)$ and $\mathcal{W}_{\lambda,\nu,s}(s, 2)$, respectively. The corresponding Hausdorff image convergence holds, and each incoming datum retains its own arc $I_{\lambda,\nu}$.

Equality of entrance constants is necessary for common interior contact selection; these estimates do not establish sufficiency.

Corollary 3.4 (Belt-coefficient sensitivity)**.** *Fix $\lambda, \theta, \nu, \mathbf{n}$ and a sufficiently small $0 < s \le s_0$ under the hypotheses of Theorem A. The entrance constant $\chi_{\lambda,\theta}(k)$ is strictly increasing in k. Thus any two distinct fixed belt coefficients satisfy the exclusion in Theorem C*(i)*. Fix $k_0 \in I$ and $0 < \alpha < 2$. Along any sequence $\varepsilon_m \downarrow 0$ for which $f_{\varepsilon_m}^{(k_0)} \to \mathcal{W}_{\lambda,\nu,s}(s, \alpha)$ in d_{F}, every other fixed $k \in I$ satisfies*

	terminal pair limit	*ordered-path limit*
$k < k_0$	$(d_*, 3)$	$\mathcal{W}(d_*, 3) = \mathcal{W}(s, 0)$
$k > k_0$	$(s, 2)$	$\mathcal{W}(s, 2)$.

The corresponding Hausdorff image convergence also holds.

Proposition 7.1(i) gives the explicit threshold sequences and their touchdown counts in Section 7; its parts (ii) and (iii) prove Theorem C. The belt corollary follows from the strict order of its entrance constant; the same selection law applies to the locally constructed profile curves in Corollary 7.7. Section 8 explains how the observation rule changes common prefix selection.

The complete physical path does not determine every additionally scaled observable: Corollary E.1 realizes an interval of limiting effective-gap velocities $v + z/\varepsilon$ above the same release endpoint path. Lemma F.1 identifies the sole nontrivial duplication of terminal parameters, and Corollary F.2 obtains the corresponding quotient-circle description for terminal states, ordered paths and compact images, at fixed λ, ν, s in the full physical state.

The proof must start with the original physical preparation and end with whole curves. We first state the entrance and return estimates, then obtain one precise comparison from the actual third touchdown. The realization and selection arguments use that same comparison.

4 The original entrance and the complete local return

The local estimates have three distinct roles. Proposition 4.1 derives the third-touchdown state and its entrance constant from the prescribed physical data. Proposition 4.2 separates the regularity of the exact reference return, the compliant value error, and the conditions under which its output can be used as a new input. The proofs of these estimates occupy Appendices A–D.

Section 5 uses these inputs to compare the complete return states through the required event indices. Section 6 then reconstructs the ordered paths and realizes contact prefixes by varying compliance at a fixed actual contact index. Finally, Section 7 compares the residual entrance phase with the contact stopping window. This order keeps the local estimates, the long return comparison and the selection argument separate, while preserving their common physical trajectory.

4.1 Normalized touchdown coordinates and the finite-scale reference

The return sends an input scale δ to $D \asymp \delta^{3/4}$. Under the bounds below, its tangential correction has size $O(\delta) = O(D^{4/3})$. The normal correction has size $O(\delta D) = O(D^{7/3})$ after the actual touchdown equation eliminates the larger release-velocity terms. These cancellations, proved in Appendix D, motivate the powers 4/3 and 7/3 in the normalized state.

Use the slips $u_r, u_0 = u_r + r$ defined in Section 2, and put $r_0 = c_\lambda/3$. At a touchdown with $p = -a\delta < 0$, the independent coordinates are

$$\begin{gathered} y = z = \varepsilon\delta^2\zeta, \qquad u_r = d_\lambda\delta + d_\lambda\delta^{4/3}\eta_U^r, \qquad v = -r_0\delta^2 - r_0\delta^{7/3}\eta_W, \\ \xi^r = (\zeta, \eta_U^r, \eta_W), \qquad W = 1 + \delta^{1/3}\eta_W. \end{gathered} \tag{4.1}$$

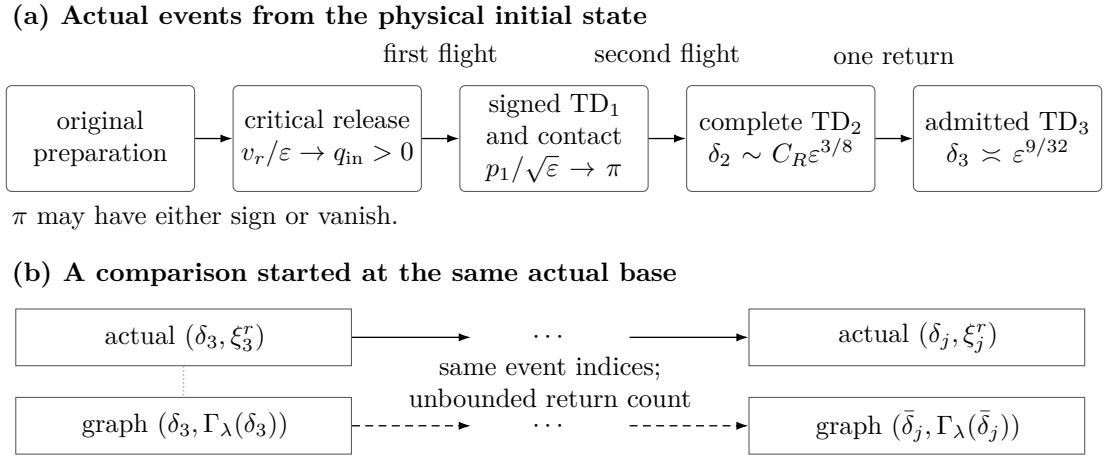


Fig. 2: From the physical preparation to the same-base return comparison. (a) Both preparatory flights and the intervening signed contact are generated by the original trajectory. (b) The graph starts at the actual third-touchdown base, with a small complete fibre discrepancy. The actual normal memory is retained; the two starting states are not identical. Arrows and spacing are schematic, not physical time.

If $p(\phi_\delta) = -a\delta$ on the chosen positive-cosine branch, the physical inverse is

$$(\phi, y, \omega, v, z) = \left(\phi_\delta, \varepsilon\delta^2\zeta, \frac{u_r + r - U_*}{\cos\phi_\delta}, v, \varepsilon\delta^2\zeta\right). \tag{4.2}$$

Thus b and $\eta_B = (-b - c_\lambda\delta)/(c_\lambda\delta^{4/3})$ are dependent coordinates. Fix a closed ball $\mathcal{K}$ in the independent coordinates with 0 and all $\xi_{*,\lambda}$ in its interior. Lemma C.3 bounds the original second-touchdown independent tuples uniformly. Choose M_{in} once so that its ball contains these tuples and $\mathcal{K}$ strictly in its interior. This choice does not change the fixed slip threshold M_{sl}. Fix $0 < \gamma_{\mathrm{cl}} < \frac{1}{2}\min_\lambda r_0$. The physical input domain is

$$|\xi^r| \le M_{\mathrm{in}}, \quad r_0 W - \zeta \ge \gamma_{\mathrm{cl}} > 0, \quad 0 < \varepsilon \le C_0\delta^2, \quad 0 \le r \le A\delta^{4/3}, \qquad e = \varepsilon\delta^{-3/2}. \tag{4.3}$$

The closing margin is $\dot{z} - v = \delta^2(r_0 W - \zeta)$. There is no positive lower bound on e or r. Taylor expansion in the chart (p, y, u_0, v) gives

$$\eta_B = \frac{g_b}{c_\lambda}\left(\eta_U^r + \frac{r}{d_\lambda\delta^{4/3}}\right) + O(\delta^{2/3}), \qquad g_b = -d_\lambda b_u(G), \tag{4.4}$$

where b_u is the derivative with respect to u_0. The identity $-ab_p(G) + d_\lambda b_u(G) = -c_\lambda$ supplies the leading cancellation; see Lemma D.1.

The limiting return is defined at a fixed positive input scale δ, with $\varepsilon, r \to 0$. Physical displacement and memory then vanish, but their normalized values still determine the release speed. The later expansion $\delta \downarrow 0$ is used to construct the graph and its phase. To define this exact return $T_{0,\lambda} = (f_0, h_0)$, set $r = 0$ and use identical values of the independent coordinates ξ. Put $\sigma = \sqrt{\delta}$ and $h_\delta = \mu A(\phi_\delta) - B(\phi_\delta)$. Use the compression clock $\tau_c = \sqrt{\delta}(t - t_{\rm in})/\varepsilon$ and scalings $(y, z) = \varepsilon\delta^{3/2}(Y, Z)$, $v = \delta^2 V_c$ and $u_0 = \delta\mathcal{U}$. With primes denoting τ_c derivatives, the saturated compression reference solves

$$\begin{gathered} Y' = V_c, \quad \sigma Z' = Y - 2Z, \quad V_c' = -a(Z - Y), \quad \mathcal{U}' = -h_\delta(Z - Y), \\ (Y, Z, V_c, \mathcal{U})(0) = (\sigma\zeta, \sigma\zeta, -r_0 W, d_\lambda(1 + \delta^{1/3}\eta_U)). \end{gathered} \tag{4.5}$$

At its first arrest $\mathcal{U} = 0$, switch to the elapsed release clock $\tau_r = (t - t_{\rm arrest})/\varepsilon$ and velocity $v = \delta^{3/2}V$. Thus $V = \sqrt{\delta}V_c$ at arrest, while Y and Z keep their compression scalings. With primes now denoting τ_r derivatives, the inherited normal motion $N = (Y, V, Z)$ solves

$$N' = A_D N, \qquad A_D = \begin{pmatrix} 0 & 1 & 0 \\ -1/2 & 0 & 1/2 \\ 1 & 0 & -2 \end{pmatrix}, \qquad N(0) = (Y, \sigma V_c, Z)_{\rm arrest}. \tag{4.6}$$

If R_0 is its first force zero and $Q = V(R_0)$, reconstruct the physical release as $(p, y, u_0, v, z) = (-a\delta, 0, 0, \delta^{3/2}Q, 0)$. Its exact free mechanical flow to the first subsequent touchdown defines $T_{0,\lambda}$. This retains the finite $\sqrt{\delta}$ compression and its inherited memory; the zero physical memory is introduced only in the release reconstruction.

For the initial state $(-2, 0, -1)$, Lemma B.1 gives $Y = Z = -\gamma_\#$ and $V = \nu_\#$ at the first release, with $\nu_\# > \gamma_\# > 0$. Define

$$\begin{gathered} q_\lambda = \nu_\#\sqrt{r_0 d_\lambda/h + a d_\lambda^2/(2h^2)}, \qquad K_\lambda = \sqrt{6q_\lambda/c_\lambda}, \\ \varkappa_W = \tfrac{1}{2} + \frac{3g_b}{2c_\lambda}, \qquad \xi_{*,\lambda} = K_\lambda^{-4/3}(0, -1, -\varkappa_W). \end{gathered} \tag{4.7}$$

4.2 Preparation of the original trajectory

Preparation serves two purposes. Its third-touchdown state must be accurate enough for repeated return comparison and for the entrance phase in Section 5. The whole preceding trajectory must also approach the prescribed incoming arc, so that later returns can be attached to it. We first record the state estimates and then the bounds on the intervening motion.

The capture and critical-passage estimates give a first release with $v_r = \varepsilon(q_{\rm in} + o(1))$, negative normal memory and a strict opening margin. Here $q_{\rm in} > 0$ depends only on (λ, θ), while $p_r/\sqrt{\varepsilon} \to H = ak/\gamma$ and $b_r/\sqrt{\varepsilon} \to -\ell H$ (Theorem A.4). With $c = c_\lambda$, the preparation constants are

$$\begin{gathered} cT_f^2 + 3\ell H T_f = 6q_{\rm in}, \quad T_f > 0, \qquad \pi = H - aT_f, \\ S = d_\lambda T_f, \quad Q = T_f(\ell H/2 + cT_f/3), \quad A_{\rm reb}^2 = QS/h - \pi S^2/(2h^2), \\ R = \nu_\# A_{\rm reb}, \qquad C_R = \sqrt{6R/c}. \end{gathered} \tag{4.8}$$

The scale $r = k\sqrt{\varepsilon}$ balances this first flight: with $t = \sqrt{\varepsilon}\,T$, the terms $v_r t$, $b_r t^2/2$ and $-ct^3/6$ in (C.3) all have order $\varepsilon^{3/2}$. The smaller memory value still determines opening on $t = O(\varepsilon)$ through $q_{\mathrm{in}} + Y_r^0 > 0$. Thus the same original preparation supplies the positive release coefficient and inherited memory for both exceptional flights. At the first touchdown, $p_1 = p(t_1)$ satisfies $p_1/\sqrt{\varepsilon} \to \pi$. Here $\sqrt{\varepsilon}$ is the preparation scale, and π may be positive, zero or negative. Its arrest margin is

$$2hQ - \pi S = T_f\{\gamma H + (2hc/3 + ad_\lambda)T_f\} > 0.$$

The signed contact theorem applies with $r = k\sqrt{\varepsilon}$ and yields $v_L = \varepsilon^{3/4}(R + o(1))$ and $y_L = z_L = -\varepsilon^{7/4}(\gamma_{\#}A_{\mathrm{reb}} + o(1))$. In particular both R and $R - \gamma_{\#}A_{\mathrm{reb}}$ have common positive lower bounds.

The ensuing touchdown occurs after an elapsed time T and has scale $D = -p(T)/a = C_R\varepsilon^{3/8}(1 + o(1))$. To obtain its complete tuple, one must eliminate the release velocity before dividing by the finer normal scale. A separate release-velocity approximation would be amplified by that division. At the actual root $y(T) = z(T)$, direct integration gives

$$-v(T) - \frac{c}{3}D^2 = \frac{y_L - z(T)}{T} - \frac{1}{2}b_L T + \frac{c}{3}(T^2 - D^2) - \frac{1}{2T}\int_0^T (T^2 - t^2)(\dot{b}(t) + c)\,\mathrm{d}t. \quad (4.9)$$

The tangential and acceleration numerators are $O(\sqrt{\varepsilon})$, and this normal numerator is $O(\varepsilon^{7/8})$. Their denominators $D^{4/3}, D^{4/3}, D^{7/3}$ have exactly these orders. Thus the actual second touchdown has a bounded full tuple, negative memory, a strict closing margin, and $r/D^{4/3} = kC_R^{-4/3}(1 + o(1))$; see Lemma C.3.

Proposition 4.1 (The original prepared entrance)**.** *Under the standing model convention, all the following estimates are uniform on the prescribed compact label classes. The original trajectory has an admitted third touchdown with*

$$\begin{aligned} &\delta_2 = C_R\varepsilon^{3/8}(1 + o(1)), \qquad \delta_3 \asymp \varepsilon^{9/32}, \quad r/\delta_3^{4/3} = O(\varepsilon^{1/8}), \\ &\left|\log\frac{\delta_3}{K_\lambda\delta_2^{3/4}}\right| + |\xi_3^r - \xi_{*,\lambda}| + |\eta_{B,3} - (g_b/c)\eta_{U,*,\lambda}| \le C\varepsilon^{3/32}. \end{aligned} \quad (4.10)$$

Here $\eta_{U,*,\lambda}$ is the slip component of $\xi_{*,\lambda}$. The third touchdown retains the full cone, inverse branch, negative normal memory, strict closing margin and all inequalities of (4.3)*, so it is a new admitted input. The preparation scale $\sqrt{\varepsilon}$ is followed by touchdown scales $\varepsilon^{3/8}$ and $\varepsilon^{9/32}$; only at the last does the normalized state approach $\xi_{*,\lambda}$. All estimates are uniform on the fixed compact label classes and are obtained from the original physical initial state.*

The entrance constant

$$\chi_\lambda(\mathfrak{a}) = \chi_{\lambda,\theta}(k) = 3\log K_\lambda - \tfrac{3}{4}\log C_R(\lambda, k, \theta), \qquad \mathfrak{a} = (k, \theta, \nu, \mathbf{n}), \quad (4.11)$$

is continuous, bounded and independent of $\nu, \mathbf{n}$. Moreover,

$$\log(K_\lambda^4/\delta_3^{\mathfrak{a}}) = \tfrac{9}{32}\log(1/\varepsilon) + \chi_\lambda(\mathfrak{a}) + r_{\mathfrak{a}}^{\log}(\varepsilon), \qquad \sup_{\mathfrak{a}} |r_{\mathfrak{a}}^{\log}(\varepsilon)| \to 0. \quad (4.12)$$

On each fixed closed incoming rigid subarc the original motion approaches that prescribed arc. For each fixed small $\eta > 0$, its remaining prefix from $p = \eta$ through TD_3 *is contained in $G + B_{C\eta+o(1)}$ in the full physical state. The motion has positive p up to the descending-slip threshold, and its mechanical distance from G tends uniformly to zero from that threshold through* TD_3.

Proof. Lemma C.3 and Corollary C.4 give the complete second input and its uniform scale asymptotic. Applying Theorems D.4 and D.5 to this input, as in Corollary C.5, gives (4.10) and the stated admissibility of the returned state. The normalized first-release data are continuous and have common positive margins by Lemma A.2 and Theorem A.4. The formulas (4.8) therefore give common positive bounds and continuity for C_R. Set

$$e_{\mathfrak{a}}(\varepsilon) = \log \frac{\delta_2^{\mathfrak{a}}}{C_R(\mathfrak{a})\varepsilon^{3/8}}.$$

The second-touchdown root estimates are uniform, so $\sup_{\mathfrak{a}} |e_{\mathfrak{a}}| \to 0$. Taking logarithms in (4.10) gives $r_{\mathfrak{a}}^{\log} = -\frac{3}{4}e_{\mathfrak{a}} + O(\varepsilon^{3/32})$, proving (4.12). The second-touchdown remainder needs no specified rate.

On a closed incoming arc $p \geq \eta$, the unilateral capture estimate in the proof of Lemma A.1 bounds the full scaled normal state through any initial departures and recontacts. Capture takes a common finite fast time, produces an $O(\varepsilon)$ slow increment, and is followed by an $O(\varepsilon)$ integrated slow-field discrepancy from the prescribed saturated rigid orbit; see (A.14). Gronwall's inequality gives convergence to that particular rigid subarc. Throughout it $|y| + |z| = O_\eta(\varepsilon^2)$ and $|v| = O_\eta(\varepsilon)$, so the fast excursion collapses to its original physical starting point.

For the remaining incoming segment, the entry estimates (A.15) control the normal state and its effect on the slow variables up to the descending-slip threshold. They apply on every stopped interval before that threshold, including the final $p \asymp \sqrt{\varepsilon}$ annulus. The weak-coordinate bounds and analytic chart put this segment in $G + B_{C\eta+o(1)}$; it remains at positive p until that threshold. The stopped comparison in Theorem A.4 controls every prefix through its first force zero, with mechanical distance $O(\sqrt{\varepsilon} + \varepsilon(1 + |\log \varepsilon|))$ from G. Proposition C.2 gives duration and mechanical state distance $O(\sqrt{\varepsilon})$ through the first exceptional touchdown. The ensuing contact uses the first application of Theorem B.2, with scale $\sqrt{\varepsilon}$ and $r = O(\sqrt{\varepsilon})$; its complete prefix bound is $O(\sqrt{\varepsilon})$. The second exceptional flight has duration and mechanical state distance from G both $O(\varepsilon^{3/8})$. The full-prefix estimates leading to Corollary C.4 control these passages and supply the actual complete TD_2 cone. Apply the contact prefix estimate (B.15) and Lemma D.2 to the $\mathrm{TD}_2 \to \mathrm{TD}_3$ return at $\delta_2 \asymp \varepsilon^{3/8}$. Its mechanical distance from G is $O(\varepsilon^{9/32})$ and its memory tends to zero. These bounds include both exceptional flights, their contacts and the passage to TD_3, and prove the stated original-prefix localization.

□

4.3 The return estimate and its domains

Let $\|\cdot\|_{1,*}$ include values and first derivatives with respect to $\delta\partial_\delta$ and ξ. The following conclusions collect Proposition D.3 and Theorems D.4–D.5. Write the actual return as

$$T_{\varepsilon,r}(\delta,\xi^r) = (D,\xi^{r,+}) = \big(f_{\varepsilon,r}(\delta,\xi^r), h^r_{\varepsilon,r}(\delta,\xi^r)\big).$$

The reference derivatives construct the invariant graph; the value error controls the actual iterates; the output conditions allow the next return. All constants are uniform on the fixed compact model classes and the prescribed finite input bounds.

Proposition 4.2 (The complete local return)**.** *For sufficiently small δ, the following statements hold.*

(i) Reference regularity. *On a bounded convex auxiliary fibre domain Ω containing the fixed independent input ball and $\mathcal{K}$, the exact reference has a C^1 extension with*

$$\begin{aligned}\left\|\log\frac{f_0}{K_\lambda\delta^{3/4}}\right\|_{1,*} + \|h_0-\xi_{*,\lambda}\|_{1,*} &\le C\delta^{1/4},\\ |D_\xi\log f_0| + |D_\xi h_0| &\le C\delta^{1/3}.\end{aligned} \tag{4.13}$$

The reference map and its state derivatives are jointly continuous in λ. The memory component of h_0 vanishes.

(ii) Compliant value approximation. *On the physical domain* (4.3)*, the actual first contact–flight return exists, retains its negative memory and transverse first-event margins, and satisfies*

$$\left|\log\frac{f_{\varepsilon,r}(\delta,\xi^r)}{f_0(\delta,\xi^r)}\right| + |h^r_{\varepsilon,r}(\delta,\xi^r) - h_0(\delta,\xi^r)| \le C(e+r/\delta). \tag{4.14}$$

(iii) Admissibility of the returned state. *There are common radii $0<\delta_{\mathrm{reset}}<\delta_{\mathrm{g}}<\delta_{\mathrm{call}}$ such that a physical input satisfying* (4.3) *with $\delta\le\delta_{\mathrm{reset}}$ has output D satisfying*

$$\begin{aligned}&\delta<D<\delta_{\mathrm{g}},\quad \xi^{r,+}\in\operatorname{int}\mathcal{K},\quad \varepsilon<C_0D^2,\quad r\le AD^{4/3},\\ &\left|\log\frac{D}{K_\lambda\delta^{3/4}}\right| + |\xi^{r,+}-\xi_{*,\lambda}| \le C(\delta^{1/4}+e+r/\delta).\end{aligned} \tag{4.15}$$

It has $u_r^+>0$, $v^+<0$, $y^+=z^+<0$, the same inverse branch, a bounded dependent coordinate and the strict closing margin. Another return requires $D\le\delta_{\mathrm{reset}}$. The outer radius may be reduced first, with the reset radius chosen afterwards.

The auxiliary extension in (4.13) continues the signed linear reference even when initial force is negative. For physical inputs, the proof of (4.14) first compares the actual return with a reference R^r reconstructed at slip $u_{0,L}=r$. This costs Ce. The references R^r and

T_0 share the entire finite-scale compression and inherited release, and their free flights differ by Cr/δ. Finally the actual output is recoded by

$$h^r_{\varepsilon,r} = h^0_{\varepsilon,r} - \frac{r}{d_\lambda D^{4/3}}(0,1,0).$$

The paired free-flight estimate has already established $D^{4/3} \asymp \delta$, so this last cost is also Cr/δ. The contact comparison retains both tail moments and the inherited normal state. In the elapsed release time ρ of Appendix B, put $w = u_r/\varepsilon$, $L = Z - Y$ and $\vartheta = B/A$. The normal-velocity projection is

$$P = \widehat{V} - e\vartheta w, \qquad \widetilde{N} = (Y, P, Z),$$
$$\widetilde{N}' = A_D\widetilde{N} + eF, \qquad F = (\vartheta w, b - \vartheta a_T - \vartheta' w, 0).$$

where primes denote ρ derivatives and $\vartheta' = \varepsilon\vartheta_\phi\dot{\phi}$. The exact identity $C - B^2/A = 1/2$ cancels the term containing w' when the normal and slip equations are combined. Thus F is bounded on the stopped boxes without estimating the stiff slip derivative. The full stopped-prefix estimates and first-release brackets are proved in (B.33)–(B.34). Only the exact reference return is differentiated in (4.13); (4.14) compares values of the actual compliant map.

5 The exact graph and comparison of actual returns

The logarithmic form of the exact limiting return contracts the complete independent state. We use it to construct an invariant graph and a normalized scalar coordinate that records progress along its returns. This coordinate fixes the explicit compliance sequences in Theorem B. The corresponding finite-scale contact and free curves retain the order of motion needed for path comparison.

Proposition 5.1 (An exact graph and normalized phase)**.** *After decreasing the outer radius δ_{g}, there are a common radius $0 < \delta_0 < \delta_{\mathrm{g}}$ and a jointly continuous graph $\Gamma_\lambda : (0, \delta_{\mathrm{g}}] \to \mathcal{K}$, with zero memory component, such that*

$$\begin{aligned} |\Gamma_\lambda(d) - \xi_{*,\lambda}| &\le Cd^{1/3}, \\ |\Gamma_\lambda(d_2) - \Gamma_\lambda(d_1)| &\le C\max(d_1,d_2)^{1/3}|\log(d_2/d_1)|. \end{aligned} \tag{5.1}$$

Its base map $F_\lambda(d) = f_{0,\lambda}(d, \Gamma_\lambda(d))$ is strictly increasing, $F_\lambda(d) > d$, and $\sup_\lambda F_\lambda(\delta_0) < \delta_{\mathrm{g}}$. For $0 < d \le \delta_0$,

$$T_{0,\lambda}(d, \Gamma_\lambda(d)) = (F_\lambda(d), \Gamma_\lambda(F_\lambda(d))). \tag{5.2}$$

On the outer interval there is a jointly continuous phase $\mathcal{S}_\lambda$, strictly decreasing and bi-Lipschitz in $\log d$, such that, with $\rho_c = 3/4$,

$$\mathcal{S}_\lambda(F_\lambda(d)) = \rho_c\mathcal{S}_\lambda(d), \qquad \mathcal{S}_\lambda(d) = \log(K_\lambda^4/d) + O(d^{1/3}). \tag{5.3}$$

The equation holds on the inner interval. The normalization fixes its germ at zero on this graph. All bounds are uniform in λ.

Proof. Put $x = \log(K_\lambda^4/\delta)$ and $\eta = \xi - \xi_{*,\lambda}$. Proposition D.3 gives

$$\widehat{T}_\lambda(x,\eta) = (\rho_c x + r_\lambda(x,\eta), H_\lambda(x,\eta)), \qquad |r| + |H| + |Dr| + |DH| \le Ce^{-x/4}. \tag{5.4}$$

For each λ the domain $[X,\infty)\times(\Omega - \xi_{*,\lambda})$ is convex. In the norm $\|(x,\eta)\| = \max\{|x|, \|\eta\|_\infty\}$ its derivative row sums are at most $\rho_c + Ce^{-X/4}$. Increase X to obtain

$$\|\widehat{T}_\lambda(z) - \widehat{T}_\lambda(\widetilde{z})\| \le q\|z - \widetilde{z}\|, \qquad q = 7/8. \tag{5.5}$$

The base segment here is logarithmic, hence geometric interpolation of its endpoint scales. It lies in the auxiliary domain of the exact reference; actual returns will be compared by (4.14) at their physical endpoints.

We seek a curve $V_\lambda(S) = (S + u_\lambda(S), v_\lambda(S))$, $S \ge S_0$, whose parameter is multiplied by ρ_c under the return:

$$V_\lambda(\rho_c S) = \widehat{T}_\lambda(V_\lambda(S)) \qquad (S \ge S_0/\rho_c).$$

Thus S will be the phase coordinate. To solve for the value at S, evaluate the right-hand side at S/ρ_c. The error $e^{-x/4}$ in (5.4) then has size $e^{-S/(4\rho_c)} = e^{-S/3}$, which determines the weight below. On continuous pairs for $S \ge S_0$ set $\|Z\|_0 = \sup_{S\ge S_0} e^{S/3}(|u(S)| + \|v(S)\|_\infty)$, and, writing $t = S/\rho_c$, define

$$(\mathcal{P}Z)(S) = \big(\rho_c u(t) + r_\lambda(t + u(t), v(t)), H_\lambda(t + u(t), v(t))\big).$$

On any fixed norm ball, all arguments lie in the common auxiliary domain for large S_0. Since $e^{S/3 - t/4} = 1$ and $e^{S/3 - t/3} = e^{-S/9}$,

$$\|\mathcal{P}0\|_0 \le C, \qquad \|\mathcal{P}Z - \mathcal{P}\widetilde{Z}\|_0 \le (\rho_c e^{-S_0/9} + C_M e^{-4S_0/9})\|Z - \widetilde{Z}\|_0.$$

Choose $M \ge \max(1, 2C)$ and then S_0 so the last coefficient is below $1/2$. The fixed point satisfies

$$u, v = O(e^{-S/3}), \qquad V_\lambda(S) = \widehat{T}_\lambda(V_\lambda(S/\rho_c)). \tag{5.6}$$

The Picard iterates from zero are C^1. Their weighted derivative norms $A_n = \sup e^{S/3}(|u_n'| + \|v_n'\|_\infty)$ obey

$$A_{n+1} \le C + (e^{-S_0/9} + C_M e^{-4S_0/9})A_n$$

by the chain rule and (5.4). They are uniformly bounded. Passing to the uniform limit in the integral inequality gives

$$|u(b) - u(a)| + \|v(b) - v(a)\|_\infty \le C\int_a^b e^{-t/3}\,\mathrm{d}t \quad (S_0 \le a < b). \tag{5.7}$$

Finite iterates are jointly continuous in (λ, S) on compact intervals, and their common contraction tail preserves this continuity. This argument uses only first state derivatives of the limiting return.

For larger S_0, choose $\eta < 1/2$ with $Ce^{-S_0/3} \le \eta$. Writing $x = S + u_\lambda(S)$, (5.7) gives

$$(1-\eta)|S_2 - S_1| \le |x_2 - x_1| \le (1+\eta)|S_2 - S_1|. \tag{5.8}$$

Thus $S \mapsto x$ is strictly increasing and bi-Lipschitz. On each fixed positive base interval this also gives two-sided Lipschitz control between the phase and the base itself. Define

$$\delta_\lambda(S) = K_\lambda^4 e^{-S-u_\lambda(S)}, \quad \Gamma_\lambda(\delta_\lambda(S)) = \xi_{*,\lambda} + v_\lambda(S), \quad \mathcal{S}_\lambda(\delta_\lambda(S)) = S. \tag{5.9}$$

Choose $\delta_{\mathrm{g}} < \min_\lambda \delta_\lambda(S_0)$ and $\delta_0 < \min_\lambda \delta_\lambda(S_0/\rho_c)$, within the limiting-return radii. Reduce δ_0 so that $\sup_\lambda F_\lambda(\delta_0) < \delta_{\mathrm{g}}$. Equation (5.6) gives $F_\lambda(\delta_\lambda(S)) = \delta_\lambda(\rho_c S) > \delta_\lambda(S)$ on the inner interval. Both factors are strictly decreasing in their arguments, so F_λ is strictly increasing there. It remains strictly increasing on the outer interval after reducing its radius: (4.13) and (5.7) give a positive lower bound for its logarithmic difference quotient. The same estimates give (5.1) and (5.3). The memory component of the fibre map vanishes, hence so does that of the graph. For two coordinates with the stated normalization on this graph, their difference satisfies $E(d) = \rho_c^j E(F_\lambda^{-j}(d)) \to 0$, proving uniqueness of the normalized germ. The unconstrained part of the outer interval is only a representative of that germ. □

The graph's logarithmic Lipschitz constant on $(0, \delta_{\mathrm{g}}]$ is $L_\Gamma \le C\delta_{\mathrm{g}}^{1/3}$. Reduce this outer radius, and then δ_0, so that $q(1 + L_\Gamma) < 1$. With the common graph domains fixed, choose δ_{reset} last as in Theorem D.5, and take $s_0 > 0$ with $2s_0 < \min(\delta_0, \delta_{\mathrm{reset}})$. All reference curves with input in a fixed positive compact subset of $(0, 2s]$ are then available in the same construction.

5.1 The ordered one-return curve

We spell out the finite-scale curve used in the path comparison. Let $p(\phi_d) = -ad$ on the selected branch and put $h_d = \mu A(\phi_d) - B(\phi_d) > 0$. Write $\Gamma_\lambda(d) = (0, \Gamma_U(d), \Gamma_W(d))$ and define

$$\begin{aligned} U_d &= d_\lambda d\{1 + d^{1/3}\Gamma_U(d)\}, \qquad V_d^{\mathrm{in}} = -r_0 d^2\{1 + d^{1/3}\Gamma_W(d)\}, \\ V_d^{\mathrm{a}} &= V_d^{\mathrm{in}} - (ad/h_d)U_d, \qquad \mathcal{E}_\lambda(d,u,v) = (\phi_d, 0, (u - U_*)/\cos\phi_d, v, 0). \end{aligned} \tag{5.10}$$

Here u denotes the tuned slip argument of the embedding. Run the exact reference compression at $\sigma = \sqrt{d}$:

$$Y' = V_c, \quad \sigma Z' = Y - 2Z, \quad V_c' = -a(Z - Y), \quad \mathcal{U}' = -h_d(Z - Y), \tag{5.11}$$

from $(Y, Z, V_c, \mathcal{U}) = (0, 0, V_d^{\mathrm{in}}/d^2, U_d/d)$. At its first arrest, start the inherited reference $N_d' = A_D N_d$, $N_d = (Y_d, V_d^{\mathrm{n}}, Z_d)$, with $N_d(0) = (Y, \sigma V_c, Z)$ at arrest. Let R_d be its first force zero. The first-event margins from Section B give

$$Z_d(R_d) = Y_d(R_d) < 0, \quad V_d^{\mathrm{n}}(R_d) > 0, \quad V_d^{\mathrm{n}}(R_d) + Z_d(R_d) > 0. \tag{5.12}$$

Put $V_d^{\mathrm{rel}} = d^{3/2} V_d^{\mathrm{n}}(R_d)$ and $\mathcal{R}_d = \mathcal{E}_\lambda(d, 0, V_d^{\mathrm{rel}})$. Write $\Phi_{\mathrm{f}}(\ell; \mathcal{R}_d)$ for the analytic free mechanical flow, with zero memory appended. Its first subsequent touchdown has elapsed time T_d and scale $F_\lambda(d)$.

The continuous ordered curve is

$$\mathcal{C}_{\lambda,d}(\alpha)=\begin{cases}\mathcal{E}_\lambda(d,(1-\alpha)U_d,V_d^{\mathrm{in}}-(ad/h_d)\alpha U_d), & 0\le\alpha\le 1,\\ \mathcal{E}_\lambda(d,0,d^{3/2}V_d^{\mathrm{n}}((\alpha-1)R_d)), & 1\le\alpha\le 2,\\ \Phi_{\mathrm{f}}((\alpha-2)T_d;\mathcal{R}_d), & 2\le\alpha\le 3.\end{cases}\tag{5.13}$$

The compression invariant $V_c-(a/h_d)\mathcal{U}=V_d^{\mathrm{in}}/d^2-(a/h_d)U_d/d$ gives $d^{3/2}V_d^{\mathrm{n}}(0)=V_d^{\mathrm{a}}$, so the first join agrees. The second join is $\mathcal{R}_d$. For $0<d\le\delta_0$, graph invariance gives

$$\mathcal{C}_{\lambda,d}(3)=\mathcal{C}_{\lambda,F_\lambda(d)}(0),\qquad \sup_\alpha|\mathcal{C}_{\lambda,d}(\alpha)-G|\le Cd^{3/4}.\tag{5.14}$$

The contact pieces have instantaneous physical duration in this reference curve; their parameters preserve the order of compression and release. They are not extra free solutions. At every fixed positive compact input interval, including the enlarged intervals used to scan $[s/2,2s]$, the event maps, free flow and graph bound make (5.13) Lipschitz in its input, uniformly in λ. The corresponding constants may depend on that interval.

The same radius choice makes $U_d>0$. Consequently the contact curve is injective on $[0,2]$: the tuned slip $(1-\alpha)U_d$ strictly decreases on compression, while on release it is zero and the normal velocity satisfies

$$\frac{\mathrm{d}V_d^{\mathrm{n}}}{\mathrm{d}\tau}=\tfrac{1}{2}(Z_d-Y_d)>0\qquad(0\le\tau<R_d).\tag{5.15}$$

Positivity of the active force gives the strict inequality, and the two pieces meet only at their common arrest point. We will use this fact both to identify an interior stopping target and to separate complete path images.

5.2 The free-section parameter

Along the free part, $\dot p=-a+O(d+\ell)<-a/2$ after the common radius is reduced. Thus p strictly decreases from $-ad$ to $-aF_\lambda(d)$. Fix $0<s\le s_0$ and put $d_*=F_\lambda^{-1}(s)$; this inverse is defined since F is continuous and strictly increasing from zero and $F(s)>s$. For $d_*\le d\le s$, the equation

$$p(\Phi_{\mathrm{f}}(\ell_s(d);\mathcal{R}_d))=-as\tag{5.16}$$

has exactly one root in $[0,T_d]$. Its derivative in ℓ is bounded away from zero, so its continuous dependence, including the endpoints, follows either from the inverse function theorem or directly from strict monotonicity and compactness. Hence

$$j_s(d)=2+\ell_s(d)/T_d,\qquad j_s(d_*)=3,\qquad j_s(s)=2\tag{5.17}$$

is the continuous parameter in (3.4). On contact the section value is instead the constant $p=-ad$; therefore its section points have $d=s$ and $0\le\alpha\le 2$. Together these observations give exactly the terminal domain $\mathcal{A}_s$ of (3.5).

5.3 Growth, the error budget and the first section

Fix all compact classes and the finite cones in Theorem D.5. Reduce the graph outer radius first, and then take its compatible reset cutoff, so that

$$0 < \delta_{\mathrm{reset}} < \delta_{\mathrm{g}} < \delta_{\mathrm{call}}, \qquad q_{\mathrm{g}} := (1 + L_\Gamma)q < 1,$$

where $q < 1$ is the full logarithmic contraction in (5.5) and L_Γ is the logarithmic Lipschitz constant of the graph on the outer interval. Choose a fixed $\Delta < \delta_{\mathrm{reset}}$ below every inner input cutoff. It is also chosen so small that

$$c_D \Delta^{-1/4} > 2, \qquad c_D \delta^{3/4} \le \delta^+ \le C_D \delta^{3/4} \quad (0 < \delta \le \Delta), \tag{5.18}$$

and the full-prefix tubes below lie in a common neighborhood $\mathcal{U}_G$ with $p_\phi \omega < -a/2$. These choices precede the choice of the observation section.

Write $\mathcal{B}_\lambda(d)$ for the value of the dependent coordinate $\eta_B = (-b - c_\lambda d)/(c_\lambda d^{4/3})$ at the graph state $(d, \Gamma_\lambda(d))$. It is obtained by substituting that state into the physical inverse (4.2) with $\varepsilon = r = 0$ and evaluating b. In the notation of Appendix D, this is $\mathcal{B}_{0,0}(d, \Gamma_\lambda(d))$; the corresponding Lipschitz estimate is (D.18).

Lemma 5.2 (Stopped cascade and its complete state budget)**.** *For each fixed $0 < h \le \Delta$, the actual returns from* TD_3 *can be generated, on the original trajectory, up to*

$$N_h = \min\{n \ge 3 : \delta_n \ge h\}.$$

This index is finite, tends to infinity as $\varepsilon \downarrow 0$, and every input used before it is below the reset and inner cutoffs. The terminal output remains in the graph outer domain. The complete cone, inverse branch, negative memory and strict first-event margins are retained. Put

$$e_n = \varepsilon \delta_n^{-3/2}, \quad \omega_n = e_n + r/\delta_n, \quad B_n = \|\xi_n^r - \Gamma_\lambda(\delta_n)\|_\infty.$$

For $3 \le m \le N_h$,

$$B_m \le q_{\mathrm{g}}^{m-3} B_3 + C \sum_{n=3}^{m-1} q_{\mathrm{g}}^{m-1-n} \omega_n, \tag{5.19}$$

$$\left| \log \frac{\delta_{n+1}}{F_\lambda(\delta_n)} \right| \le q B_n + C\omega_n \quad (3 \le n < N_h), \tag{5.20}$$

$$\Omega_\varepsilon := \sup_{3 \le n < N_h} \omega_n \le C(\varepsilon^{37/64} + \varepsilon^{7/32}) \longrightarrow 0. \tag{5.21}$$

If an actual and a limiting chain are compared in full logarithmic state, their distance satisfies, through the first output at which either input chain leaves the common inner domain,

$$E_m \le q^{m-3} E_3 + C \sum_{n=3}^{m-1} q^{m-1-n} \omega_n. \tag{5.22}$$

At any sequence of positive-scale inputs whose indices tend to infinity, the dependent acceleration coordinate also tends to its graph value. Thus this convergence concerns the complete normalized touchdown state, including memory.

Proof. Proposition 4.1 gives the actual entrance

$$\delta_3 \asymp \varepsilon^{9/32}, \qquad \varepsilon \le C_0\delta_3^2, \qquad r \le A\delta_3^{4/3}, \qquad x_3 = \tfrac{9}{32}\log(1/\varepsilon) + O(1), \quad x = \log(K_\lambda^4/\delta). \tag{5.23}$$

The admissibility of the returned state in Theorem D.5 and (5.18) give strict scale growth. Since ε and $r = k\sqrt{\varepsilon}$ stay fixed on each trajectory, both admissibility inequalities propagate to the next input. The return estimate is applied again when its new input is below $h \le \Delta$. All events are the actual first release and subsequent first touchdown, with positive gap between them. If there were infinitely many inputs below h, (5.18) would give $\delta_n \ge 2^{n-3}\delta_3$, a contradiction. Conversely every fixed finite index has $\delta_n \to 0$, by (5.23) and the growth upper bound. Consequently $N_h \to \infty$ uniformly on the compact classes.

The direct value comparison in Theorem D.4 is made at identical values of the independent normalized touchdown coordinates and gives a logarithmic output defect at most $C\omega_n$. Integrating the derivative bound for the limiting map along the logarithmic joining segment gives $E_{n+1} \le qE_n + C\omega_n$, and hence (5.22). Only the limiting map is differentiated. To compare with the graph, take its input at the actual base δ_n. Graph invariance and the same argument give

$$\max\left\{\left|\log\frac{\delta_{n+1}}{F_\lambda(\delta_n)}\right|, \|\xi_{n+1}^r - \Gamma_\lambda(F_\lambda(\delta_n))\|_\infty\right\} \le qB_n + C\omega_n.$$

Both output bases are in the outer graph interval. Correcting the graph base there costs $L_\Gamma(qB_n + C\omega_n)$, so $B_{n+1} \le q_{\mathrm{g}}B_n + C\omega_n$. Iteration proves (5.19). The bounded entrance error and the two geometric sums give bounds independent of the number of returns. Scale growth also gives

$$\Omega_\varepsilon \le \varepsilon\delta_3^{-3/2} + r/\delta_3 \le C(\varepsilon^{37/64} + \varepsilon^{7/32}),$$

which proves (5.21).

To compare complete states, recover the acceleration row from the same physical inverse. At a common base d, the actual and graph arguments differ by

$$\begin{aligned} &\Delta p = 0, \qquad \Delta u_0 = d_\lambda d^{4/3}(\eta_U^r - \Gamma_U(d)) + r, \\ &\Delta v = -r_0 d^{7/3}(\eta_W - \Gamma_W(d)), \qquad \Delta y = \varepsilon d^2\zeta. \end{aligned} \tag{5.24}$$

Smoothness of that inverse and division by $c_\lambda d^{4/3}$ give

$$|\eta_B - \mathcal{B}_\lambda(d)| \le C\{B_n + r/d^{4/3} + \varepsilon d^{2/3}|\zeta|\}. \tag{5.25}$$

On any fixed positive-scale compact interval this tends to zero when $n \to \infty$ and $\varepsilon \to 0$. The memory component of the graph vanishes, while the actual values at touchdown remain exactly $y = z = \varepsilon d^2\zeta$. No normal coordinate is reset. □

We record the prefix estimates needed to turn this state budget into a first-crossing and path argument. The contact estimates (B.15), followed by Lemma D.2, give, for every

admitted input $\delta \le \Delta$ and every actual prefix,

$$\begin{gathered}\sup_{\text{contact}} |X-G| \le C\delta, \qquad t_L - t_{\mathrm{TD}} \le C\varepsilon/\sqrt{\delta}, \\ \sup_{\text{contact}} (|y|+|z|) \le C\varepsilon\delta^{3/2}.\end{gathered} \tag{5.26}$$

$$\begin{gathered}T := t_+ - t_L \le H\delta^{3/4}, \qquad |X(t_L+\ell)-G| \le C(\delta+\ell), \\ z(t_L+\ell) = z_L e^{-\ell/\varepsilon} < 0 \qquad (0 \le \ell \le T).\end{gathered} \tag{5.27}$$

In particular, since $|b_L| \le C\delta$ and $b(t_L+\ell) = O(\delta+\ell)$, integration of $\dot v = b$ and $\dot y = v$ gives

$$\begin{gathered}|z|/\varepsilon \le C\delta^{3/2}, \qquad |v| \le C(\delta^{3/2} + \delta\ell + \ell^2), \\ |y| \le C(\varepsilon\delta^{3/2} + \delta^{3/2}\ell + \delta\ell^2 + \ell^3) \le C\delta^{9/4}.\end{gathered} \tag{5.28}$$

The contact velocity estimate uses $e/\sqrt{\delta} \le C_0$ in the weighted compression bounds. Thus these estimates remain valid without a positive lower bound on e.

Lemma 5.3 (Original preparation and the first physical section)**.** *The whole actual curve from time zero through* TD_3 *tends to the prescribed incoming arc* $I_{\lambda,\nu}$ *in the completed ordered distance.*

There is a common $s_0 > 0$ *such that for fixed* $0 < s \le s_0$ *the first crossing* t_s *of* $p = -as$ *from the original initial state exists and is transverse. With* $N = N_s$*,*

$$t_{N-1} < t_s \le t_N, \qquad c_s := (s/C_D)^{4/3} \le d_\varepsilon = \delta_{N-1} < s, \tag{5.29}$$

and the complete same-base defect

$$\kappa_\varepsilon = |\xi^r_{N-1} - \Gamma_\lambda(d_\varepsilon)| + |\eta_{B,N-1} - \mathcal{B}_\lambda(d_\varepsilon)| \longrightarrow 0. \tag{5.30}$$

All statements here are uniform on the fixed compact label classes.

Proof. Fix a small incoming cutoff $\eta > 0$. Proposition 4.1 gives convergence to the corresponding rigid prefix. Transverse alignment on that closed subarc gives an increasing pairing. The remaining rigid tail lies in $G + B_{C\eta}$, while the proposition puts the remaining original prefix through TD_3 in $G + B_{C\eta+o(1)}$. Pair these tails in their order and then let $\eta \downarrow 0$; this proves the ordered incoming limit.

For the cascade up to N_s, take the supremum of the local bounds:

$$\sup_{[t_3,t_N]} |X-G| \le Cs^{3/4}, \qquad \sup_{[t_3,t_N]} |z|/\varepsilon \le Cs^{3/2}. \tag{5.31}$$

There is no sum over returns. Choose $0 < s_0 < \Delta/2$ so this tube lies in $\mathcal{U}_G$. The exact identity

$$\dot p = p_\phi(\phi)\omega < -a/2 \tag{5.32}$$

then holds on every contact and free prefix and at their joins. The entrance has $\delta_3 < s$, whereas $\delta_N \ge s$, giving a unique crossing between the stated events. On the original closed

incoming prefix p is positive; the part from its small cutoff to the threshold also has positive p. From the threshold through TD_3, Proposition 4.1 gives $\sup|p| = o(1)$. For fixed s, this excludes an earlier crossing of $-as$. Hence the crossing is the first one from the original initial data. Only after the tube has been established, integration of (5.32) also gives $t_s - t_3 \le 2(s - \delta_3)$.

The last growth upper bound gives $d_\varepsilon \ge c_s$. The last-input index tends to infinity, so (5.19) and (5.25) give (5.30) on $[c_s, s]$. □

Corollary 5.4 (Localization of an actual contact stop)**.** *Fix $0 < s \le s_0$. If an actual touchdown generated by the return estimates has $\delta_j \in [s/2, 2s]$ and the first section time satisfies $t_j < t_s \le t_{L,j}$, then*

$$0 < s - \delta_j \le C_s\varepsilon, \qquad 0 < t_s - t_j \le C_s\varepsilon. \tag{5.33}$$

The constant is uniform on the prescribed compact label classes.

Proof of Corollary 5.4. On the fixed band $[s/2, 2s]$, (5.26) gives $t_{L,j} - t_j \le C_s\varepsilon$. Since $\dot p$ is bounded and satisfies (5.32) throughout this contact,

$$0 < s - \delta_j = -\frac{1}{a}\int_{t_j}^{t_s} \dot p \, \mathrm{d}t \le C_s(t_s - t_j) \le C_s\varepsilon.$$

The same contact bound gives the asserted time estimate. □

For a first section time during contact j, Lemma 5.3 gives $j = N_s - 1$ and $\delta_j \in [c_s, s)$. The same contact-time argument on this fixed positive band gives $s - \delta_j = O_s(\varepsilon)$, so the input eventually lies in $[s/2, 2s]$. Thus (5.33) applies to every actual contact stop, without assuming the band in advance.

5.4 Sharp comparison from the actual third touchdown

Fix λ and a compact auxiliary class $\mathfrak{K}$ within the standing ranges. Put $A_0 = 9/32$, $\rho_c = 3/4$ and $L = \log(1/\varepsilon)$. The scalar phase contracts by ρ_c; the complete-state contraction is $q = 7/8$. The entrance state is already $O(\varepsilon^{3/32})$ from the graph at its own base, so its complete-state budget resolves the residual entrance constant through all required returns.

Lemma 5.5 (Same-index phase accuracy)**.** *Uniformly for $\mathfrak{a} \in \mathfrak{K}$ and $3 \le n \le N_s^{\mathfrak{a}}$,*

$$\mathcal{S}_\lambda(\delta_n^{\mathfrak{a}}) = \rho_c^{n-3}\{A_0 L + \chi_\lambda(\mathfrak{a}) + r_{\mathfrak{a}}(\varepsilon)\} + O(\varepsilon^{3/32}), \qquad L = \log(1/\varepsilon), \tag{5.34}$$

where the entrance remainder satisfies $\sup_{\mathfrak{a}\in\mathfrak{K}}|r_{\mathfrak{a}}(\varepsilon)| \to 0$. The same comparison holds through any prescribed index $j = j(\varepsilon)$ whose graph chain started at the actual base $\delta_3^{\mathfrak{a}}$ has its jth base below $3s/2$, even if the observation time precedes j.

Proof. The uniform logarithmic entrance in (4.12) and the exact-phase expansion give

$$\mathcal{S}_\lambda(\delta_3^{\mathfrak{a}}) = A_0 L + \chi_\lambda(\mathfrak{a}) + r_{\mathfrak{a}}(\varepsilon), \qquad r_{\mathfrak{a}} = r_{\mathfrak{a}}^{\log} + O(\varepsilon^{3/32}), \tag{5.35}$$

where the remainder tends to zero uniformly. We preserve its parameter dependence by starting at the actual base.

Start the exact graph chain at the actual δ_3, with fibre $\Gamma_\lambda(\delta_3)$. The sharp third-entry estimate (4.10) and the graph bound give

$$E_3 = B_3 \le |\xi_3^r - \xi_{*,\lambda}| + |\Gamma_\lambda(\delta_3) - \xi_{*,\lambda}| = O(\varepsilon^{3/32}). \tag{5.36}$$

The equality $E_3 = B_3$ uses the identical starting base. The two-chain budget (5.22) gives

$$E_n \le q^{n-3} B_3 + C \sum_{i=3}^{n-1} q^{n-1-i} \omega_i, \qquad \sup_i \omega_i \le C(\varepsilon^{37/64} + \varepsilon^{7/32}).$$

The geometric sum is bounded by 8. Thus

$$E_n = O(\varepsilon^{3/32}) \tag{5.37}$$

through every comparison within those domains, without a factor counting the returns. All constants are common on $\mathfrak{K}$, since the entrance, return and graph bounds are common. In particular the bound persists at fixed positive scales.

Let $\bar{N}_{2s}$ be the first graph index whose base is at least $2s$. Compare until $M = \min(N_s, \bar{N}_{2s})$. Before M both inputs lie below $2s < \Delta$ and the inner cutoffs; the estimate also applies to their first output M in the outer domain. If $\bar{N}_{2s} < N_s$, the graph base at M is at least $2s$ while the actual base is less than s. Their logarithmic distance is then at least $\log 2$, contradicting (5.37). Thus the comparison covers all $n \le N_s$. No further iteration at an output outside the input domain has been used.

For the last assertion, generate the original trajectory until N_{2s}, as permitted by Lemma 5.2. The graph bases through j are below $3s/2$, since $F(d) > d$. If the actual chain reached $2s$ at or before j, its first such output would have logarithmic distance at least $\log(4/3)$ from the graph base. This again contradicts (5.37). The actual events through j therefore exist and admit the stated comparison on the underlying trajectory.

The graph phase at n is exactly $\rho_c^{n-3}\mathcal{S}_\lambda(\delta_3)$. The two-sided bound (5.8) transfers the logarithmic base error (5.37) to phase. Substitution of (5.35) proves (5.34). □

6 Actual realization and complete stopped paths

The comparison in Section 5 controls actual touchdown states and the first physical section. We now recover the whole motion between those events. On fixed positive input bands we pair all contact and flight prefixes. A continuous scan of one unchanged contact index realizes its stopping phase; matching finitely many last returns and controlling the entire earlier tail then gives the complete path.

6.1 Ordered comparison on every positive-scale compact set

The realization argument scans $[s/2, 2s]$, including inputs above the observation scale. We therefore compare whole returns on any positive-scale compact interval where the return estimates apply.

Lemma 6.1 (A whole return and all its prefixes)**.** *Let $K \Subset (0, \delta_{\mathrm{reset}})$ lie below the inner return cutoffs. Consider actual inputs with scales $d \in K$, generated within the return domains, and with complete same-base defect $\kappa = |\xi^r - \Gamma_\lambda(d)| + |\eta_B - \mathcal{B}_\lambda(d)| \to 0$. There is an increasing homeomorphism $h : [0,3] \to [t_0, t_1]$ for their actual return, with*

$$\begin{gathered} h(0) = t_0, \quad h(1) = t_{\mathrm{sl}}, \quad h(2) = t_L, \quad h(3) = t_1, \\ \sup_{\alpha \in [0,3]} |\mathfrak{X}_\varepsilon(h(\alpha)) - \mathcal{C}_{\lambda,d}(\alpha)| \le C_K(\varepsilon + r + \kappa). \end{gathered} \tag{6.1}$$

Here t_{sl} is the actual first descending $u_r/\varepsilon = M_{\mathrm{sl}}$ event, t_L the actual first release and t_1 the next first touchdown. On compression h is fixed by

$$J(h(\alpha)) = \alpha J_{\mathrm{sl}}, \quad J(t) = \int_{t_0}^{t} \lambda_N \, \mathrm{d}t, \quad J_{\mathrm{sl}} = J(t_{\mathrm{sl}}), \qquad 0 \le \alpha \le 1; \tag{6.2}$$

on release and flight it is

$$h(1+q) = t_{\mathrm{sl}} + q(t_L - t_{\mathrm{sl}}), \qquad h(2+q) = t_L + q(t_1 - t_L), \quad 0 \le q \le 1. \tag{6.3}$$

The same estimate pairs every stopped prefix in both directions. Moreover, in the normal coordinates $y, z = \varepsilon d^{3/2}(Y, Z)$, $v = d^{3/2}V$, at the actual first release,

$$\left| v_L - d^{3/2} V_d^{\mathrm{n}}(R_d) \right| + \left| z_L/\varepsilon - d^{3/2} Z_d(R_d) \right| \le C_K(\varepsilon + r + \kappa). \tag{6.4}$$

The flight durations have a common positive lower bound on K.

Proof. Write $\chi = \varepsilon + r + \kappa$. On the fixed positive compact set, $e = \varepsilon/d^{3/2} = O_K(\varepsilon)$ and all weighted normal estimates give unweighted estimates with constants depending on K. By (5.26), the contact lasts $O_K(\varepsilon)$, and its coefficients have bounded drift. For $t \le t_{\mathrm{sl}}$, put $f_\theta(w) = 1 - \Phi_\theta(w)$ and $\mathcal{H}_\theta(w) = \int_{M_{\mathrm{sl}}}^{w} f_\theta(q) \, \mathrm{d}q$. The profile tail gives $0 \le \mathcal{H}_\theta \le C$. The fixed slip threshold of Appendix B gives $H_\theta(u_r/\varepsilon, X) \ge \kappa_H > 0$ on this interval, and $\dot{u}_r = a_T - H_\theta(u_r/\varepsilon, X)\lambda_N$. Since $J(t) \le C_K$, the identity for $\varepsilon \mathcal{H}_\theta(u_r/\varepsilon)$ implies on every prefix

$$0 \le \int_{t_0}^{t} f_\theta(u_r/\varepsilon)\lambda_N \, \mathrm{d}t \le \kappa_H^{-1} \left\{ \int_{t_0}^{t} f_\theta(u_r/\varepsilon) a_T \, \mathrm{d}t - \varepsilon [\mathcal{H}_\theta(u_r/\varepsilon)]_{t_0}^{t} \right\} \le C_K \varepsilon. \tag{6.5}$$

The two exact velocity rows consequently give, simultaneously on all compression prefixes,

$$u_r(t) = u_r(t_0) - h_d J(t) + O_K(\varepsilon), \qquad v(t) = v(t_0) - a d J(t) + O_K(\varepsilon). \tag{6.6}$$

Coefficient drifts multiply a bounded impulse, while the free-field integrals and the unsaturated terms are controlled as above. At t_{sl}, $u_r = \varepsilon M_{\mathrm{sl}}$, so $J_{\mathrm{sl}} = u_r(t_0)/h_d + O_K(\varepsilon) \geq j_K > 0$. The force is positive on (t_0, t_L); hence J/J_{sl} is continuous and strictly increasing through compression. This proves that (6.2) is a homeomorphism and, by (6.6), gives precisely the linear impulse parameter of $\mathcal{C}_{\lambda,d}|_{[0,1]}$. Reconstructing $\omega = (u_r + r - U_*)/\cos\phi$ and using $|y| + |z| \leq C_K\varepsilon$ proves (6.1) there.

From the actual M_{sl} event use elapsed normal time $\tau = (t - t_{\mathrm{sl}})/\varepsilon$. The full inherited-state estimate (B.9) first compares to the inherited reference for that actual normalized touchdown input. Move this reference to the graph input afterwards. On K, its input sensitivity and (B.10) give

$$|(Y, V, Z)(\tau) - N_d(\tau)| + |L_\varepsilon(\tau) - L_d(\tau)| + |L'_\varepsilon(\tau) - L'_d(\tau)| \leq C_K\chi \tag{6.7}$$

on every stopped release prefix. The positive-force interval and strict first-root bracket yield

$$R_\varepsilon = (t_L - t_{\mathrm{sl}})/\varepsilon, \qquad |R_\varepsilon - R_d| \leq C_K\chi.$$

For $1 \leq \alpha \leq 2$, compare at actual time $(\alpha - 1)R_\varepsilon$ and then move only the linear reference to $(\alpha - 1)R_d$. The strip bound gives $u_0 = r + O_K(\varepsilon)$, and the remaining physical coordinates follow from the chart and (6.7). This proves the release portion of (6.1), including its moving endpoint, and also (6.4). In particular $|z_L| \leq C_K\varepsilon$. The opening margins give $\min(v_L, v_L + z_L/\varepsilon) \geq v_K > 0$.

On the analytic free extensions the mechanical field is smooth and independent of belt translation. Gronwall, on their common compact time interval, compares the actual-release and graph-release flows with error $C_K\chi$. Their exact effective gap is

$$g(\ell) = z_L(1 - e^{-\ell/\varepsilon}) + \int_0^\ell v(q)\ \mathrm{d}q. \tag{6.8}$$

To exclude an additional root next to release, use $|b| \leq B_K$:

$$g'(\ell) \geq \min(v_L, v_L + z_L/\varepsilon) - B_K\ell > 0$$

on a fixed short opening interval. Thereafter the reference gap has a positive minimum away from its terminal root. In a common terminal bracket its velocity has a negative margin, and the memory derivative is uniformly negligible because that bracket is bounded away from release. The actual gap has opposite endpoint signs and negative derivative there. Its first subsequent zero therefore is the actual touchdown, with $|T_\varepsilon - T_d| \leq C_K\chi$. This also gives a positive common lower bound for T_ε. Moving the reference time from qT_ε to qT_d proves the flight portion of (6.1). The three clocks agree at their joins and are strictly increasing. Restricting the same pairing to any prefix proves both image inclusions as well as its ordered comparison. □

Apply this lemma to the last input in (5.30) and write $\alpha_\varepsilon = h_\varepsilon^{-1}(t_s)$. Compactness gives subsequences $(d_\varepsilon, \alpha_\varepsilon) \to (d, \alpha)$, and the actual section equation gives

$$p(\mathcal{C}_{\lambda,d}(\alpha)) = -as. \tag{6.9}$$

The contact has $p = -ad$, and the free part decreases strictly from $-ad$ to $-aF_\lambda(d)$. Its complete solution set is therefore

$$\mathcal{A}_s = \{(d, j_s(d)) : F_\lambda^{-1}(s) \le d \le s\} \cup (\{s\} \times [0, 2]). \tag{6.10}$$

This includes a crossing at the next touchdown even if the varying reference curve has not itself crossed before its own endpoint. Every terminal prefix converges in order, since

$$\sup_{q\in[0,1]} |\mathfrak{X}_\varepsilon(h_\varepsilon(q\alpha_\varepsilon)) - \mathcal{C}_{\lambda,d}(q\alpha)| \longrightarrow 0. \tag{6.11}$$

For $\alpha = 0$ the limiting piece is constant, as allowed by the completed ordered topology.

6.2 Actual realization of the terminal domain

Reduce s_0 once more so that $2s_0$ lies below all inner input cutoffs and, for $0 < s \le s_0$,

$$2s < \Delta, \qquad (2s/c_D)^{4/3} < s/3, \qquad c_D(s/2)^{3/4} > s. \tag{6.12}$$

Lemma 6.2 (A continuous sweep of one actual contact)**.** *For each fixed complete label and every sufficiently large j there is an interval $J_j = [a_j, c_j]$, $c_j \to 0$, on which the same indexed actual touchdown, contact and next return exist continuously, and*

$$\delta_j(a_j) = s/2, \qquad \delta_j(c_j) = 2s, \qquad s/2 \le \delta_j(\varepsilon) \le 2s \quad (\varepsilon \in J_j). \tag{6.13}$$

Every earlier controlled touchdown is below $s/3$, and the next touchdown is above s. The ordered comparison (6.1) *holds uniformly on J_j with error tending to zero as $j \to \infty$.*

Proof. At positive compliance the positive-part vector field is locally Lipschitz and depends continuously on compliance. The prescribed initial data are continuous. The strict opening, closing, first M_{sl} and first-release brackets consequently give continuous dependence of any fixed finite sequence of the actual events. The needed local constants are only on sets bounded away from zero in compliance and input scale. Positive gap between successive first events prevents a change of the indexed passage.

Fix a common $\varepsilon_* > 0$ so that the original entrance and its admission hold for $0 < \varepsilon < \varepsilon_*$. For fixed j, the entrance and growth upper bound imply $\delta_j(\varepsilon) \to 0$; all preceding inputs eventually satisfy the return hypotheses. Let $O_j \subset (0, \varepsilon_*)$ be the open set on which the events through TD_j are generated with $\delta_n < \Delta$ for $3 \le n < j$. Its component adjacent to zero is $(0, b_j)$. These components are nested. For every fixed $\varepsilon_0 \in (0, \varepsilon_*)$, finite exit at that compliance excludes ε_0 from O_j for sufficiently large j; hence $b_j \to 0$. In particular $b_j < \varepsilon_*$ for all sufficiently large j, so its right endpoint is not the artificial end of the total compliance interval. At such a positive endpoint b_j, the finitely many scales have a positive lower bound and their full states and strict margins stay in compact sets. Extend the finite events to that endpoint successively. Inputs equal to Δ are allowed because $\Delta < \delta_{\mathrm{reset}}$. No earlier input $\delta_n(b_j)$, $n \le j - 2$, can equal Δ: its next output would be larger than Δ,

contradicting the left limiting inequality for that next input. If also $\delta_{j-1}(b_j) < \Delta$, all finite events and strict input inequalities would continue past b_j. Hence

$$\delta_{j-1}(b_j) = \Delta, \qquad \delta_j(b_j) > \Delta > 2s.$$

Only this endpoint call uses Δ; its outer output is not used as a new input.

Take the first compliance $c_j < b_j$ at which $\delta_j = 2s$ and the last preceding compliance a_j at which $\delta_j = s/2$. These first and last choices give (6.13) on the whole interval by continuity. Its scanned input is itself below Δ, so the return estimates apply to the next return throughout. The growth lower bound and (6.12) give

$$\delta_{j-1} \le (2s/c_D)^{4/3} < s/3, \qquad \delta_{j+1} \ge c_D(s/2)^{3/4} > s.$$

Earlier scales are smaller by strict growth. Finally the budget gives $B_j \le q_{\mathrm{g}}^{j-3} B_3 + C\Omega_\varepsilon \to 0$ uniformly on J_j; (5.25) recovers the full defect on $[s/2, 2s]$. Lemma 6.1 applies on this entire positive compact set, proving the last assertion. □

Proposition 6.3 (Every terminal pair is realized)**.** *For every fixed complete label,*

$$\mathrm{Clust}_{\varepsilon\downarrow 0}(d_\varepsilon, \alpha_\varepsilon) = \mathcal{A}_s.$$

For fixed λ, s and $d \in (d_*, s)$, $d_* = F_\lambda^{-1}(s)$, the explicit sequence

$$\varepsilon_m(d) = \exp\{-\tfrac{32}{9}(3/4)^{-m}\mathcal{S}_\lambda(d)\} \tag{6.14}$$

realizes $(d, j_s(d))$ uniformly on the fixed compact $(k, \theta, \nu, \mathbf{n})$ class. Contact and endpoint realization is asserted for each fixed complete label and target; Proposition 7.1 supplies explicit common sequences for the two free endpoints.

Proof. Fix first $\alpha \in (0, 2]$. On J_j define $t_{j,\alpha}$ by (6.2) if $0 < \alpha \le 1$, and by the first formula of (6.3) if $1 \le \alpha \le 2$. These actual event times are continuous in compliance. In the first case this follows from continuity of the flow and integral, and the strictly increasing inverse of J/J_{sl} with $J_{\mathrm{sl}} \ge j_s > 0$. The two definitions agree at $\alpha = 1$. The continuous function

$$G_j(\varepsilon) = p(\mathfrak{X}_\varepsilon(t_{j,\alpha}(\varepsilon))) + as$$

has signs

$$G_j(a_j) = as/2 + O_s(a_j) > 0, \qquad G_j(c_j) = -as + O_s(c_j) < 0 \tag{6.15}$$

for large j. Indeed contact takes $O_s(\varepsilon)$ and $\dot{p}$ is bounded throughout its actual tube. Take a root $\varepsilon_j \in J_j$. Since $\alpha > 0$, $t_{j,\alpha} > t_j$; strict decrease of p gives $\delta_j < s$. The predecessor and successor bounds in the sweep exclude both neighboring last-input indices: $N_s = j + 1$, and the root is the first physical section time. Moreover $\delta_j = s + O_s(\varepsilon_j) \to s$ and the stopping phase equals α exactly. The comparison lemma proves its whole terminal prefix limit. By passing to later indices the compliances can be made strictly decreasing.

For $\alpha_i \downarrow 0$, choose sufficiently late terms from these positive-target sequences with decreasing compliance and all base and comparison errors below $1/i$. This realizes $(s, 0)$.

Solving the equation directly at $\alpha = 0$ would instead give $\delta_j = s$ and $N_s = j$, with the predecessor as last input; the positive-target diagonal avoids that change of index.

For the free targets set $j = m+3$ at (6.14). The exact graph chain started at the actual δ_3 has phase at j tending uniformly to $\mathcal{S}_\lambda(d)$, by (5.35). Its bases through j are eventually below $3s/2$. Lemma 5.5 therefore gives the actual events through this index and

$$\begin{aligned} \mathcal{S}_\lambda(\delta_j) = \mathcal{S}_\lambda(d) + \rho_c^m\{\chi_\lambda(\mathfrak{a}) + r_{\mathfrak{a}}(\varepsilon_m)\} + O(\varepsilon_m^{3/32}) \longrightarrow \mathcal{S}_\lambda(d), \\ E_j = O(\varepsilon_m^{3/32}). \end{aligned} \tag{6.16}$$

Thus $\delta_j \to d$ and its complete independent state tends to the graph, uniformly. Equation (5.25) recovers the dependent acceleration row. The preceding graph base tends to $F_\lambda^{-1}(d) < d_*$, so the preceding actual inputs stay a fixed distance below s.

This output is separately checked as a new input: it is close to $d < s$, below the reset cutoff, retains the full cone and closing margins, and satisfies the propagated compliance and detuning inequalities. Its actual next return has output tending to $F(d) > s$. The preceding scales are below $d_* < s$, so $N_s = m + 4$ and $d_\varepsilon \to d$. Every limit of its stopping phase satisfies (6.9). Since $d < s$, the unique solution is the free parameter $j_s(d)$. This proves uniform convergence: a failure of uniformity would give a varying compact-label subsequence and a different phase limit, contrary to the same equation and unique free root. The complete tuple and ordered prefix comparisons are uniform for the same reason. For either free endpoint, a diagonal of interior targets and successively late terms realizes that endpoint with the complete label fixed.

We have realized the entire vertical contact segment and free branch in (6.10). The reverse inclusion was already proved by (6.9); this proves the stated equality of cluster sets. □

6.3 Backward matching and the whole stopped path

Lemma 6.4 (Every fixed number of last returns)**.** *If $(d_\varepsilon, \alpha_\varepsilon) \to (d, \alpha)$, then for every fixed M and $0 \le m \le M$,*

$$\delta_{N_s-1-m} \longrightarrow b_m = F_\lambda^{-m}(d). \tag{6.17}$$

The complete normalized touchdown state tends to the graph at each b_m. The M corresponding full actual returns converge in order to $\mathcal{C}_{\lambda,b_m}$, and the last prefix converges to $\mathcal{C}_{\lambda,d}|_{[0,\alpha]}$.

Proof. The last scale is bounded below by c_s. Repeated use of the growth upper bound gives a positive lower bound for each fixed finite number of predecessors, depending on M, s. Their indices tend to infinity. The graph budget and the dependent-row recovery therefore give complete graph attraction at each such input. Given the limit b_{m-1}, every limit b of its predecessor satisfies $F_\lambda(b) = b_{m-1}$ by (5.20). Strict increase of F_λ makes this limit unique, equal to $F_\lambda^{-1}(b_{m-1})$. Compactness removes further extraction and proves (6.17). Lemma 6.1 applies to these finitely many positive-scale compact intervals; continuity in their bases gives the fixed reference returns. Equation (6.11) treats the last prefix. □

Proof of Theorems A and B. Lemma 5.3 gives the first transverse section, and Lemma 5.2 gives $N_s \to \infty$. Take a sequence with terminal pair tending to (d, α). The pair lies in $\mathcal{A}_s$ by (6.9). Fix a depth M, and put $b_M = F^{-M}(d)$. For sufficiently small compliance all actual inputs earlier than TD_{N_s-1-M} are at most $2b_M$. Their whole return segments are therefore in the same ball $G + B_{C(2b_M)^{3/4}}$, by (5.26)–(5.28). This is a supremum over all early returns. The analogous infinite reference tail in L_d is in $G + B_{Cb_M^{3/4}}$.

For a fixed incoming cutoff $\eta > 0$, Lemma 5.3 aligns the incoming prefixes and puts the remaining original-to-TD_3 curve in $G + B_{C\eta+o(1)}$. Match in their order the entire intervening actual curve and the remaining incoming reference tail followed by the small infinite reference cascade. Their common ball gives an error at most $C\eta + Cb_M^{3/4} + o(1)$, regardless of how many actual returns it contains. Match the remaining M full returns and final prefix by Lemma 6.4, and join their increasing pairings at the event endpoints. Their finite maximum error tends to zero. Constant limit pieces are handled by nondecreasing continuous surjections, approximated by increasing homeomorphisms as in the definition of the completed distance. Consequently

$$\limsup_{\varepsilon \to 0} d_{\mathrm{F}}(f_\varepsilon, \mathcal{W}_{\lambda,\nu,s}(d, \alpha)) \le C\eta + Cb_M^{3/4}. \tag{6.18}$$

First let $\varepsilon \to 0$ at fixed M, η, and then let $\eta \downarrow 0$, $M \to \infty$. Inverse contraction gives $b_M \to 0$, uniformly for $d \in [d_*, s]$ and the mechanical compact class. This proves the whole ordered convergence. Since $d_H \le d_{\mathrm{F}}$, it also proves the two-sided Hausdorff convergence.

The map $(d, \alpha) \mapsto \mathcal{W}_{\lambda,\nu,s}(d, \alpha)$ is continuous on $\mathcal{A}_s$: for fixed M use continuity of the finitely many last reference pieces and their terminal prefix, then bound both omitted tails uniformly by $C \sup_{d \in [d_*, s]} (F^{-M}(d))^{3/4}$. Its image is compact. Every sequence tending to zero has a convergent terminal-pair subsequence, giving the upper cluster inclusion. At a positive compliance limit, continuous dependence and the transverse section give continuity of the stopped curve. These two cases prove relative compactness of the sufficiently small compliance family. Proposition 6.3 gives the reverse inclusion from actual compliances with the same complete label. This proves the exact cluster classification for paths and for their images.

For the common free sequence, the entrance estimate, the budgets, the incoming comparison, the fixed-depth comparison and both small tail bounds are uniform on $(k, \theta, \nu, \mathbf{n})$ in the fixed compact class. The incoming prefix is always compared with its own $I_{\lambda,\nu}$. Thus (6.18) gives the asserted uniform whole-path convergence. The same finite-depth comparison and small-tail bounds apply whenever the terminal pairs converge uniformly on the compact auxiliary class. We use this observation for the explicit endpoint sequences in Proposition 7.1. The free sequence fixes λ, s, d; the contact roots above are taken for each fixed complete label and target. □

Corollary 6.5 (Different limits from the same full initial state)**.** *Fix λ, k, θ, ν and take $\mathbf{n} = 0$. The common full initial state is $(\phi_\nu, 0, \omega_\nu, 0, 0)$. There are two compliance sequences along which the stopped paths tend respectively to $\mathcal{W}_{\lambda,\nu,s}(s, 0)$ and $\mathcal{W}_{\lambda,\nu,s}(s, 1)$. Their terminal velocities and compact path images differ.*

Here the belt still follows $U_b = U_* - k\sqrt{\varepsilon}$.

Proof of Corollary 6.5. With $\mathbf{n} = 0$, (2.15) is the stated fixed full initial point. Theorem A realizes the two targets with the remaining label fixed. Put $U_s = d_\lambda s(1 + s^{1/3}\Gamma_U(s)) > 0$ and $h_s = \mu A(\phi_s) - B(\phi_s) > 0$, where $p(\phi_s) = -as$. The compression curve in (5.11) gives

$$\omega_{(s,0)} - \omega_{(s,1)} = \frac{U_s}{\cos\phi_s} > 0, \qquad v_{(s,0)} - v_{(s,1)} = \frac{as}{h_s}U_s > 0.$$

The first image meets $p = -as$ only at its admission point; the second contains the different arrest point on that section. Hence their compact path images differ. □

7 Selection by the residual entrance phase

Section 5 derives the entrance phase and its actual same-index accuracy; Section 6 proves convergence and realization of the complete ordered paths. We now compare the surviving entrance difference with the actual fixed-angle stopping window. This gives one selection law, followed by its belt and profile applications.

Fix λ, a sufficiently small $0 < s \le s_0$, and put

$$A_0 = 9/32, \qquad \rho_c = 3/4, \qquad S_s = \mathcal{S}_\lambda(s) > 0, \qquad d_* = F_\lambda^{-1}(s).$$

The entrance constant $\chi_\lambda(\mathfrak{a})$ is defined in (4.11); it is continuous and bounded on the prescribed compact auxiliary classes.

7.1 Selection by the entrance constant

Proposition 7.1 (Selection by the entrance constant)**.** *Under the standing hypotheses the following assertions hold.*

(i) *Let $\mathfrak{K}$ be any nonempty compact auxiliary class within the prescribed classes. For any $\chi_* \in \mathbb{R}$, define*

$$\varepsilon_m(\chi_*) = \exp\{-A_0^{-1}\rho_c^{-m}S_s + A_0^{-1}\chi_*\}, \qquad j = m + 3. \tag{7.1}$$

This sequence strictly decreases to zero. For every fixed $\mathfrak{a} \in \mathfrak{K}$ with $\chi_\lambda(\mathfrak{a}) \neq \chi_*$, its eventual touchdown count and its terminal-pair and ordered-path limits are

	N_s	*terminal pair*	*ordered path*
$\chi_\lambda(\mathfrak{a}) < \chi_*$	j	$(d_*, 3)$	$\mathcal{W}_{\lambda,\nu,s}(s, 0)$
$\chi_\lambda(\mathfrak{a}) > \chi_*$	$j+1$	$(s, 2)$	$\mathcal{W}_{\lambda,\nu,s}(s, 2)$.

For every fixed $\Delta_\chi > 0$, the index conclusions hold simultaneously for all sufficiently large m, and the parameter, ordered-path and Hausdorff limits are uniform, on each of the compact subclasses

$$\{\mathfrak{a} \in \mathfrak{K} : \chi_\lambda(\mathfrak{a}) \le \chi_* - \Delta_\chi\}, \qquad \{\mathfrak{a} \in \mathfrak{K} : \chi_\lambda(\mathfrak{a}) \ge \chi_* + \Delta_\chi\}.$$

Each incoming datum is compared with its own arc $I_{\lambda,\nu}$. No conclusion at equality is asserted.

(ii) *For any two fixed auxiliary labels $\mathfrak{a}_1, \mathfrak{a}_2$ in the prescribed classes with $\chi_\lambda(\mathfrak{a}_1) \neq \chi_\lambda(\mathfrak{a}_2)$, there is a positive compliance threshold, depending on this fixed pair, below which their two first section times cannot both occur after touchdown and at or before its following release. In particular there is no common compliance sequence selecting contact interiors for both labels, either in terminal parameters or in ordered paths, even at different phases.*

(iii) *Suppose that, for a fixed label $\mathfrak{a}_0$ and $0 < \alpha_0 < 2$, a sequence $\varepsilon_m \downarrow 0$ satisfies*

$$f_{\varepsilon_m}^{\mathfrak{a}_0} \longrightarrow \mathcal{W}_{\lambda,\nu_0,s}(s, \alpha_0) \qquad \text{in } d_{\mathrm{F}}.$$

Along this same given sequence, every fixed label $\mathfrak{a}$ with unequal entrance constant has the following limits:

	terminal pair; ordered path
$\chi_\lambda(\mathfrak{a}) < \chi_\lambda(\mathfrak{a}_0)$	$(d_*, 3)$; $\mathcal{W}_{\lambda,\nu,s}(s, 0)$
$\chi_\lambda(\mathfrak{a}) > \chi_\lambda(\mathfrak{a}_0)$	$(s, 2)$; $\mathcal{W}_{\lambda,\nu,s}(s, 2)$.

This assertion uses the given reference sequence; it makes no claim about its touchdown count. The corresponding Hausdorff image convergence holds, and each path retains its own incoming arc $I_{\lambda,\nu}$. Equality of the entrance constants is necessary for common interior contact selection; its sufficiency is not determined by these estimates.

Taking $\chi_* > \max_{\mathfrak{K}} \chi_\lambda$ or $\chi_* < \min_{\mathfrak{K}} \chi_\lambda$ gives explicit common sequences for the admission or release endpoint, respectively. The threshold need not be attained by a label. These sequences use the exact phase at s, as the free-target sequence in Theorem B uses the exact phase at its target. The scale comparison is

$$\varepsilon \ll 1/\log(1/\varepsilon) \ll 1 :$$

the retained entrance separation vanishes, but exceeds the actual contact stopping window. Figure 3 shows the retained phase gap and the actual stopping window.

The following two lemmas connect this phase comparison to actual stopping events and complete path limits.

Lemma 7.2 (Identification of an interior contact target)**.** *For fixed labels, if $f_{\varepsilon_m} \to \mathcal{W}(s, \alpha)$ in d_{F}, with $0 < \alpha < 2$, then $(d_{\varepsilon_m}, \alpha_{\varepsilon_m}) \to (s, \alpha)$.*

Proof. Every subsequence has a further convergent terminal-pair subsequence, by Theorem A. If its limit is $(d, \beta) \in \mathcal{A}_s$, that theorem gives the path limit $\mathcal{W}(d, \beta)$. Ordered convergence preserves endpoints, hence $\mathcal{C}_d(\beta) = \mathcal{C}_s(\alpha)$. An interior free endpoint has positive normal gap; on contact $y = z = 0$. The two free endpoints are contact admission and release. The contact endpoint map is injective on $[0, 2]$ by (5.15), including the arrest point

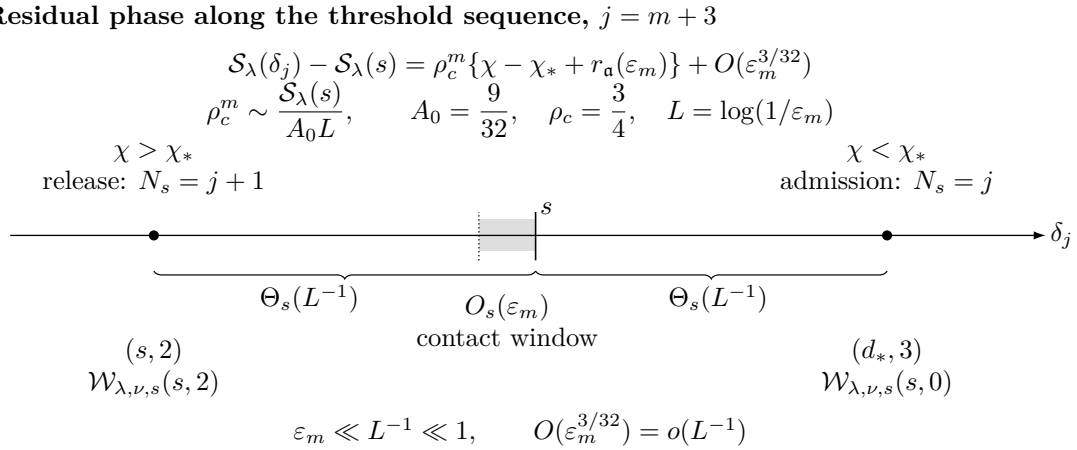


Fig. 3: Residual phase versus contact resolution, with λ, s fixed, and $\chi = \chi_\lambda(\mathfrak{a})$. The exact phase decreases with the base, giving opposite signs for the phase and base gaps. On compact label subclasses separated from χ_* by a fixed margin, the base gaps have order L^{-1}, larger than both the comparison error and the necessary contact-stop window $0 < s - \delta_j \le C_s\varepsilon_m$. The displayed pairs are the actual terminal-pair limits. The admission-side touchdown is read on the original trajectory continued past observation. Positions and widths are schematic; no label attaining χ_* is assumed.

$\beta = 1$. An interior contact point is neither admission nor release, so neither free endpoint nor the repeated admission representation can give this endpoint. The only possible pair is (s, α), proving convergence of the full parameter sequence. □

Lemma 7.3 (Endpoint test at a fixed touchdown index)**.** *Suppose that along a compliance sequence the actual jth touchdown is defined within the common input domain, $j \to \infty$, and $\delta_j \to s$. If $(\delta_j - s)/\varepsilon \to +\infty$, then $N_s = j$, the terminal pair tends to $(d_*, 3)$ and the ordered path tends to $\mathcal{W}_{\lambda,\nu,s}(s, 0)$. If $(\delta_j - s)/\varepsilon \to -\infty$, then $N_s = j+1$, the pair tends to $(s, 2)$ and the path tends to $\mathcal{W}_{\lambda,\nu,s}(s, 2)$. The index assertions are eventual. All limits, including the Hausdorff image limits, are uniform on a compact auxiliary class whenever the hypotheses are uniform there.*

Proof. The growth bounds give a positive lower bound for the predecessor base. Its complete state tends to the graph by Lemma 5.2; the actual return and strict increase of F_λ then give $\delta_{j-1} \to F_\lambda^{-1}(s) = d_* < s$. For a positive gap at index j, the predecessor contact stays below the section: its base remains a fixed distance below s and its contact drift is $O_s(\varepsilon)$. Since $\delta_j > s$, the first crossing is in its following free flight, before touchdown j. Thus $N_s = j$ and $d_\varepsilon \to d_*$. The unique solution of the reference section equation at d_* is the free endpoint 3.

For a negative gap, contact j changes the base coordinate by $O_s(\varepsilon)$, less than $s - \delta_j$, so it cannot reach the section. Its input is below the inner cutoffs and has all the admission margins. Apply the actual return estimate at this input; its next output tends to $F_\lambda(s) > s$. The first crossing is in that free flight, with $N_s = j + 1$, $d_\varepsilon \to s$ and $\alpha_\varepsilon > 2$. The reference section equation at s has the unique solution 2 on $[2, 3]$. This use of the next return requires no iteration at its output.

Strict decrease (5.32) and the original-prefix control in Lemma 5.3 exclude earlier crossings. Theorem A gives the path limits, using $\mathcal{W}(d_*, 3) = \mathcal{W}(s, 0)$ at admission. For uniform limits, all base bounds, contact drifts and complete-state budgets above are uniform. A varying-label subsequence can have only the stated unique free root. Match any fixed finite number of last returns uniformly and put the entire preceding cascade in its common small ball, as in (6.18). The incoming prefix is compared with its own $I_{\lambda,\nu}$. This gives uniform ordered convergence, and (3.2) gives the uniform image convergence. □

Proof of Proposition 7.1. Write $L = \log(1/\varepsilon)$. For (i), the exponent in (7.1) strictly decreases to $-\infty$, and $A_0 L = \rho_c^{-m} S_s - \chi_*$. The graph chain started at the actual $\delta_3^{\mathfrak{a}}$ has phase at $j = m + 3$ equal to

$$S_s + \rho_c^m \{\chi_\lambda(\mathfrak{a}) - \chi_* + r_{\mathfrak{a}}(\varepsilon_m)\}.$$

The bounded entrance constants and uniform remainder in (5.35) put these graph bases uniformly near s; all bases through j are eventually below $3s/2$. Lemma 5.5 therefore continues the actual events through this same index for every label and gives

$$\mathcal{S}_\lambda(\delta_j^{\mathfrak{a}}) - S_s = \rho_c^m \{\chi_\lambda(\mathfrak{a}) - \chi_* + r_{\mathfrak{a}}(\varepsilon_m)\} + O(\varepsilon_m^{3/32}). \tag{7.2}$$

Here $\rho_c^m \sim S_s/(A_0 L)$ and the last error is $o(\rho_c^m)$. On either compact subclass separated from χ_* by $\Delta_\chi > 0$, the phase difference has the sign of $\chi_\lambda(\mathfrak{a}) - \chi_*$ and absolute value between

$c\Delta_\chi \rho_c^m$ and $C\rho_c^m$, uniformly. Strict decrease and the two-sided Lipschitz bound (5.8) give the opposite signed base gap of the same order. Since $\varepsilon_m = o(\rho_c^m)$, Lemma 7.3 proves all index, terminal-pair and path assertions, including uniformity and image convergence. Every fixed unequal label belongs to such a separated subclass. At equality the sign is not determined.

For the event exclusion in (ii), suppose instead that two fixed unequal labels both stop during contact along a common sequence tending to zero, at touchdown indices j_1, j_2. The contact window (5.33) and the sharp phase give, for $i = 1, 2$,

$$\rho_c^{j_i-3}\{A_0 L + \chi_\lambda(\mathfrak{a}_i) + o(1)\} = S_s + O_s(\varepsilon^{3/32}).$$

Thus $\rho_c^{j_i-3} A_0 L \to S_s$. Division gives $\rho_c^{j_1-j_2} \to 1$; because the index difference is an integer, eventually $j_1 = j_2 = j$. Subtracting the two equations gives

$$\rho_c^{j-3}\{\chi_\lambda(\mathfrak{a}_1) - \chi_\lambda(\mathfrak{a}_2) + o(1)\} = O_s(\varepsilon^{3/32}).$$

Multiplication by L contradicts $\rho_c^{j-3} L \to S_s/A_0 > 0$ and the unequal constants. Consequently the exclusion holds below a positive threshold for this fixed pair. Interior terminal-pair limits eventually give actual contact stops; Lemma 7.2 gives the same implication for interior ordered-path limits.

For (iii), the identification lemma puts the $\mathfrak{a}_0$ stops in contact. Set $j = N_s^{\mathfrak{a}_0} - 1$. Then

$$\delta_j^{\mathfrak{a}_0} = s + O_s(\varepsilon), \qquad \rho_c^{j-3} \sim \frac{S_s}{A_0 L}.$$

For any other fixed $\mathfrak{a}$, the graph chain started at its own actual $\delta_3^{\mathfrak{a}}$ has phase at this index tending to S_s. Its bases through j are eventually below $3s/2$. The continuation in Lemma 5.5 therefore compares the actual events through the same index, even after that trajectory's observation time. Subtracting the reference relation gives

$$\mathcal{S}_\lambda(\delta_j^{\mathfrak{a}}) - S_s = \frac{S_s}{A_0 L}\{\chi_\lambda(\mathfrak{a}) - \chi_\lambda(\mathfrak{a}_0) + o(1)\} + O_{\mathfrak{a},\mathfrak{a}_0,s}(\varepsilon^{3/32}). \tag{7.3}$$

The signed base gap is opposite to the fixed entrance difference and has size $\Theta(L^{-1})$. Lemma 7.3 gives the terminal-pair and ordered-path limits in (iii), with the stated incoming arcs and image convergence. It is applied at the index fixed by the reference label, not at the prescribed index of (i), so no touchdown-count formula for the given sequence is asserted. □

Proof of Theorem C. Proposition 4.1 constructs the entrance constant from the physical preparation and proves its continuity and boundedness. The same-index comparison and actual stopping-window estimates lead to Proposition 7.1. Its part (ii) gives assertion (i) of the theorem, and part (iii) gives assertion (ii). The common compact classes and each label's incoming arc are exactly those of Section 2. □

Remark 7.4 (Phase and stopping resolution). The integer-index argument uses the following general hypotheses. Let $L = \log(1/\varepsilon)$, $A > 0$, $S_* > 0$ and $0 < \rho < 1$. Suppose that at the relevant integer event indices $n \in \mathbb{N}_0$, for each fixed label η,

$$P_n(\varepsilon, \eta) = \rho^n\{AL + \chi(\eta) + r_\eta(\varepsilon)\} + R_n(\varepsilon, \eta), \qquad r_\eta \to 0, \quad R_n = o(L^{-1}),$$

where the estimate for R_n is uniform over all relevant event indices for that fixed label. If the event requires $|P_n - S_*| \le w_\eta(\varepsilon) = o(L^{-1})$, two fixed labels with unequal χ cannot both meet it for arbitrarily small common ε. The ratio and subtraction argument in the proof of Proposition 7.1(ii) applies with n_i in place of $j_i - 3$. The actual preparation, same-index accuracy and stopping window needed here were established in Sections 4 and 5.

7.2 Belt ordering of the entrance constant

Theorem C reduces belt dependence to the order of the prepared coefficient C_R. The following strict order gives Corollary 3.4. Throughout this subsection λ, θ are fixed; the mechanical coefficient is denoted by d_λ as in Section 2.

Proposition 7.5 (Belt order of the entrance constant)**.** *Let $C_R(k)$ be the second-touchdown coefficient in* (4.8)*, and let K_λ be the return coefficient in* (4.7)*. Then $C_R(k)$ strictly decreases on I. Hence $\chi_{\lambda,\theta}(k)$ in* (4.11) *strictly increases.*

Proof. The normalized critical passage determines $q = q_{\mathrm{in}} > 0$ from (λ, θ) alone; see Lemma A.2 and Theorem A.4. The normal-release constant $\nu_\#$ is determined by the fixed linear problem (B.8), independently of the original labels. Put $H = ak/\gamma$ and $t = T_f(k)$ in (4.8), where A_{reb} is the amplitude A_s of (C.11). Implicit differentiation of the flight equation gives

$$t'(k) = -\frac{3\ell(a/\gamma)t}{2ct + 3\ell H} < 0.$$

Substitution in the amplitude first yields

$$A_{\mathrm{reb}}^2 = \frac{d_\lambda}{2h^2}\left[\gamma H t^2 + (2hc/3 + ad_\lambda)t^3\right].$$

Eliminating $H = (6q - ct^2)/(3\ell t)$ and using $\gamma = h\ell - d_\lambda$, we obtain

$$A_{\mathrm{reb}}^2 = \frac{d_\lambda}{2h^2}\left[\frac{2\gamma q}{\ell}t + \left(\frac{hc}{3} + \frac{d_\lambda c}{3\ell} + ad_\lambda\right)t^3\right]. \tag{7.4}$$

Every coefficient is positive. Hence the amplitude, R and C_R strictly decrease with k, which proves the order of χ.

□

The profile dependence requires a different argument, because a slip level can be visited more than once. We next construct a single-visit perturbation with nonzero release-speed response.

Proof of Corollary 3.4. For the exclusion, take $\mathfrak{a}_i = (k_i, \theta, \nu, \mathbf{n})$. Proposition 7.5 gives unequal entrance constants for distinct fixed k_i; Proposition 7.1(ii) therefore gives the event, terminal-pair and ordered-path exclusions. For a given reference sequence, apply Proposition 7.1(iii) with $\mathfrak{a}_0 = (k_0, \theta, \nu, \mathbf{n})$ and $\mathfrak{a} = (k, \theta, \nu, \mathbf{n})$. The strict increase of $\chi_{\lambda,\theta}(k)$ gives admission for $k < k_0$ and release for $k > k_0$, with the same incoming arc and the corresponding Hausdorff image convergence. □

7.3 Friction-profile response

The belt order does not by itself establish sensitivity to the friction profile. That question is decided by the normalized critical passage. A slip level can be visited several times, so a change of the profile does not provide an independently chosen forcing at each visit. We use an incoming slip interval visited only once before first release and prove that its release-speed response cannot vanish in every admissible direction.

Proposition 7.6 (Profile response and entrance order)**.** *Fix a mechanical parameter satisfying the critical-rate and sign conditions of Section 2, and an admissible C^4 profile Φ^{b}. Write $q_{\mathrm{in},\lambda}[\Phi] = q_{\mathrm{in}}$ for the normalized first-release velocity in Lemma A.2. There is $H > Q_\Phi$ such that, for every nonempty open interval $J \subset (H, \infty)$, there are a nonzero $\psi \in C_c^\infty(J)$ and $\vartheta_{\max} > 0$ for which*

$$\{\Phi^{\mathrm{b}} + \vartheta\psi : |\vartheta| \le \vartheta_{\max}\}$$

is an admissible compact profile class and $D_\Phi q_{\mathrm{in},\lambda}[\Phi^{\mathrm{b}}](\psi) \neq 0$. Constructing the curve at $\lambda_c = (7/3, 0)$, we may orient it and choose a sufficiently small relative closed mechanical box Λ_{fr} about λ_c and a closed parameter interval so that

$$\partial_\vartheta q_{\mathrm{in},\lambda}[\Phi_\vartheta] \ge c_{\mathrm{fr}} > 0 \quad \text{on } \Lambda_{\mathrm{fr}} \times [-\vartheta_{\max}, \vartheta_{\max}], \qquad \Phi_\vartheta = \Phi^{\mathrm{b}} + \vartheta\psi. \tag{7.5}$$

For each fixed $\lambda \in \Lambda_{\mathrm{fr}}$ and $k \in I$, $\chi_{\lambda,\vartheta}(k)$ strictly decreases with ϑ.

Proof. On the baseline critical contact write $L = Z - Y$, let τ_r be its first release, and abbreviate $C = C_G$, $m = \mu A_G$ and $f = 1 - \Phi^{\mathrm{b}}$. The normalization gives $U \to +\infty$ and $U' \to -\gamma < 0$ as $\tau \to -\infty$. Choose $\tau_0 < \tau_r$ with $U' < -\gamma/2$ on $(-\infty, \tau_0]$, and take

$$H > \max\{Q_\Phi, \max_{[\tau_0, \tau_r]} U\}.$$

Every level $u > H$ is visited exactly once before first release; denote its inverse time by $\tau(u)$.

Let $F_{\Phi^{\mathrm{b}}}$ be the contact extension of (A.20), obtained by replacing L_+ by L. In state order (Y, V, Z, U) put $A(\tau) = D_W F_{\Phi^{\mathrm{b}}}(\overline{W}(\tau))$ and solve

$$\zeta' = -A^T\zeta, \qquad \zeta(\tau_r) = e_V = (0, 1, 0, 0)^T, \qquad Q = C\zeta_V + m\zeta_U.$$

For an admissible compactly supported variation, Lemma A.3 gives the exact response

$$D_\Phi q_{\mathrm{in},\lambda}[\Phi^{\mathrm{b}}](\psi) = -\int_{-\infty}^{\tau_r} LQ\psi(U)\, d\tau. \tag{7.6}$$

A controllability calculation prevents Q from vanishing on an interval. Indeed,

$$A = A_0 + b_0 r^T, \qquad A_0 = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 1 & 0 & -2 & 0 \\ h & 0 & -h & 0 \end{pmatrix}, \tag{7.7}$$
$$b_0 = (0, C, 0, m)^T, \qquad r = (-f, 0, f, f'L)^T, \qquad Q = b_0^T \zeta,$$
$$\det[b_0, A_0 b_0, A_0^2 b_0, A_0^3 b_0] = -hC^4 \neq 0.$$

If Q vanished on an open contact interval, then $\zeta' = -A_0^T \zeta$ there. Successive differentiation of $b_0^T \zeta = 0$ would give $(A_0^j b_0)^T \zeta = 0$ for $j = 0, 1, 2, 3$. The determinant in (7.7) would force $\zeta = 0$, contradicting its nonzero terminal value by linear uniqueness. Thus Q is not identically zero on any open contact interval; no analyticity is needed.

Take a nonempty open $J \subset (H, \infty)$. Strict monotonicity, the mean-value theorem and continuity of the derivative give an open subinterval where $(\Phi^{\mathrm{b}})' > 0$. Its time preimage contains a point with $Q \neq 0$. We may therefore choose a closed $K \Subset J$ on which $(\Phi^{\mathrm{b}})'$ has a positive lower bound and $Q(\tau(u))$ has a fixed nonzero sign. Changing variables in (7.6) gives

$$D_\Phi q_{\mathrm{in},\lambda}[\Phi^{\mathrm{b}}](\psi) = -\int_K \frac{L(\tau(u))Q(\tau(u))}{|U'(\tau(u))|}\psi(u)\, du.$$

A nonzero nonnegative smooth bump supported inside K has nonzero response. Reverse its sign if needed to make the response positive.

For sufficiently small $|\vartheta| \leq \vartheta_{\max}$, the profile derivative stays positive on K, while the profile is unchanged elsewhere. Its zero, range and saturation limits are preserved. The core strip and (F3) are unchanged because $K \subset (Q_\Phi, \infty)$; (F4) holds with a common enlarged constant since the perturbation has fixed compact support. The closed parameter curve is continuous in the weighted C^4 metric, and hence compact.

For the common mechanical range choose this curve first at λ_c. The model restrictions provide a common mechanical box for the fixed compact class. Uniform incoming tails give a common finite time before the support is reached. At each fixed λ, all profiles have exactly their baseline's state at that time, so their shape derivative starts at zero. The baseline state is continuous in λ by Lemma A.2. Continuous dependence of the finite-time state and linear variation, together with the transverse first-release event, makes $\partial_\vartheta q_{\mathrm{in},\lambda}[\Phi_\vartheta]$ jointly continuous in (λ, ϑ). Its positive value at $(\lambda_c, 0)$ gives (7.5) after restricting the box and parameter interval. No derivative with respect to λ is needed.

Finally keep λ, k fixed, put $H_k = ak/\gamma$, and let $t = t(q) > 0$ solve

$$ct^2 + 3\ell H_k t = 6q, \qquad t_q = \frac{6}{2ct + 3\ell H_k} > 0.$$

The first amplitude identity in the proof of Proposition 7.5 gives $A_{\mathrm{reb}}^2 = d_\lambda B(t)/(2h^2)$, where

$$B(t) = \gamma H_k t^2 + (2hc/3 + a d_\lambda) t^3.$$

As $R = \nu_{\#} A_{\mathrm{reb}}$ and $C_R = \sqrt{6R/c}$,

$$\partial_q \chi = -\frac{3}{16} \frac{B'(t)}{B(t)} t_q < 0. \tag{7.8}$$

Here $\nu_{\#}$ and K_λ are independent of the original profile. Combining (7.5) and (7.8) proves the strict entrance order. □

Corollary 7.7 (Friction-profile sensitivity at a fixed belt coefficient)**.** *Let Φ^{b} be an admissible C^4 baseline profile, and fix compact belt and physical-preparation classes as in Section 2. There exist a nonzero $\psi \in C_c^\infty((0,\infty))$, $\vartheta_{\max} > 0$, a sufficiently small relative closed mechanical box Λ_{fr} about $\lambda_c = (7/3, 0)$, and $s_{\mathrm{fr}} > 0$ such that*

$$\Phi_\vartheta = \Phi^{\mathrm{b}} + \vartheta\psi, \qquad \vartheta \in \Theta_{\mathrm{fr}} = [-\vartheta_{\max}, \vartheta_{\max}]$$

is a compact class satisfying (F1)–(F4). With this curve as the profile class, the common model restrictions and Theorem A apply on Λ_{fr} for $0 < s \le s_{\mathrm{fr}}$. Orient the curve so that its normalized first-release speed $q_{\mathrm{in},\lambda}[\Phi_\vartheta]$ increases with ϑ for every $\lambda \in \Lambda_{\mathrm{fr}}$.

Fix $\lambda \in \Lambda_{\mathrm{fr}}$, $k \in I$, ν, $\mathbf{n}$ and $0 < s \le s_{\mathrm{fr}}$. In every system retain $U_b = U_* - k\sqrt{\varepsilon}$ and vary only the fixed profile parameter.

(i) *For every two distinct fixed $\vartheta_1, \vartheta_2 \in \Theta_{\mathrm{fr}}$ there is $\varepsilon_0 > 0$, depending on this pair and the fixed data, such that for $0 < \varepsilon < \varepsilon_0$ the two actual first section times cannot both occur after touchdown and at or before its following release. Consequently no common sequence $\varepsilon_m \downarrow 0$ has terminal-pair limits (s, α_i) for both profiles with $0 < \alpha_i < 2$. There is likewise no common sequence with ordered limits $\mathcal{W}_{\lambda,\nu,s}(s, \alpha_i)$, even if $\alpha_1 \ne \alpha_2$.*

(ii) *Fix $\vartheta_0 \in \Theta_{\mathrm{fr}}$ and $0 < \alpha < 2$. Along any sequence $\varepsilon_m \downarrow 0$ for which $f_{\varepsilon_m}^{(\vartheta_0)} \to \mathcal{W}_{\lambda,\nu,s}(s, \alpha)$ in d_{F}, every other fixed $\vartheta \in \Theta_{\mathrm{fr}}$ satisfies*

	terminal pair limit	*ordered-path limit*
$\vartheta > \vartheta_0$	$(d_*, 3)$	$\mathcal{W}(d_*, 3) = \mathcal{W}(s, 0)$
$\vartheta < \vartheta_0$	$(s, 2)$	$\mathcal{W}(s, 2)$.

This is the locally constructed profile application of Theorem C: the curve and its common mechanical range are constructed together; no nonconstancy along a prescribed profile family is assumed. Each profile is fixed before the compliance limit. With $\mathbf{n} = 0$ the full physical initial state is the same for all profiles and compliances, while the belt still follows the stated joint scaling. Theorem A supplies the reference sequences in part (ii); the ordered-path limits also give the corresponding Hausdorff image convergence by (3.2).

Proof of Corollary 7.7. Construct the curve at λ_c by Proposition 7.6, choose the common model range for this compact class, and restrict it as in (7.5). Fix the labels in the corollary. Equation (7.8) shows that their entrance constant strictly decreases with ϑ. Apply Proposition 7.1(ii) with $\mathfrak{a}_i = (k, \vartheta_i, \nu, \mathbf{n})$. Part (ii) gives both the terminal-pair and ordered-path exclusions. For the given reference sequence, Proposition 7.1(iii) gives admission for $\vartheta > \vartheta_0$ and release for $\vartheta < \vartheta_0$. Theorem A supplies actual reference sequences, and the same comparison gives Hausdorff image convergence. □

7.4 Return count and contact-root asymptotics

Corollary 7.8 (Double-logarithmic return count)**.** *The actual touchdown count satisfies*

$$N_s(\varepsilon) = \frac{\log\log(1/\varepsilon)}{\log(4/3)} + O_s(1). \tag{7.9}$$

Proof. The last input belongs to $[c_s, s)$, on which the positive phase has fixed positive upper and lower bounds. Formula (5.34) at $n = N_s - 1$ therefore gives $\rho_c^{N_s-4} L \asymp_s 1$. Taking logarithms proves the claim. □

For any actual contact-root family ε_j with the complete label fixed, write $C_R = C_R(\lambda, k, \theta)$ and let $j \to \infty$ be the contact number. The same formula and (5.33) give

$$\begin{aligned} \log(1/\varepsilon_j) &= \tfrac{32}{9}\rho_c^{-(j-3)}\mathcal{S}_\lambda(s) - \tfrac{32}{9}\chi_\lambda(\mathfrak{a}) + o(1), \\ \varepsilon_j &= K_\lambda^{32/3} C_R^{-8/3} \exp\{-\tfrac{32}{9}\rho_c^{-(j-3)}\mathcal{S}_\lambda(s)\}(1 + o(1)). \end{aligned} \tag{7.10}$$

Indeed $\rho_c^{-(j-3)} = O(L_j)$, so division of the shadow error costs only $O(L_j \varepsilon_j^{3/32}) = o(1)$. Different interior phases share this asymptotic formula; their individual selection is supplied by the actual scan of Lemma 6.2.

8 Observation rules for complete paths

Three observation rules act differently on a complete ordered path. A downstream angle retains a whole contact and a following free prefix; first hitting the contact angle cuts at admission and is discontinuous at that path; a fixed local patch crossed strictly by the contact selects any prescribed interior prefix.

8.1 Common paths beyond a complete contact

Write f_ε^τ for the actual path stopped at its first crossing of $p = -a\tau$, with the same normalization of its parameter interval as in (3.8). The return curve and phase are unchanged. Since $F_\lambda(s) > s$, there is $s < \sigma \le s_0$ with $\sigma < F_\lambda(s)$ for every $0 < s < s_0$. After a common restriction with $2s \le s_0$ and $F_\lambda(s) > 2s$, one may take $\sigma = 2s$.

Corollary 8.1 (Common paths beyond a complete contact)**.** *Fix λ and $0 < s < \sigma \le s_0$ with $\sigma < F_\lambda(s)$. The common sequence* (3.10) *with target $d = s$ satisfies*

$$\sup_{(k,\theta,\nu,\mathbf{n})} d_{\mathrm{F}}\big(f^\sigma_{\varepsilon_m(s)}, \mathcal{W}_{\lambda,\nu,\sigma}(s, j_\sigma(s))\big) \longrightarrow 0.$$

The corresponding image convergence is uniform in d_{H}. Here $2 < j_\sigma(s) < 3$, so the common limiting path includes the entire contact $\mathcal{C}_{\lambda,s}|_{[0,2]}$ and a positive-length prefix of its following free flight.

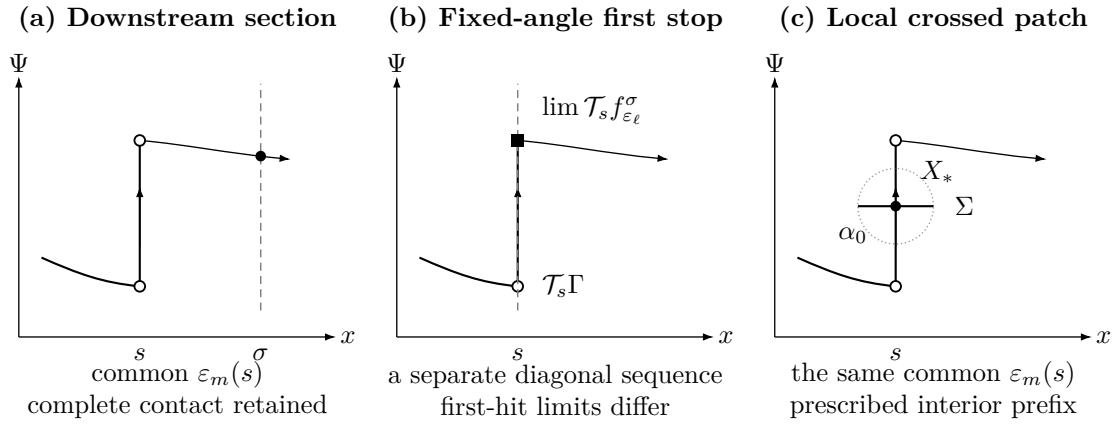


Fig. 4: Three observations of one limiting contact, schematically projected to $x = -p/a$ and $\Psi = v - c_\Sigma u_0$ (Corollary 8.3), with $s < \sigma < F_\lambda(s)$. Open circles mark admission and release; arrows indicate curve order. (a) A downstream section retains the whole contact along the common sequence $\varepsilon_m(s)$. (b) The limit path Γ first reaches $x = s$ at admission, while actual first-stop paths from a separate diagonal sequence tend to release (square). (c) The fixed local patch crossed strictly at $X_* = \mathcal{C}_{\lambda,s}(\alpha_0)$ selects that prefix along the same sequence as (a). The dotted circle indicates localization only; the physical patch and its separation from earlier prefixes are established in full state space. Other coordinates and distinct incoming arcs are suppressed, not identified.

Proof. The inequalities on σ give $F_\lambda^{-1}(\sigma) < s < \sigma$. Thus $d = s$ is an interior free target for the σ section. Apply Theorem B at that section. Its sequence depends on λ, d, but not on the section, so it is precisely (3.10) with $d = s$. Strict decrease of p along the free flight and $s < \sigma < F_\lambda(s)$ give $2 < j_\sigma(s) < 3$. The flight duration T_s is positive, and the retained free duration $(j_\sigma(s) - 2)T_s$ is therefore positive. □

8.2 First-angle truncation

For an oriented path $g : [0,1] \to \mathbb{R}^5$ that starts above and reaches the section $p = -as$, define its first-hit truncation by

$$\mathcal{T}_s g = g|_{[0,\tau_s(g)]}, \qquad \tau_s(g) = \min\{q \in [0,1] : p(g(q)) = -as\},$$

reparameterized onto $[0,1]$. This operation is well defined in the zero-distance quotient of (3.1). Indeed, two zero-distance paths admit a common representation $g \circ a = \widetilde{g} \circ b$, with a, b continuous nondecreasing surjections. Such a representation is obtained by taking a limit of the increasing comparison graphs and parameterizing the limiting monotone relation by the sum of its coordinates. At the first section point of the common representation, a and b reach the first section parameters of g and $\widetilde{g}$, respectively. Restricting a, b there gives a common representation of their prefixes. Thus only inserted pauses can differ, and the prefixes have zero ordered distance.

Corollary 8.2 (Discontinuity of first-hit truncation)**.** *Under the same restrictions on λ, s, σ, fix a complete label $\mathfrak{l} = (\lambda, k, \theta, \nu, \mathbf{n})$ and put*

$$\Gamma = \mathcal{W}_{\lambda,\nu,\sigma}(s, j_\sigma(s)).$$

There is a sequence $\varepsilon_\ell \downarrow 0$ for this label such that

$$d_{\mathrm{F}}(f^\sigma_{\varepsilon_\ell}, \Gamma) \longrightarrow 0,$$
$$d_{\mathrm{F}}(\mathcal{T}_s f^\sigma_{\varepsilon_\ell}, \mathcal{W}_{\lambda,\nu,s}(s,2)) \longrightarrow 0,$$

whereas $\mathcal{T}_s\Gamma = \mathcal{W}_{\lambda,\nu,s}(s,0)$ in the ordered quotient. These two stopped limit paths have different endpoints. Consequently $\mathcal{T}_s$ is discontinuous at Γ, even on the subspace consisting of Γ and actual paths stopped at the σ section.

Proof. Choose $d_\ell \uparrow s$ with $F_\lambda^{-1}(\sigma) < d_\ell < s$. Each d_ℓ is an interior free target for both the s and σ sections. The explicit sequences supplied by Theorem B at these two sections are identical:

$$\varepsilon_m(d_\ell) = \exp\{-\tfrac{32}{9}(3/4)^{-m}\mathcal{S}_\lambda(d_\ell)\}.$$

For each ℓ, take m_ℓ sufficiently large that both ordered-path errors are below $1/\ell$, and also that $\varepsilon_{m_\ell}(d_\ell) < 1/\ell$ and, when $\ell > 1$, $\varepsilon_{m_\ell}(d_\ell) < \varepsilon_{m_{\ell-1}}(d_{\ell-1})/2$. Write $\varepsilon_\ell = \varepsilon_{m_\ell}(d_\ell)$.

Continuous dependence of the finite-scale return curves and their free-section roots gives

$$j_\sigma(d_\ell) \longrightarrow j_\sigma(s), \qquad j_s(d_\ell) \longrightarrow j_s(s) = 2.$$

The complete-path continuity proved after (6.18), applied separately at the fixed sections s and σ, therefore gives

$$\mathcal{W}_{\lambda,\nu,\sigma}(d_\ell, j_\sigma(d_\ell)) \longrightarrow \Gamma,$$
$$\mathcal{W}_{\lambda,\nu,s}(d_\ell, j_s(d_\ell)) \longrightarrow \mathcal{W}_{\lambda,\nu,s}(s, 2).$$

The two diagonal error bounds prove the two asserted actual limits. The actual s section is reached before the σ section, and its prefix is exactly $\mathcal{T}_s f^{\sigma}_{\varepsilon_\ell} = f^{s}_{\varepsilon_\ell}$ in the ordered quotient.

The incoming arc has $p \geq 0$. Every return preceding the base s in L_s has $p > -as$ except that the last reaches $-as$ at its final touchdown, the admission point $\mathcal{C}_{\lambda,s}(0)$. Hence Γ first reaches this section at that admission point, and its first-hit prefix is $\mathcal{W}_{\lambda,\nu,s}(s, 0)$. The admission slip is $U_s > 0$ from (5.10), while the release slip is zero. In particular,

$$|\mathcal{C}_{\lambda,s}(2) - \mathcal{C}_{\lambda,s}(0)| \geq \frac{U_s}{|\cos\phi_s|} > 0.$$

The ordered distance bounds the distance between terminal states, so the limits of the truncated paths and of the long paths cannot commute. This proves the claimed discontinuity. □

8.3 Fixed local observation patches

The angular section contains the whole limiting contact. A local patch can cross it at any prescribed interior point: the next construction uses slip on compression, normal velocity on release, and the two one-sided crossings at arrest.

Corollary 8.3 (Common contact prefixes at a local observation surface)**.** *Fix λ, $0 < s < \sigma \leq s_0$ with $\sigma < F_\lambda(s)$, and $\alpha_0 \in (0, 2)$. Put*

$$X_* = \mathcal{C}_{\lambda,s}(\alpha_0), \qquad h_s = \mu A(\phi_s) - B(\phi_s) > 0,$$

and choose $c_\Sigma > as/h_s$. Define on physical states

$$u_0(\mathfrak{X}) = \omega\cos\phi + U_*, \qquad \Psi(\mathfrak{X}) = v - c_\Sigma u_0(\mathfrak{X}).$$

There is $r_\Sigma > 0$ such that the closed observation patch

$$\Sigma = \{\mathfrak{X} : |\mathfrak{X} - X_*| \leq r_\Sigma, \quad \Psi(\mathfrak{X}) = \Psi(X_*)\}$$

has the following property. For all sufficiently large m, every actual trajectory at $\varepsilon = \varepsilon_m(s)$ in the prescribed compact auxiliary label class first meets Σ before reaching $p = -a\sigma$, and this first meeting occurs during the actual contact whose base tends to s. If f^{Σ}_{ε} is the path stopped at this first meeting and reparameterized onto $[0, 1]$, then along the common sequence (3.10) *with $d = s$,*

$$\sup_{(k,\theta,\nu,\mathbf{n})} d_{\mathrm{F}}\big(f^{\Sigma}_{\varepsilon_m(s)}, \mathcal{W}_{\lambda,\nu,s}(s, \alpha_0)\big) \longrightarrow 0.$$

The image convergence is also uniform in d_{H}. The patch is independent of ε and the auxiliary labels; each incoming datum retains its own arc $I_{\lambda,\nu}$.

Proof. The exact contact formulas (5.10) and (5.13) give, on compression,

$$\frac{\mathrm{d}}{\mathrm{d}\alpha}\Psi(\mathcal{C}_{\lambda,s}(\alpha)) = (c_\Sigma - as/h_s)U_s > 0 \qquad (0 < \alpha < 1).$$

On release $u_0 = 0$, and (4.6) gives

$$\frac{\mathrm{d}}{\mathrm{d}\alpha}\Psi(\mathcal{C}_{\lambda,s}(\alpha)) = \tfrac{1}{2}s^{3/2}R_s(Z_s - Y_s)((\alpha-1)R_s) > 0 \qquad (1 < \alpha < 2).$$

The active force is positive before its first zero. These two formulas and the continuous join show that Ψ is strictly increasing along the entire reference contact. At $\alpha_0 = 1$ we use the two one-sided strict crossings.

The compact tail L_s meets $p = -as$ only at $\mathcal{C}_{\lambda,s}(0)$, whose Ψ value is smaller than $\Psi(X_*)$. Hence $X_* \notin L_s$. Choose $r_\Sigma > 0$ so that the closed $2r_\Sigma$ ball about X_* avoids L_s and is contained in $\{p < 0\}$. Every incoming arc lies in $\{p \geq 0\}$, so this choice also separates the patch from all incoming arcs uniformly.

Choose $\eta > 0$ with $0 < \alpha_- := \alpha_0 - \eta < \alpha_+ := \alpha_0 + \eta < 2$ so small that $\mathcal{C}_{\lambda,s}([\alpha_-, \alpha_+])$ lies in the open r_Σ ball. At the two ends $\Psi - \Psi(X_*)$ has opposite signs. The earlier contact segment $\mathcal{C}_{\lambda,s}([0, \alpha_-])$ has strictly smaller Ψ value and is a positive distance from the closed patch. Together with the preceding tail and incoming-arc separation, this gives a uniform positive distance between Σ and the entire preceding reference prefix.

Corollary 8.1 and its ordered pairings give uniform convergence of the actual paths through this contact. More specifically, Lemma 6.1 pairs its compression and release with $\mathcal{C}_{\lambda,s}|_{[0,2]}$, after using the convergence of the actual base to s. These pairings align the earlier prefixes as in the proof of Theorem A. Thus, for large m, the actual trajectory cannot meet Σ before the paired point α_-, while the whole paired bracket stays in the open r_Σ ball and its end values of $\Psi - \Psi(X_*)$ have opposite signs. The intermediate value theorem supplies a meeting inside this bracket. Since Σ is closed, the nonempty set of meeting times up to the end of the bracket is compact and has a first element. It lies between the paired points α_- and α_+, within this actual contact.

Restrict the ordered pairings at that first meeting. Their errors tend to zero uniformly in the auxiliary labels, and the reference stopping parameter lies in $[\alpha_0 - \eta, \alpha_0 + \eta]$. Uniform continuity of $\mathcal{C}_{\lambda,s}$, followed by $\eta \downarrow 0$, proves the asserted prefix convergence. The image assertion follows from $d_{\mathrm{H}} \leq d_{\mathrm{F}}$. □

Appendix F identifies the single terminal-parameter duplication and proves that the full-state accumulation families are circles. Appendix E records the interval of scaled effective-gap velocities above the release endpoint path.

9 Discussion

The selection comparison uses three inputs: an entrance phase with a bounded correction and a uniform vanishing remainder, a same-index return error smaller than $1/\log(1/\varepsilon)$,

and localization of the actual stopping event to a narrower window. Together they explain why parameter dependence disappears for a free target but determines on which side of a contact the first crossing occurs. The abstract criterion in Remark 7.4 records this last comparison; the original preparation, admissible return domains and sharp full-state budget establish its hypotheses here. For complete paths one also needs convergence of the incoming motion, ordered comparison of any fixed number of final returns, and a common shrinking neighborhood for the whole earlier tail.

The normal filter, the 3/4 return exponent, the 9/32 entrance exponent and the signs ordering χ come from the specified mechanical and constitutive laws. Compactly supported profile variations share an exact incoming tail, so their critical-passage response reduces to a finite-time variational problem (Lemma A.3). Controllability and the single-visit high-slip interval then supply a direction with nonzero response, with a common mechanical box and section chosen for the constructed compact curve. The profile tails and mechanical rate margins also enter the preparation estimates. The prescribed compact β range makes the preparation estimates uniform; it is the range treated by the present proof. The signed first contact permits $r = O(\delta)$, whereas the preparatory flights put later inputs in $r \le A\delta^{4/3}$. A different detuning scale or normal law would require its own entrance and return estimates before the selection comparison could be applied.

The observation results distinguish a longer path from the first fixed-angle prefix that it contains. A strict local crossing gives continuous prefix selection, while the angle section contains the whole limiting contact and has a discontinuous first-hit operation. The additional scaled-memory and path-identification results retain information that can be lost by projecting the state or discarding the contact order.

The selection results compare fixed labels. Whether belt or profile parameter differences of order $\varepsilon \log(1/\varepsilon)$ yield a transition law remains open. The absolute shadowing estimate used here does not control such parameter increments.

A From physical preparation to the first release

The incoming normal equilibrium loses uniform attraction as p tends to zero. Negative detuning moves the relevant slip transition to $p \asymp \sqrt{\varepsilon}$, where normal dissipation and variation of its energy metric have comparable sizes. We follow the prescribed rigid trace by coupling a normal energy with its exact weak coefficient. A first graph correction has identically zero force defect. The resulting energy estimate controls the outer range by dissipation and the critical annulus by a bounded Gronwall factor. Although the weak coefficient loses accuracy there, conversion back to the physical state multiplies its error by p^{β}. This gives the complete normal state needed for the first release.

The compact classes are those of Section 2. In particular, $r = -\eta = k\sqrt{\varepsilon}$, $k \in I = [k_-, k_+] \Subset (0, \infty)$, and all labels, including the original fast triple, are fixed before $\varepsilon \downarrow 0$. Constants below are common on these classes. Write $\gamma = -g_* > 0$ and $\rho = b/p$.

A.1 The weak coordinate and a finite normal energy

The rigid first integral. Let $\alpha_1 = F_{\mathrm{free}}p$, $\alpha_2 = F_{\mathrm{free}}b$ and $\alpha_3 = -Q_{\mathrm{sat}}b$, restricted to $y = v = 0$ when written as functions of (p, b). The rigid saturated field is

$$\begin{aligned} \dot p = \alpha_1, \qquad \dot b = \alpha_2 + \alpha_3 b/p, \\ F(p,\rho) = \frac{\alpha_2 + (\alpha_3 - \alpha_1)\rho}{p}, \qquad \mathscr{D} = \alpha_1\partial_p + F\partial_\rho. \end{aligned} \tag{A.1}$$

Here $\alpha_1(G) = -a$, $\alpha_2(G) = -c$, $\alpha_3(G) = -a\beta$ and $c = a(\beta-1)\ell$. The scalar equation for ρ has the regular-singular form

$$p\rho_p = H(p,\rho), \qquad H = \frac{\alpha_2(p,p\rho) + (\alpha_3-\alpha_1)(p,p\rho)\rho}{\alpha_1(p,p\rho)}, \qquad H(0,\rho) = (\beta-1)(\rho+\ell).$$

To construct its analytic weak curve, put $s = \bar\rho + \ell$. Its equation is $(p\partial_p - (\beta-1))s = pJ(p,s)$, with J analytic. In absolutely convergent power series the inverse diagonal operator divides coefficient n by $n-(\beta-1)$, $n \ge 1$. These divisors stay away from zero uniformly for $2 < \beta_- \le \beta \le \beta_+ < 5/2$. On a sufficiently small common disk multiplication by p therefore makes the fixed-point equation for s a contraction. This gives $\bar b = p\bar\rho = -\ell p + O(p^2)$.

For $d_b = \rho - \bar\rho$, subtraction gives exactly $d_b' = ((\beta-1)/p + A_0(p,d_b))d_b$, where A_0 is analytic and bounded: $\partial_\rho H - (\beta-1)$ is divisible by p. Thus $d_b = p^{\beta-1}\zeta$ satisfies $\zeta' = A_0(p, p^{\beta-1}\zeta)\zeta$, a regular equation down to $p = 0$ on each bounded ζ set. Define $\mathcal{C}(p,b)$ as its limiting value there. Its integral equation and variational equation give

$$\begin{aligned} b(p,C) = \bar b(p) + p^\beta\{C + O(p)\}, \qquad |\nabla_{p,b}\mathcal{C}| \le Cp^{-\beta}, \\ |b(p,C) - b(p,\widetilde C)| \le Cp^\beta |C - \widetilde C|, \end{aligned} \tag{A.2}$$

$$\rho = -\ell + O(p) + O(p^{\beta-1}), \qquad \mathscr{D}\rho = O(1), \qquad \mathscr{D}^2\rho = O(p^{\beta-3}). \tag{A.3}$$

For the first line, the variation in ζ and its inverse are bounded by e^{Cp}; conversion from $\zeta = (b-\bar b)/p^\beta$ gives the displayed derivatives. For the second, differentiate the regular equation along the rigid field. These are values of rigid derivatives at points of a bounded coefficient wedge, not derivatives of an actual fast transient. In particular $|F_p| + |F_\rho| \le C/p$ there. The exact first integral equals the prescribed C_1 on the prescribed rigid trace.

A degenerating normal energy. The normal decay rate tends to zero with p. We therefore seek a Lyapunov norm whose square decays at a rate proportional to p for the frozen normal equation in fast time. Converting this norm to displacement and memory will cost $p^{-1/2}$, whereas the velocity estimate remains uniform. These component weights determine how the normal error enters the weak coordinate estimate below.

Set $N = (Y, V, Z) = (y/\varepsilon^2, v/\varepsilon, z/\varepsilon^2)$ and define

$$M_o(p) = \begin{pmatrix} 0 & 1 & 0 \\ -p & 0 & p \\ 1 & 0 & -2 \end{pmatrix}, \quad e_{\mathrm{eq}} = (2,0,1)^T, \quad q_N = (-1,0,1)^T, \quad a_1 = (-1,2,-1)^T. \tag{A.4}$$

The polynomial of M_o is z^3+2z^2+pz+p. Its simple root $f=-2+p/4+O(p^2)$ is analytic; write the remaining factor as $z^2-a_cz+d_c$, where $a_c=-p/4+O(p^2)$ and $d_c=p/2+O(p^2)$. The polynomial projector

$$P_f=\frac{M_o^2-a_cM_o+d_cI}{f^2-a_cf+d_c},\qquad P_c=I-P_f,$$
$$c_2=P_c(0,2,-1/2)^T,\quad c_1=(M_o-a_cI)c_2,\quad e_f=P_f(0,0,1)^T.$$

defines an analytic invertible matrix $T_o=(c_1,c_2,e_f)$. At zero its columns are e_{eq}, $(0,2,-1/2)^T$ and $(0,0,1)^T$; their determinant is 4, and

$$T_o^{-1}M_oT_o=\operatorname{diag}(A_c,f),\qquad A_c=\begin{pmatrix}0&1\\-d_c&-h_c\end{pmatrix},\qquad h_c=-a_c.$$

Put $S_c=\left(\begin{smallmatrix}d_c & h_c/2\\ h_c/2 & 1\end{smallmatrix}\right)$ and $\mathbb{S}=T_o^{-T}\operatorname{diag}(S_c,1)T_o^{-1}$. The exact identity $A_c^TS_c+S_cA_c=-h_cS_c$ and $f<-1$ give

$$\begin{aligned}M_o^T\mathbb{S}+\mathbb{S}M_o&\le -c_0p\mathbb{S},\\ |w_1|+|w_3|+|q_N^Tw|&\le Cp^{-1/2}W,\qquad |w_2|\le CW,\end{aligned}\tag{A.5}$$

where $W=\|w\|_{\mathbb{S}}$; also $\|e_{\mathrm{eq}}\|_{\mathbb{S}}\le C\sqrt{p}$ and $\|a_1\|_{\mathbb{S}}+\|(0,1,0)^T\|_{\mathbb{S}}\le C$. Indeed $S_c\asymp\operatorname{diag}(p,1)$; the first and third columns of T_o have second component $O(p)$, giving the stronger w_2 bound. To differentiate the metric, write $D_o=\operatorname{diag}(S_c,1)$, $\zeta_o=T_o^{-1}w$ and $B_o=T_o^{-1}T_o'$. Then

$$w^T\mathbb{S}'w=\zeta_o^TD_o'\zeta_o-\zeta_o^T(B_o^TD_o+D_oB_o)\zeta_o,\qquad |w^T\dot{\mathbb{S}}w|\le C|\dot p|p^{-1}W^2.$$

This estimate uses the actual $\dot p$.

Coupled tracking of the normal and weak coordinates. The first graph correction is

$$G_1=\varepsilon a_1F,\qquad N=e_{\mathrm{eq}}\rho+G_1+w.\tag{A.6}$$

It satisfies $M_oG_1=\varepsilon e_{\mathrm{eq}}F$ and $q_N^TG_1=0$ exactly. To keep the finite-profile feedback, evaluate $A_j=\alpha_j$ at the actual mechanical state $\Xi(p,p\rho,\varepsilon^2(2\rho+G_{1,1}+w_1),\varepsilon(G_{1,2}+w_2))$, and set

$$d_\varepsilon=\{A_2+(A_3-A_1)\rho\}/p,\qquad \mathcal{T}_v=\mu B(1-\Phi_\theta(u/\varepsilon))\lambda_N,$$
$$\mathcal{T}_b=\mu(1-\Phi_\theta(u/\varepsilon))\lambda_N(\partial_{\dot q}b)M^{-1}Q_T.$$

With $E=G_1+w$ the exact equations are

$$\dot p=A_1,\qquad \dot\rho=d_\varepsilon-(A_3/p)q_N^TE+\mathcal{T}_b/p,$$
$$\varepsilon\dot E=M_oE-\varepsilon e_{\mathrm{eq}}d_\varepsilon+\varepsilon(A_3/p)e_{\mathrm{eq}}q_N^TE+\mathcal{T}_ve_V-\varepsilon e_{\mathrm{eq}}\mathcal{T}_b/p,\quad e_V=(0,1,0)^T.$$

Subtracting the derivative of G_1 gives, without truncation,

$$\begin{aligned}\varepsilon\dot{w} = M_o w &+ \varepsilon\frac{A_3}{p}(e_{\rm eq}+G_{1,\rho})q_N^T w \\ &- \varepsilon(e_{\rm eq}+G_{1,\rho})(d_\varepsilon - F) - \varepsilon G_{1,p}(A_1-\alpha_1) - \varepsilon^2 a_1 \mathscr{D}^2\rho \\ &+ \mathcal{T}_v e_V - \frac{\varepsilon}{p}(e_{\rm eq}+G_{1,\rho})\mathcal{T}_b.\end{aligned} \tag{A.7}$$

In particular both profile terms remain in the normal equation.

Here are the estimates needed to use this identity. They hold for $p \geq h\sqrt{\varepsilon}$, with any fixed $h > 0$, on partial intervals with bounded weak coefficient, strict normal-force bounds, $W \leq \rho_0\sqrt{p}$, and $a/2 \leq -\dot{p} \leq 2a$. From (A.3), $G_1 = O(\varepsilon)$, $\|G_{1,p}\|_{\mathbb{S}} + \|G_{1,\rho}\|_{\mathbb{S}} \leq C\varepsilon/p$, and $\|e_{\rm eq} + G_{1,\rho}\|_{\mathbb{S}} \leq C_h\sqrt{p}$. The last bound uses $\varepsilon/p^{3/2} \leq h^{-3/2}\varepsilon^{1/4}$. Superscript G denotes $w = 0$. Since its physical y, v are $O(\varepsilon^2)$, $|A_j^G - \alpha_j| \leq C\varepsilon^2$ and $|d_\varepsilon^G - F| \leq C\varepsilon^2/p$. The four saturated inhomogeneous terms in (A.7) have energy bounds, respectively,

$$C\varepsilon^2 p^{\beta-3}, \qquad C\varepsilon^3 p^{-1/2}, \qquad C\varepsilon^4 p^{-2}, \qquad C\varepsilon^4 p^{-1}, \tag{A.8}$$

when ordered as $-\varepsilon^2 a_1\mathscr{D}^2\rho$, $-\varepsilon e_{\rm eq}(d_\varepsilon^G - F)$, $-\varepsilon G_{1,\rho}(d_\varepsilon^G - F)$ and $-\varepsilon G_{1,p}(A_1^G - \alpha_1)$. Relative to $\varepsilon^2 p^{\beta-3}$, the last three bounds have ratios $\varepsilon p^{5/2-\beta}$, $\varepsilon^2 p^{1-\beta}$ and $\varepsilon^2 p^{2-\beta}$. For $h\sqrt{\varepsilon} \leq p \leq p_i < 1$ and the prescribed compact range $\beta < 5/2$, these are uniformly small. Thus each term is bounded by $C_h\varepsilon^2 p^{\beta-3}$. At $\beta = 5/2$ these three ratios are, respectively, $O(\varepsilon)$, $O_h(\varepsilon^{5/4})$ and $O_h(\varepsilon^{7/4})$ on the same interval. These bounds describe the endpoint of this forcing calculation; the proof below uses the prescribed compact range. The physical differences from this graph are $(\varepsilon^2 w_1, \varepsilon w_2)$, so $|A_j - A_j^G| \leq C\varepsilon W$ and $|d_\varepsilon - d_\varepsilon^G| \leq C\varepsilon W/p$. The three homogeneous terms have bounds $C_h\varepsilon W/p$, $C_h\varepsilon^2 p^{-1/2}W$ and $C\varepsilon^3 W/p$. A value bound $1 - \Phi_\theta(u/\varepsilon) \leq CR_\Phi$ gives for the complete profile forcing $C_h R_\Phi(1 + \varepsilon p^{-1/2} + \varepsilon^2 p^{-2}) \leq C_h R_\Phi$. The metric derivative and (A.5) consequently give

$$\varepsilon D^+ W \leq (-c_0 p + C_h\varepsilon/p)W + C_h\varepsilon^2 p^{\beta-3} + C_h R_\Phi. \tag{A.9}$$

We retain the positive metric term when crossing the critical annulus. Finally the rigid field annihilates $\mathcal{C}$. At equal (p, b), its difference from the actual field is

$$\delta\dot{p} = A_1 - \alpha_1, \quad \delta\dot{b} = (A_2 - \alpha_2) + (A_3 - \alpha_3)\rho - A_3 q_N^T w + \mathcal{T}_b, \quad |A_j - \alpha_j| \leq C(\varepsilon^2 + \varepsilon p^{-1/2}W).$$

Thus (A.2) yields the coupled transport bound

$$\left|\frac{d\mathcal{C}_\varepsilon}{dp}\right| \leq C\{\varepsilon^2 p^{-\beta} + p^{-\beta-1/2}W + R_\Phi p^{-\beta}\}. \tag{A.10}$$

A.2 Capture and the actual descending threshold

Lemma A.1 (The actual incoming threshold)**.** *The physical initial states reach a common fixed section $p = p_i > 0$ in positive-slip contact. On its continuous contact passage there*

is a unique first descending crossing t_e of $u(t_e) = \gamma\varepsilon T$, where $T = m|\log\varepsilon|$ and m is a sufficiently large fixed constant. If p_{stop} is the zero of the prescribed detuned rigid slip, then

$$p_{\mathrm{stop}} = ak\sqrt{\varepsilon}/\gamma + O(\varepsilon), \qquad p_e = p_{\mathrm{stop}} + a\varepsilon T + R_e, \qquad R_e = o(\varepsilon), \tag{A.11}$$

$$b_e = -\ell p_{\mathrm{stop}} + o(\sqrt{\varepsilon}), \quad |X_e - G| = O(\sqrt{\varepsilon}), \quad \lambda_N(t_e) = \ell + O(\sqrt{\varepsilon}), \quad V_e = O(\varepsilon), \tag{A.12}$$

$$E_{N,e} := |Y_e + 2\ell| + |V_e| + |Z_e + \ell| = O(\sqrt{\varepsilon}). \tag{A.13}$$

All assertions are uniform on the fixed compact classes.

Proof. For capture, freeze $p > 0$, $b < 0$ on a closed incoming rigid arc and put $\ell_f = -b/p$. The frozen fast equations and proper energy are

$$Y' = V, \quad V' = b + pL, \quad Z' = -Z - L, \quad L = (Z - Y)_+,$$
$$\mathcal{E}_f = \frac{V^2}{2p} + \frac{Z^2 + L^2}{2} + \ell_f Y, \qquad \mathcal{E}_f' = -(Z')^2.$$

Minimization over Z gives potential $\ell_f Y$ for $Y \geq 0$ and $Y^2/4 + \ell_f Y$ for $Y < 0$, so sublevels are uniformly compact on the fixed positive ranges of p, ℓ_f. In an invariant zero-dissipation set, $L = -Z$. If $L > 0$, then $Y = 2Z$, $V = 0$, and $V' = 0$ gives $L = \ell_f$; if $L = 0$, then $Z = 0$, $Y \geq 0$, but $Y'' = b < 0$ cannot remain there. The invariance principle therefore gives attraction to $(-2\ell_f, 0, -\ell_f)$, including initially free states. The Hurwitz matrix (A.4) gives a common local exponential estimate on this fixed arc. Continuous dependence and a finite cover of the parameter and initial-state compact sets give a common finite fast capture time. Over it the actual slow drift is $O(\varepsilon)$ and the closed-arc slip margin makes the profile error $O(e^{-c/\varepsilon})$, so the actual trajectory enters the strict-contact neighborhood too.

Subtract the moving equilibrium there. Its physical-time derivative is bounded on the fixed arc; the local Lyapunov inequality gives $|N - N_*(p, b)| \leq Ce^{-c(t-t_c)/\varepsilon} + C\varepsilon$. The time integral of the slow-field discrepancy is $O(\varepsilon)$. Gronwall and alignment at a transverse section, leaving a fixed positive time after capture, give

$$\mathcal{C}_\varepsilon(p_i) = C_1 + O(\varepsilon), \quad W(p_i) = O(\varepsilon), \quad t_\varepsilon(p_i) - t_{\mathrm{rig},\nu}(p_i) = O(\varepsilon). \tag{A.14}$$

The graph correction is $O(\varepsilon)$ at this section. Capture before it may contain departures and recontacts.

We now stop on partial intervals by $|\mathcal{C}_\varepsilon - C_1| \leq 1$, $W \leq \rho_0\sqrt{p}$, strict force bounds, the regular chart and $a/2 \leq -\dot{p} \leq 2a$, adding the slip bounds below. Choose p_i, ρ_0 small, then K_L large, and finally ε small. On $p \geq K_L\sqrt{\varepsilon}$ the positive term in (A.9) is absorbed, and the rigid slip slope is γ/a. Requiring $u \geq c_{\mathrm{sl}}p$ gives a profile tail smaller than every fixed power of ε. Integration in the actual decreasing p direction gives

$$W(p) \leq H_\varepsilon(p) + C\varepsilon^2 p^{\beta-4} + CR_\Phi/p, \qquad H_\varepsilon = CW(p_i)e^{-c(p_i^2 - p^2)/\varepsilon}. \tag{A.15}$$

Indeed the forcing integral is $C\varepsilon\int_p^{p_i} e^{-c(s^2-p^2)/\varepsilon} s^{\beta-3}\,ds$; $s^2 - p^2 \geq 2p(s-p)$ bounds it by $C\varepsilon^2 p^{\beta-4}$. Substitution into (A.10) gives

$$|\mathcal{C}_\varepsilon(p) - C_1| \leq C\{\varepsilon + \varepsilon^2 p^{1-\beta} + \varepsilon^2 p^{-7/2} + R_\Phi p^{-\beta-1/2}\}. \tag{A.16}$$

The principal transport integral is $\varepsilon^2 \int_p^{p_i} s^{-9/2}\,ds = O(\varepsilon^2 p^{-7/2})$. The weighted integral of H_ε is $O(\varepsilon W(p_i)) = O(\varepsilon^2)$: above $p_i/2$ its width is $O(\varepsilon)$, and below $p_i/2$ the fixed exponential absorbs every endpoint power. Thus the safe endpoint has $W = O(\varepsilon^{\beta/2})$ and $\mathcal{C}_\varepsilon - C_1 = O(\varepsilon^{1/4})$. These bounds improve the coefficient and relative-energy stops. The rigid slip on the bounded weak wedge is at least $3\gamma p/(4a)$ after reducing p_i; its physical normal correction is $o(p)$ and $k\sqrt{\varepsilon}/p \le k_+/K_L$. The large choice of K_L improves the slip stop. The force, chart and orientation bounds improve too, so the endpoint is reached.

On $h_I\sqrt{\varepsilon} \le p \le K_L\sqrt{\varepsilon}$, where $h_I = ak_-/(2\sup\gamma) > 0$, stop additionally at the actual threshold $u = \gamma\varepsilon T$. Choose

$$\kappa_{\text{tail}} = \alpha \inf\gamma/2, \qquad N_{\text{tail}} = \kappa_{\text{tail}} m > \beta_+/2 + 3/4. \tag{A.17}$$

The coarse energy radius makes the actual and graph slips differ by $O(\varepsilon)$, so both are at least $(\inf\gamma)\varepsilon T/2$ before this stop. Hence $R_\Phi \le C\varepsilon^{N_{\text{tail}}}$. Retain the positive C_h/p in the equation for W. Since $\int C_h\,dt/p \le C_h \log(K_L/h_I)$ and $\Delta t \le C\sqrt{\varepsilon}$, its bounded Gronwall factor gives

$$W \le C\{\varepsilon^{\beta/2} + \varepsilon^{N_{\text{tail}}-1/2}\} = O(\varepsilon^{\beta/2}).$$

The three contributions to weak transport on the annulus are $O(\varepsilon^{(5-\beta)/2})$, $O(\varepsilon^{1/4}) + O(\varepsilon^{N_{\text{tail}}-\beta/2-1/4})$, and $O(\varepsilon^{N_{\text{tail}}+(1-\beta)/2})$. They give $\mathcal{C}_\varepsilon - C_1 = O(\varepsilon^{1/4})$ throughout; (A.16) also holds there up to a fixed factor. The component weights now yield $w_1, w_3, q_N^T w = O(\varepsilon^{\beta/2-1/4}) = o(\sqrt{\varepsilon})$, $w_2 = O(\varepsilon^{\beta/2}) = o(\varepsilon)$. Together with $G_{1,V} = 2\varepsilon F$, these prove the normal bounds in (A.13) on every partial annulus. In particular all energy, coefficient, force, chart and orientation stops are strictly improved.

Along a rigid trace, division of $\dot{u}_0 = g_* + O(p) + O(p^{\beta-1})$ by $\dot{p} = -a + O(p)$ gives $u_{0,\text{rig},\nu} = \gamma p/a + O(p^2) + O(p^\beta)$. Detuning subtracts $k\sqrt{\varepsilon}$ without changing that trace. This proves the formula for p_{stop}. At equal p, the inverse weak-coordinate estimate gives on the annulus

$$u - u_{\eta,\text{rig},\nu}(p) = O(\varepsilon^2 + p^\beta \varepsilon^{1/4}) = o(\varepsilon). \tag{A.18}$$

Also $\dot{u} = g_* + O(\sqrt{\varepsilon}) + O(\varepsilon^{N_{\text{tail}}}) < -\inf\gamma/2$. At the lower annular boundary the rigid slip is at most $-k_-\sqrt{\varepsilon}/2 + O(\varepsilon)$, contradicting the positive slip stop and (A.18). The threshold is therefore reached first and is crossed strictly once. Earlier slip margins exclude a preceding crossing on this continuous passage. Expansion at the same rigid zero gives more explicitly

$$|R_e| \le C\{\varepsilon^2 + \varepsilon^{\beta/2+1/4} + \varepsilon^{3/2}T + \varepsilon^2 T^2\} = o(\varepsilon). \tag{A.19}$$

If the profile tail is kept separately it adds $C\varepsilon^{N_{\text{tail}}-1/4} = o(\varepsilon)$. This proves the lemma. □

A.3 The critical passage and its first force zero

The threshold sets $\tau = (t - t_e)/\varepsilon - T$ and $U = u/\varepsilon$, so $U(-T) = \gamma T$ exactly. Its normal error is $E_{N,e} = O(\sqrt{\varepsilon})$, including Z, and $(1+T)^2 E_{N,e} \to 0$. The power two matters because the saturated critical propagator has a quadratic neutral part.

The limiting normal–slip equations are

$$\begin{aligned} Y' = V, \qquad Z' = -Z - L_+, \qquad L = Z - Y, \\ V' = C_G(1 - \Phi_\theta(U))L_+, \\ U' = d_\lambda + [-h + \mu A_G(1 - \Phi_\theta(U))]L_+. \end{aligned} \tag{A.20}$$

with normalization $(Y, V, Z, U - g_*\tau) \to (-2\ell, 0, -\ell, 0)$ as $\tau \to -\infty$.

Lemma A.2 (The normalized critical passage)**.** *This normalization determines a unique solution $\overline{W}$ on $\mathbb{R}$, with error $O(e^{\omega_c \tau})$ as $\tau \to -\infty$, for common $\omega_c > 0$. Its finite first force zero τ_r satisfies*

$$Y_r^0 = Z_r^0 < 0, \qquad q_{\mathrm{in}} = V_r^0 > 0, \qquad U_r^0 > 0, \qquad q_{\mathrm{in}} + Y_r^0 > 0. \tag{A.21}$$

The solution, first-zero time and release data depend continuously on (λ, θ); the four strict quantities have common positive lower bounds on the fixed compact classes.

Proof. On a far left interval use positive contact. Define $If(\tau) = \int_{-\infty}^{\tau} f(s)\, ds$ and $Kf(\tau) = \int_{-\infty}^{\tau} e^{-2(\tau - s)} f(s)\, ds$. The equations with their incoming limits are equivalent to

$$\begin{aligned} Y + 2\ell = IV, \quad Z + \ell = KIV, \quad L = \ell + (K - \mathrm{Id})IV, \\ U = g_*\tau - hI(L - \ell) + (\mu A_G / C_G)V, \\ V = I[C_G(1 - \Phi_\theta(U))L]. \end{aligned}$$

In $\|f\|_\omega = \sup_{\tau \le -T_0} e^{-\omega\tau}|f(\tau)|$, where $0 < \omega < \min(2, \alpha \inf \gamma/2)$, the operators I, K have norms $1/\omega, 1/(2 + \omega)$. The first four formulas reconstruct the state boundedly from V. On a fixed ball, sufficiently large common T_0 makes $L > 0$ and U lie in the positive tail. Both $1 - \Phi_\theta(U)$ and its U derivative are then bounded by $Ce^{\alpha \inf \gamma \tau}$. The last formula is a self-map contraction, with constant $Ce^{-\alpha \inf \gamma T_0} < 1$. It gives the exponential tail. Any solution with the stated incoming limits first has exponentially small V'; integration from its zero incoming velocity gives these same formulas and places it in the contraction ball. This proves uniqueness. The locally Lipschitz positive-part field has at most linear growth, so continuation is global.

The filter formula $Z(\tau) = e^{-(\tau - s)} Z(s) - \int_s^\tau e^{-(\tau - \xi)} L_+(\xi)\, d\xi$ keeps $Z < 0$. At $U = 0$ its derivative is $d_\lambda + B_G L_+ > 0$, so U stays positive. Until a first force zero, $V' > 0$, hence $V > 0$. If force remained positive forever, $Y' = V$ would eventually make $Y > 0$, contrary to $Y < Z < 0$. A finite first zero exists. There $L' = -Z - L_+ - V \le 0$. If $L' = 0$, differentiation from the contact side gives $L''(\tau_r^-) = Z_r < 0$, forcing $L < 0$ just before its first zero. Thus $L' = -(q_{\mathrm{in}} + Y_r^0) < 0$. Uniform contraction, continuous dependence on compact intervals, and this transverse first zero prove continuous dependence of all the data. Compactness gives the common strict margins. □

Lemma A.3 (Compactly supported variations of the critical passage)**.** *Fix λ and an admissible C^4 baseline Φ^{b}. Let $K \Subset (Q_\Phi, \infty)$ be a closed interval on which $(\Phi^{\mathrm{b}})'$ has a positive lower bound, and put*

$$E_K = \{\eta \in C^4(\mathbb{R}) : \operatorname{supp} \eta \subset K\}.$$

On a sufficiently small open ball about zero in E_K, equipped with its C^4 norm, the normalized first-release time and state for $\Phi^{\mathrm{b}} + \eta$ are Fréchet C^1. For a direction $\psi \in E_K$, let z be the fixed-time variation of the contact extension, in state order (Y, V, Z, U), and write $q = q_{\mathrm{in},\lambda}[\Phi^{\mathrm{b}}] = V(\tau_r)$ and $y = Y(\tau_r) = Z(\tau_r)$. Then

$$D\tau_r[\psi] = \frac{z_Z(\tau_r) - z_Y(\tau_r)}{q + y}, \qquad Dq[\psi] = z_V(\tau_r). \tag{A.22}$$

Let $F_{\Phi^{\mathrm{b}}}$ be the contact extension of (A.20)*, obtained by replacing L_+ by L. With $b = (0, C_G, 0, \mu A_G)^T$, $A = D_W F_{\Phi^{\mathrm{b}}}(\overline{W})$ and $\zeta' = -A^T\zeta$, $\zeta(\tau_r) = e_V$, one has*

$$Dq[\psi] = -\int_{-\infty}^{\tau_r} L(\tau) b^T \zeta(\tau) \psi(U(\tau))\, d\tau. \tag{A.23}$$

Proof. Choose a finite $\tau_a < \tau_r$ so negative that $U(\tau) > \sup K + 1$ and $L(\tau) > \ell/2$ for $\tau \le \tau_a$. Every perturbation supported in K vanishes on this entire baseline tail. Attach the same value $\overline{W}(\tau_a)$ to the perturbed forward initial-value problem. The attached orbit has the prescribed normalization; the tail uniqueness in Lemma A.2 identifies it with the perturbed pullback. Conversely, any normalized perturbed orbit lies above K sufficiently far in the past, where that same uniqueness applies. Thus the incoming state at τ_a is exactly independent of the shape.

Use the contact extension through the baseline first release. Evaluation $(\eta, U) \mapsto \eta(U)$ is jointly C^1 on $E_K \times \mathbb{R}$, so finite-time ODE dependence from the common initial state is Fréchet C^1. At the event $L' = -(q+y) < 0$, and the implicit-function theorem gives a nearby C^1 zero. The exact positive incoming tail, the positive minimum of L on the preceding compact interval, and strict decrease in a small sign-changing event bracket exclude all earlier zeros. The extension therefore agrees with the true positive-part solution up to its first release.

The fixed-time variation satisfies

$$z' = Az - L\psi(U)b, \qquad z(\tau_a) = 0.$$

Differentiating $Z - Y = 0$ gives the event-time formula. Since $V'(\tau_r) = C_G[1 - \Phi^{\mathrm{b}}(U(\tau_r))]L(\tau_r) = 0$, the moving-event contribution to Dq vanishes. For the fixed state event $H = Z - Y$ and output $J = V$, with no direct time or profile dependence, the terminal covector is therefore

$$\nabla J - \frac{\nabla J \cdot F}{\nabla H \cdot F} \nabla H = e_V \neq 0, \qquad \nabla H \cdot F = -(q + y) \neq 0.$$

Pairing with the terminal adjoint gives $(\zeta^T z)' = -L\psi(U) b^T \zeta$, hence (A.23). Its integrand vanishes before τ_a, so no integrability condition for the adjoint at negative infinity is needed. Small perturbations in E_K remain admissible by the derivative margin on K, unchanged core and zero, and the finite compact-support contribution to (F4). The weighted and unweighted C^4 norms on E_K are equivalent. □

Theorem A.4 (First release of the captured passage)**.** *Let* $\sigma = \sqrt{\varepsilon}$ *and*

$$\mathcal{D}_\varepsilon = (1+T)^2(E_{N,e} + e^{-\omega_c T}) + \sigma(1+T)^3 + \varepsilon(1+T)^4 \longrightarrow 0. \tag{A.24}$$

The contact passage from p_i *has a first release* $t_r > t_e$*. Slip remains positive, and its complete release state satisfies*

$$t_r = t_e + \varepsilon\{T + \tau_r + O(\mathcal{D}_\varepsilon)\}, \quad y_r = z_r = \varepsilon^2\{Y_r^0 + O(\mathcal{D}_\varepsilon)\}, \tag{A.25}$$

$$v_r = \varepsilon\{q_{\rm in} + O(\mathcal{D}_\varepsilon)\}, \quad u_r = \varepsilon\{U_r^0 + O(\mathcal{D}_\varepsilon)\}, \quad \dot g_r = \varepsilon\{q_{\rm in} + Y_r^0 + O(\mathcal{D}_\varepsilon)\} > c_0\varepsilon, \tag{A.26}$$

$$|X_r - G| = O(\sigma), \qquad p_r/\sigma \to H := ak/\gamma, \qquad b_r/\sigma \to -\ell H. \tag{A.27}$$

The leading normal–slip data depend only on (λ, θ)*.*

Proof. The exact scaled equations have Y', Z' as in (A.20) and

$$V' = b + [p + \mu B(1 - \Phi_\theta(U))]L_+, \qquad U' = a_T + [B - \mu A + \mu A(1 - \Phi_\theta(U))]L_+.$$

Stop while (Y, V, Z) stays in a fixed enlarged bounded neighborhood of the reference normal range, $|U - \overline{U}(\tau)| \le 1$, and the mechanical state stays in a common chart. On $[-T, -T_0]$, also stop at the boundary of the convex positive-contact tail neighborhood used below. On the tail, $\overline{U}(\tau) = g_*\tau + O(e^{\rho\tau})$, so this tube and every joining segment satisfy $U \ge -\gamma\tau - C$. On every partial interval of length $T + C$, the mechanical field is bounded and $|X - G| \le CM_\varepsilon$, $M_\varepsilon = \sigma + \varepsilon(1+T)$. Subtracting (A.20) gives zero residual in Y, Z and residuals

$$b + pL_+ + (\mu B - C_G)(1 - \Phi_\theta(U))L_+, \qquad (a_T - d_\lambda) + (B - \mu A + h)L_+ + \mu(A - A_G)(1 - \Phi_\theta(U))L_+,$$

of size CM_ε. The term $b + pL_+$ is included.

On $[-T, -T_0]$, the saturated linear part is

$$M_\infty = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 1 & 0 & -2 & 0 \\ h & 0 & -h & 0 \end{pmatrix}.$$

Its homogeneous solutions have constant V, affine Y, a decaying $Z - Y/2 + V/4$, and at most quadratic U; hence $\|e^{M_\infty s}\| \le C(1+s)^2$ for $s \ge 0$. In a convex positive-contact tail neighborhood the remaining nonlinearity is $(1 - \Phi_\theta(U))L(0, C_G, 0, \mu A_G)^T$ and its derivative is $Ce^{\omega_c\tau}$. Decrease ω_c if needed so the same exponent applies here and to the pullback tail. This follows from (F4) on every joining segment. Thus the profile tail has all needed moments through degree two; no slow-time derivative of its composition is used. For $D = |W_\varepsilon - \overline{W}|$, Duhamel gives on each partial interval

$$D(\tau) \le C(1 + \tau + T)^2 a_\varepsilon + C\int_{-T}^{\tau} (1 + \tau - s)^2 e^{\omega_c s} D(s)\, ds,$$

$$a_\varepsilon \le C\{E_{N,e} + e^{-\omega_c T} + \sigma(1+T) + \varepsilon(1+T)^2\}.$$

Divide by $(1+\tau+T)^2$ and take the supremum. The integral coefficient is at most $Ce^{-\omega_c T_0}\int_0^\infty (1+s)^2 e^{-\omega_c s}\,ds < 1/2$ after increasing T_0. Therefore $\sup_{[-T,-T_0]} D \le C\mathcal{D}_\varepsilon$. This improves the scaled-state, positive-force and chart stops. On a common fixed interval from $-T_0$ past all reference releases, the positive-part field is uniformly Lipschitz. Stopped Gronwall gives the same bound and improves the remaining state stop.

The reference signed quantity $L = Z - Y$ is positive before a common small bracket of τ_r, has opposite endpoint signs there, and has a strictly negative derivative in the bracket. The exact identity $L' = -Z - L_+ - V$ transfers this derivative margin and the signs to the actual trajectory. It therefore has exactly one zero there, at $\tau_r + O(\mathcal{D}_\varepsilon)$, and no preceding zero. The threshold lemma supplies contact from p_i to t_e. At physical $u = 0$, $\dot u = a_T + B\lambda_N > 0$ on the chart, so the initial positive slip persists. At the zero, $\dot g = \varepsilon(V+Z)$; the data and strict opening follow. Finally the physical duration is $O(\varepsilon(1+T)) = o(\sigma)$. Integration of $\dot p = -a + O(M_\varepsilon)$ gives

$$p_r = p_e - a\varepsilon(T+\tau_r) + O(\varepsilon\mathcal{D}_\varepsilon + \varepsilon(1+T)M_\varepsilon).$$

The $a\varepsilon T$ term cancels before using (A.19); the b change is $O(\varepsilon(1+T))$. This proves (A.27). Uniqueness of the normalized critical solution proves the asserted dependence of the leading data. □

B Compression and release with inherited memory

The first touchdown may have positive, zero or negative p; subsequent touchdowns have $p < 0$. Their contact passages have the same two time scales. Compression reduces the incoming $O(\delta)$ slip to $O(\varepsilon)$ on the time scale $\varepsilon/\sqrt{\delta}$; the normal motion then releases on the time scale ε. The friction tail contributes different errors to velocity and displacement, so we estimate its first two time moments. A projection of the normal velocity removes the stiff slip derivative during release and carries the displacement and memory inherited from compression. The comparison retains the finite incoming scale, including the case $p_{\mathrm{in}} = 0$.

B.1 Incoming data and the finite-scale reference

Write $u_r = u_0 - r$, where the belt offset r is constant during the passage. The determinant of (p, u_0) with respect to (ϕ, ω) at G is $a\cos\phi^* = 35/136$. Thus (p, u_0, y, v) is another regular mechanical chart, on the same local branch as the chart in Section 2. In particular, b is a dependent function of these four coordinates.

Fix $C_0, C_{\mathrm{in}}, Z_*, \gamma_0 > 0$ and a compact set

$$\mathcal{K} \Subset \{(\pi, Q, S) : Q > 0,\ S > 0,\ \pi S < 2hQ\}. \tag{B.1}$$

All constants below are uniform on these sets and on the mechanical and profile classes of

Section 2. We consider actual physical inputs at time t_{in} satisfying

$$\begin{gathered} p_{\mathrm{in}} = \pi\delta, \quad v_{\mathrm{in}} = -\delta^2 Q, \quad u_{r,\mathrm{in}} = \delta S, \quad y_{\mathrm{in}} = z_{\mathrm{in}} = \varepsilon\delta^2\zeta, \\ (\pi, Q, S) \in \mathcal{K}, \qquad 0 < \varepsilon \le C_0\delta^2, \\ |\zeta| \le Z_*, \quad Q - \zeta \ge \gamma_0, \quad |X_{\mathrm{in}} - G| \le C_{\mathrm{in}}\delta. \end{gathered} \tag{B.2}$$

The inverse uses $u_0 = u_r + r$; b_{in} is evaluated at this same state. The mechanical localization in (B.2) implies $b_{\mathrm{in}} = O(\delta)$. It allows $r = O(\delta)$, as needed at the first touchdown. The smaller offset range used for subsequent returns will be imposed only in that application.

Put

$$\sigma = \sqrt{\delta}, \qquad e = \varepsilon\delta^{-3/2}, \qquad \tau = e\sigma = \varepsilon/\delta, \qquad h_{\mathrm{in}} = \mu A(\phi_{\mathrm{in}}) - B(\phi_{\mathrm{in}}). \tag{B.3}$$

Then $e \le C_0\sigma$, with no positive lower bound on e. Use compression time $s = \sigma(t - t_{\mathrm{in}})/\varepsilon$ and variables

$$(y, z) = \varepsilon\delta^{3/2}(Y, Z), \qquad v = \delta^2 V_c, \qquad u_r = \delta U, \qquad L = Z - Y. \tag{B.4}$$

The saturated reference is the signed linear system

$$\begin{gathered} \bar{Y}' = \bar{V}_c, \qquad \sigma\bar{Z}' = \bar{Y} - 2\bar{Z}, \qquad \bar{L} = \bar{Z} - \bar{Y}, \\ \bar{V}_c' = \pi\bar{L}, \qquad \bar{U}' = -h_{\mathrm{in}}\bar{L}, \\ (\bar{Y}, \bar{Z}, \bar{V}_c, \bar{U})(0) = (\sigma\zeta, \sigma\zeta, -Q, S). \end{gathered} \tag{B.5}$$

We shall prove that its first arrest T_σ, defined by $\bar{U} = 0$, occurs in positive force. From that event the inherited normal reference is

$$N_0' = A_D N_0, \quad A_D = \begin{pmatrix} 0 & 1 & 0 \\ -1/2 & 0 & 1/2 \\ 1 & 0 & -2 \end{pmatrix}, \qquad N_0(0) = (\bar{Y}, \sigma\bar{V}_c, \bar{Z})(T_\sigma). \tag{B.6}$$

Here $N_0 = (Y_0, V_0, Z_0)$ and its elapsed time is denoted by ρ. Its first force zero is R_0; its initial memory is the value $\bar{Z}(T_\sigma)$ inherited from compression.

The zero-scale compression solves $y'' = -\pi y/2$, $u' = hy/2$, with $(y, y', u)(0) = (0, -Q, S)$. Its first arrest time and amplitude are

$$\xi_s = \begin{cases} \sqrt{2/\pi}\,\arccos(1 - \pi S/(hQ)), & \pi > 0, \\ 2\sqrt{S/(hQ)}, & \pi = 0, \\ \sqrt{2/(-\pi)}\,\operatorname{arcosh}(1 + (-\pi)S/(hQ)), & \pi < 0, \end{cases} \qquad A_s^2 = \frac{QS}{h} - \frac{\pi S^2}{2h^2} > 0, \tag{B.7}$$

$$(y, y', u)(\xi_s) = (-2A_s, -Q + \pi S/h, 0).$$

The formulas extend smoothly through $\pi = 0$. For $\pi > 0$ the strict inequality in (B.1) says exactly that arrest precedes the first return of y to zero.

Lemma B.1 (Normal opening)**.** *Let $n'_\# = A_D n_\#$ and $n_\#(0) = (-2, 0, -1)$. Its force has a finite first zero $\rho_\#$, with*

$$Y_\#(\rho_\#) = Z_\#(\rho_\#) = -\gamma_\# < 0, \qquad V_\#(\rho_\#) = \nu_\# > \gamma_\# > 0. \tag{B.8}$$

Proof. While $L_\# = Z_\# - Y_\# > 0$, $V'_\# = L_\#/2 > 0$. If there were no force zero, $V_\#$ would eventually have a positive lower bound and $Y_\#$ would reach zero. Before that point, $Z_\#(\rho) = -e^{-2\rho} + \int_0^\rho e^{-2(\rho-t)} Y_\#(t)\, dt < 0$, which contradicts positive force at $Y_\# = 0$. Hence the first force zero is finite, occurs with $Y_\# = Z_\# < 0$, and has $V_\# > 0$. The identities $L'_\# = -Y_\# - 2L_\# - V_\#$ and $L''_\# = -V_\# - 2L'_\# - L_\#/2$ show that its derivative there is strictly negative: if it vanished, the second identity would make the force negative just before its first zero. At the zero, $-L'_\# = V_\# + Z_\# = \nu_\# - \gamma_\#$. □

Theorem B.2 (Contact from a complete incoming state)**.** *For all sufficiently small δ, every input* (B.2) *has the following contact passage. The constant $M_{\rm sl}$ is fixed independently of ε, δ.*

(i) Compression to $w = M_{\rm sl}$. *Positive contact persists to the first descending section $w = u_r/\varepsilon = M_{\rm sl}$. At this section,*

$$\begin{gathered} L_{\rm sl} \ge \ell_a > 0, \quad U'(s_{\rm sl}) \le -g_a < 0, \qquad |s_{\rm sl} - T_\sigma| \le C(e+\tau), \\ |(Y, \sigma V_c, Z)(s_{\rm sl}) - N_0(0)| \le Ce. \end{gathered} \tag{B.9}$$

(ii) Release and opening. *The passage then reaches its first release. There is a common bracket $0 < r_- < R_0 < r_+ < R_*$ for the inherited first release. In elapsed release time $\rho = (t - t_{\rm sl})/\varepsilon$, set $v = \delta^{3/2}\widehat{V}$. On every positive-contact prefix through the actual first force zero R_ε,*

$$\begin{aligned} |(Y, \widehat{V}, Z) - N_0| + |L - L_0| + |L' - L'_0| \le Ce, \qquad L_0 = Z_0 - Y_0, \\ |R_\varepsilon - R_0| + |(Y, \widehat{V}, Z)(R_\varepsilon) - N_0(R_0)| \le Ce, \qquad R_\varepsilon \in (r_-, r_+). \end{aligned} \tag{B.10}$$

At release, for common positive constants,

$$y_L = z_L < 0, \quad v_L > 0, \quad w_b \le u_{r,L}/\varepsilon \le W_*, \qquad v_L + z_L/\varepsilon \ge \kappa_{\rm op}\delta^{3/2} > 0. \tag{B.11}$$

(iii) Physical state, time and leading increments. *Reconstruct a reference physical release in the same inverse chart by*

$$p_{L,0} = \pi\delta, \quad u_{r,L,0} = 0, \quad u_{0,L,0} = r, \quad y_{L,0} = z_{L,0} = 0, \quad v_{L,0} = \delta^{3/2} V_0(R_0). \tag{B.12}$$

With Δ denoting actual minus reconstructed values,

$$|\Delta p_L| + |\Delta b_L| \le Ce\delta, \qquad |\Delta u_{r,L}| + |\Delta v_L| \le Ce\delta^{3/2}, \qquad |y_L| + |z_L| \le Ce\delta^3. \tag{B.13}$$

For $t_{L,0} = t_{\rm in} + (\varepsilon/\sigma)T_\sigma + \varepsilon R_0$,

$$|t_{L,\varepsilon} - t_{L,0}| \le C\{(\varepsilon/\sigma)(e+\tau) + \varepsilon e\} \le C(C_0)\varepsilon. \tag{B.14}$$

The entire physical contact prefix satisfies

$$\begin{aligned} t_L - t_{\mathrm{in}} \le C\varepsilon/\sqrt{\delta}, \quad \sup|\mathfrak{X} - G| \le C\delta, \quad \sup|p - \pi\delta| \le Ce\delta, \quad \dot{p} < -a/2, \\ \sup(|y| + |z|) \le C\varepsilon\delta^{3/2}, \quad \sup|v| \le C\delta^{3/2}, \qquad 0 \le \int_{t_{\mathrm{in}}}^{t} \lambda_N \, dt \le C\delta. \end{aligned} \tag{B.15}$$

Finally, using the actual normalized inputs in (B.7)*,*

$$\begin{aligned} v_L &= \delta^{3/2}\{A_s\nu_\# + O(\sigma)\}, & y_L = z_L &= \varepsilon\delta^{3/2}\{-A_s\gamma_\# + O(\sigma)\}, \\ q_L - q_{\mathrm{in}} &= (\varepsilon/\sigma)\{\xi_s\dot{q}_{\mathrm{in}} + O(\sigma)\}, & p_L - p_{\mathrm{in}} &= (\varepsilon/\sigma)\{-a\xi_s + O(\sigma)\}, \\ \dot{q}_L - \dot{q}_{\mathrm{in}} &= \delta(S/h)(-5/4, 1)^T + O(\delta^{3/2}), & b_L - b_{\mathrm{in}} &= a\beta\delta S/h + O(\delta^{3/2}). \end{aligned} \tag{B.16}$$

These estimates compare values of the actual compliant event map.

B.2 Proof of the contact estimates

Proof of Theorem B.2. We first obtain positive-force margins for the reference, then compare the physical compression up to a fixed small slip and along its descending tail. The tail force enters velocity directly and position after one time integration; both of its time moments are needed. We then project out the stiff slip derivative during release, retaining the inherited displacement and memory, and reconstruct the physical state and its leading increments.

Step 1: Positive reference compression. The reference filter gives the exact formulas

$$\begin{aligned} R_f := \bar{Z} - \bar{Y}/2 &= \tfrac{1}{2}\sigma\zeta e^{-2s/\sigma} - \tfrac{1}{2}\int_0^s e^{-2(s-t)/\sigma}\bar{V}_c(t)\,dt, \\ \bar{L}(s) &= -\tfrac{1}{2}\sigma\zeta(1 - e^{-2s/\sigma}) - \tfrac{1}{2}\int_0^s (1 + e^{-2(s-t)/\sigma})\bar{V}_c(t)\,dt. \end{aligned} \tag{B.17}$$

The kernel mass is at most $\sigma/2$. Its integral form and Gronwall's inequality bound the reference on every fixed finite interval, uniformly in σ, and give $R_f = O(\sigma)$. Substituting $\bar{L} = -\bar{Y}/2 + R_f$ gives an $O(\sigma)$ comparison with the zero-scale system in (B.7). Compactness of $\mathcal{K}$ gives $0 < T_- \le \xi_s \le T_+$, $A_s \ge A_- > 0$, and a common positive-force neighborhood of arrest.

Positivity close to the initial force zero needs a separate estimate. Equation (B.17) first gives $|\bar{L}(s)| \le Cs$, hence $|\bar{V}_c(s) + Q| \le Cs^2$. Choose a common small s_0 such that $\bar{V}_c \le -Q + \eta$ there, with $\eta \le \min(Q_-/2, \gamma_0/2)$. The second formula then yields

$$\bar{L}(s) \ge \min\{(Q - \eta)/2, Q - \eta - \zeta\}s \ge \min\{Q_-/4, \gamma_0/2\}s \quad (0 < s \le s_0). \tag{B.18}$$

Indeed, a negative coefficient of $\sigma(1 - e^{-2s/\sigma})$ is bounded using $1 - e^{-2s/\sigma} \le 2s/\sigma$; a positive one is discarded. Beyond s_0 the reduced positive force and the $O(\sigma)$ comparison exclude a force zero up to a fixed extension past arrest. Thus $\bar{L} \ge \kappa_L s$ on the required interval, for a common $\kappa_L > 0$, and $\bar{U}' = -h_{\mathrm{in}}\bar{L}$ gives a unique first arrest with $T_\sigma = \xi_s + O(\sigma)$.

The exact invariant gives $\bar{V}_c(T_\sigma) = -Q + \pi S/h_{\rm in}$. An amplitude calculation which does not divide by π is useful here. Put $I(s) = \int_0^s \bar{L}\,dt$, so that $\bar{V}_c = -Q + \pi I$, $I' = -\bar{Y}/2 + R_f$ and $I(T_\sigma) = S/h_{\rm in}$. Integrating $2\bar{Y}\bar{V}_c = -4\bar{V}_c I' + 4\bar{V}_c R_f$ gives

$$\bar{Y}(T_\sigma)^2 = \sigma^2\zeta^2 + \frac{4QS}{h_{\rm in}} - \frac{2\pi S^2}{h_{\rm in}^2} + 4\int_0^{T_\sigma} \bar{V}_c R_f\,ds. \tag{B.19}$$

The force margin selects its negative square root. Since $h_{\rm in} = h + O(\delta)$,

$$N_0(0) = A_s(-2, 0, -1) + O(\sigma). \tag{B.20}$$

Choose $0 < u_a < S_-/4$ small enough that the entire reference segment $-u_a/2 \le \bar{U} \le u_a$ lies in the above arrest neighborhood and has $\bar{L} \ge 4\ell_a > 0$, $-\bar{U}' \ge m_a > 0$. This choice uses the compact strict margin in (B.1). The fixed profile conditions give, on the mechanical chart,

$$H_\theta(w, X) := \mu A\Phi_\theta(w) - B \ge \kappa_H := \mu A_{\min}\varkappa_{\rm sl} > 0 \quad (w \ge M_{\rm sl}). \tag{B.21}$$

Here $A_{\min} > 0$ is the common lower bound of A on the chart; monotonicity and (F3) give $\Phi_\theta(w) - B/(\mu A) \ge \varkappa_{\rm sl}$.

Step 2: Compression to the small-slip section. On positive contact the physical equations give exactly, with $F_\theta = 1 - \Phi_\theta$,

$$\begin{aligned} &Y' = V_c, && \sigma Z' = Y - 2Z, \\ &V_c' = (p/\delta)L + (e/\delta)b + (\mu B/\delta)F_\theta(U/\tau)L, && \\ &U' = ea_T - h(\phi)L + \mu A F_\theta(U/\tau)L, \qquad && h(\phi) = \mu A - B. \end{aligned} \tag{B.22}$$

Here $\lambda_N = e^{-1}L$, $dt/ds = e\delta$ and $u_r/\varepsilon = U/\tau$. The global memory row is instead $\sigma Z' = -Z - [Z - Y]_+$; we use the displayed contact row only up to its first force zero. At the initial point $L'(0) = Q - \zeta > 0$.

For mechanical increments set $\chi = (q - q_{\rm in})/(e\delta)$ and $\mathcal{V} = (\dot{q} - \dot{q}_{\rm in})/\delta$. The exact rows are

$$\chi' = \dot{q}_{\rm in} + \delta\mathcal{V}, \qquad \mathcal{V}' = eM^{-1}f + M^{-1}(Q_N - \mu Q_T\Phi_\theta(U/\tau))L. \tag{B.23}$$

On a bounded stopped interval with bounded $\int L$, these give $\chi, \mathcal{V} = O(1)$. Smoothness of the physical chart then gives

$$|b| \le C\delta, \quad |a_T| \le C, \qquad |p - \pi\delta| + |h(\phi) - h_{\rm in}| \le Ce\delta s. \tag{B.24}$$

In particular p/δ is allowed to vary by $O(e)$.

First stop at $U = u_a$, positive-time force zero, a normal or mechanical box boundary, or the reference section bracket endpoint. On this interval $U \ge u_a$, and (F4) gives

$\delta^{-1}F_\theta(U/\tau) \le C\delta^{-1}e^{-\alpha u_a/\tau} \le C\tau^2/\delta = Ce^2 \le Ce$. Subtract (B.5) from (B.22). The error has zero initial data and satisfies

$$E_Y' = E_{V_c}, \quad \sigma E_Z' = E_Y - 2E_Z, \quad E_{V_c}' = \pi E_L + R_V, \quad E_U' = -h_{\mathrm{in}}E_L + R_U,$$
$$R_V = \frac{p-\pi\delta}{\delta}L + \frac{e}{\delta}b + \frac{\mu B}{\delta}F_\theta(U/\tau)L, \qquad R_U = ea_T - (h(\phi) - h_{\mathrm{in}})L + \mu A F_\theta(U/\tau)L. \tag{B.25}$$

Both residuals are $O(e)$. The zero-data filter has mass at most $1/2$, so integration and Gronwall give

$$|E_{V_c}(s)| + |E_U(s)| \le Ces, \qquad |E_Y(s)| + |E_Z(s)| + |E_L(s)| \le Ces^2. \tag{B.26}$$

The quadratic force error and (B.18) exclude an immediate force zero; the later reference force margin excludes the others. The estimates strictly improve the normal boxes, and (B.23) improves the mechanical boxes. The reference section bracket and its negative slip slope force the first descending section $U = u_a$. There is no earlier M_{sl} section, since $u_a/\tau > M_{\mathrm{sl}}$ for small δ.

Let $s_a, \bar{s}_a$ be these actual and reference section times. They satisfy

$$|s_a - \bar{s}_a| \le Ce, \qquad E_a = (Y, Z, V_c, U)(s_a) - (\bar{Y}, \bar{Z}, \bar{V}_c, \bar{U})(\bar{s}_a) = O(e), \qquad E_{a,U} = 0. \tag{B.27}$$

Only the reference is moved in this comparison. Its derivative is bounded: for example $\bar{Z}' = -\zeta e^{-2s/\sigma} + \sigma^{-1}\int_0^s e^{-2(s-t)/\sigma}\bar{V}_c(t)\,dt = O(1)$.

Step 3: The descending friction tail. On the tail compare times $s_a + \rho$ and $\bar{s}_a + \rho$. Put $w = U/\tau$ and $G_s = H_\theta(w, X)L - ea_T$. Stop at the first $w = M_{\mathrm{sl}}$, $L = \ell_a$, $G_s = g_a$, a box boundary, or the reference endpoint $\bar{U} = -u_a/2$. Choose $0 < g_a < \kappa_H\ell_a/4$ and then reduce δ. On every stopped prefix,

$$U' = -G_s, \quad w' = -G_s/\tau, \qquad G_s \ge (\kappa_H/2)L, \quad G_s \ge g_a, \quad L/G_s \le 2/\kappa_H. \tag{B.28}$$

Integrating with respect to U first gives $\int L\,d\rho \le C$. Consequently (B.23) and (B.24) hold with constants independent of the as yet unchosen unweighted V_c box.

For each stopped endpoint ρ_b, write $w_b' = w(\rho_b) \ge M_{\mathrm{sl}}$. The actual decreasing slip gives $d\rho = -\tau\,dw/G_s$ and $\rho_b - \rho(w) \le (\tau/g_a)(w - w_b')$. Hence (F4) gives the two actual tail moments

$$\begin{aligned} I_0 &= \int_0^{\rho_b} F_\theta(w)L\,d\rho \le C\tau\int_{w_b'}^{\infty} e^{-\alpha w}\,dw \le C\tau, \\ I_1 &= \int_0^{\rho_b} (\rho_b - \rho)F_\theta(w)L\,d\rho \le C\tau^2\int_{w_b'}^{\infty}(w - w_b')e^{-\alpha w}\,dw \le C\tau^2. \end{aligned} \tag{B.29}$$

These are changes of variable on the actual orbit.

Let S_σ be the propagator of the homogeneous system in (B.25), ordered as (Y, Z, V_c, U). The filter mass and an integral Gronwall estimate give $\|S_\sigma(\rho)\| \le C$ on the fixed horizon. For the column $v^{(c)} = S_\sigma(\rho)e_{V_c}$ the same argument, now with $v_Y^{(c)}(0) = v_Z^{(c)}(0) = v_U^{(c)}(0) = 0$,

gives $|v_{V_c}^{(c)}| \le C$ and $|v_Y^{(c)}| + |v_Z^{(c)}| + |v_U^{(c)}| \le C\rho$; also $S_\sigma(\rho)e_U = e_U$. Duhamel's formula retains $S_\sigma(\rho_b)E_a$. Applying these bounds to the two residual rows in (B.25) gives

$$\begin{aligned} |E_{V_c}| &\le C(e + \delta^{-1}I_0) \le C(e + e/\sigma), \\ |\sigma E_{V_c}| + |E_Y| + |E_Z| + |E_U| &\le C(e + \sigma\delta^{-1}I_0 + \delta^{-1}I_1 + I_0) \le Ce. \end{aligned} \tag{B.30}$$

Here $\sigma\tau/\delta = e$, $\tau^2/\delta = e^2$, and $\tau \le e$. The first line need not be $O(e)$.

The second line strictly improves the force and normal boxes, and (B.28) improves the descent margin. The preceding impulse bound improves the mechanical boxes. We may now choose the unweighted velocity radius larger than the reference bound plus $C(1 + C_0)$, and reduce δ last. This excludes its boundary by the first line. If no $M_{\rm sl}$ event occurred by the reference endpoint, actual $U \ge \tau M_{\rm sl} > 0$ would contradict $\bar{U} = -u_a/2$ and $E_U = O(e)$. Thus $M_{\rm sl}$ is the first descending crossing and no release has preceded it. The reference negative slip slope gives $|s_{\rm sl} - T_\sigma| \le C(e + \tau)$; moving only that reference gives (B.9). All preceding estimates hold on every prefix, and $\int \lambda_N\, dt = \delta \int L\, ds \le C\delta$.

Step 4: Projection through the first release. By (B.20) and Lemma B.1, $N_0 - A_s n_\#$ and its derivative are $O(\sigma)$ on a common finite interval. Choose $r_- < \rho_\# < r_+ < R_*$ so that, after reducing δ, the inherited reference satisfies

$$\begin{gathered} L_0 \ge m_{\rm pre} > 0 \ (0 \le \rho \le r_-), \quad L_0(r_+) \le -m_{\rm end} < 0, \quad L_0' \le -m_{\rm der} < 0 \ (r_- \le \rho \le r_+), \\ Y_0(R_0) = Z_0(R_0) \le -m_Y, \quad V_0(R_0) \ge m_V, \quad V_0(R_0) + Z_0(R_0) \ge m_{\rm der}. \end{gathered} \tag{B.31}$$

This proves that R_0 is the first root and $R_0 = \rho_\# + O(\sigma)$.

In actual release variables the exact positive-contact equations are

$$\begin{aligned} Y' &= \widehat{V}, & Z' &= Y - 2Z, \\ \widehat{V}' &= eb + (C - \mu B\Phi_\theta(w))L, & w' &= a_T + e^{-1}(B - \mu A\Phi_\theta(w))L, \qquad w = u_r/\varepsilon. \end{aligned} \tag{B.32}$$

At $\rho = 0$, $\widehat{V} = \sigma V_c(s_{\rm sl})$ and $w = M_{\rm sl}$. Stop at first force zero, R_*, or a normal or mechanical boundary. For $w \ge M_{\rm sl}$ the friction term is nonpositive, so $w \le W_* := M_{\rm sl} + R_* \sup|a_T|$. Since $B, a_T > 0$ and $\Phi_\theta(0) = 0$, a common $w_b \in (0, M_{\rm sl})$ satisfies $B - \mu A\Phi_\theta(w) \ge B_{\min}/2$ on $[0, w_b]$. The vector field there points upward, and hence $w_b \le w \le W_*$ throughout the stopped interval.

Set $\vartheta = B/A$ and $P = \widehat{V} - e\vartheta w$. The variable coefficients must be differentiated before estimating. The identity $C - B^2/A = 1/2$ gives exactly

$$\widetilde{N} = (Y, P, Z), \qquad \widetilde{N}' = A_D\widetilde{N} + eF, \qquad F = (\vartheta w, b - \vartheta a_T - \vartheta' w, 0), \quad \vartheta' = \varepsilon\vartheta_\phi\dot{\phi}. \tag{B.33}$$

No derivative of w remains. On the stopped boxes F is bounded. The original mechanical equation also gives, for every such prefix,

$$|q - q_{\rm sl}| \le C\varepsilon, \qquad |\dot{q} - \dot{q}_{\rm sl}| \le C\varepsilon + C\delta^{3/2}\int_0^\rho L\, dt \le C(\varepsilon + \delta^{3/2}), \tag{B.34}$$

which strictly improves the mechanical box. The initial error is $\widetilde{N}(0) - N_0(0) = \{(Y, \widehat{V}, Z)(0) - N_0(0)\} - (0, e\vartheta(0)M_{\rm sl}, 0) = O(e)$. Duhamel's formula in (B.33), followed by $\widehat{V} = P + e\vartheta w$, proves the first line of (B.10) on every stopped prefix. For the force derivative use $L' = Y - 2Z - P - e\vartheta w$, which again contains no w'. This strictly improves the normal box.

The first margin in (B.31) excludes an earlier force zero before r_-. In the stopped bracket $L' \le -m_{\rm der}/2$; if positive contact persisted to r_+, its negative reference endpoint would give a contradiction. The actual first zero is therefore in (r_-, r_+), and $|L_0(R_\varepsilon)| \le Ce$ gives $|R_\varepsilon - R_0| \le Ce$. Moving only the reference proves the remaining event comparison. At this zero $Y = Z$ and $-L' = \widehat{V} + Z$, so $\dot g = v + z/\varepsilon = \delta^{3/2}(\widehat{V} + Z) > 0$. This proves (B.11), including the negative memory term.

Step 5: Physical reconstruction and leading increments. The physical reconstruction in (B.12) uses zero physical displacement and memory, while (B.6) retains their complete scaled values. We have $u_{r,L} = \varepsilon w = O(e\delta^{3/2})$ and $y_L = z_L = O(\varepsilon\delta^{3/2}) = O(e\delta^3)$. The p drift through release is $O(\varepsilon)$, and the common inverse chart gives $|\Delta b_L| \le C(|\Delta p_L| + |\Delta u_0| + |\Delta y_L| + |\Delta v_L|)$, with $\Delta u_0 = \Delta u_r$. Thus (B.13) follows, with b_L evaluated at the outgoing state in both passages. The independent section and release clocks give (B.14).

The compression increments, its bounded horizon and impulse, and (B.34) prove every estimate in (B.15); the release impulse is $O(\delta^{3/2})$. The velocity bound uses $|v| \le C\delta^2$ in compression and the release scale thereafter. The full state includes the continuous memory coordinate at the joining point. Finally $\dot p = p_\phi \omega = -a + O(\delta)$ throughout this same prefix.

Integrating the compression slip row, using $U(s_{\rm sl}) = \tau M_{\rm sl}$, (B.29) and (B.24), gives

$$\int_0^{s_{\rm sl}} L\, ds = S/h_{\rm in} + O(e), \qquad \int_{t_{\rm in}}^{t_{\rm sl}} \lambda_N\, dt = \delta S/h + O\{\delta(e+\delta)\}. \tag{B.35}$$

The discrepancy from saturated impulse is $\delta I_0 = O(\delta\tau)$. The saturated direction at G is $(-5/4, 1)^T$ in generalized velocity, and $Q_{\rm sat} b(G) = a\beta$. The free-force impulse is $O(e\delta)$ and the release impulse is $O(\delta^{3/2})$. These facts prove the velocity and b rows of (B.16). Integrating the configuration and p rows, with $s_{\rm sl} = \xi_s + O(\sigma)$ and release duration $O(\varepsilon)$, proves their rows. The normal-state formulas in (B.16) follow from (B.20) and (B.10). All constants, boxes and event brackets are chosen after the fixed classes and C_0, and the common scale cutoff is reduced last. □

B.3 Parameter dependence of the contact reference

To differentiate the limiting return, we let the initial displacement and memory of the linear contact reference vary independently.

Lemma B.3 (Parameter dependence of the limiting contact)**.** *In* (B.5)*, replace* $h_{\rm in}$ *by an independent* h_0 *near* h *and the initial data by* $(y_0, z_0, -Q, S)$*. On a neighborhood of the compact data* $(0, \pi, h, 0, 0, Q, S)$*, the first arrest, its inherited normal state, and the first inherited release have* C^2 *extensions in* $(\sigma, \pi, h_0, y_0, z_0, Q, S)$ *to* $\sigma = 0$*, with uniform bounds*

through order two and continuous dependence on λ. If $\mathcal{Q} = V_0(R_0)$ denotes the release velocity, then

$$\mathcal{Q} = \nu_{\#}\sqrt{QS/h_0 - \pi S^2/(2h_0^2)} + O(\sigma + |y_0| + |z_0|). \tag{B.36}$$

The same error bound holds after one derivative in (π, h_0, Q, S); derivatives in (σ, y_0, z_0) are uniformly bounded. Compression on this auxiliary neighborhood is the signed linear continuation: its force need not be positive near time zero. Its arrest is the first zero of U, and the subsequent inherited release is the first zero of its normal force. No physical closing condition is imposed on the auxiliary initial data.

Proof. Set $R = \bar{Z} - \bar{Y}/2$. The normal compression rows become

$$(Y, V, R)' = (V, -\pi Y/2 + \pi R, -V/2 - 2R/\sigma).$$

A slow invariant plane $R = mY + nV$ is determined by

$$2m + \sigma\pi n(m - 1/2) = 0, \qquad 2n + \sigma(m + \pi n^2 + 1/2) = 0.$$

The implicit function theorem gives analytic m, n, uniformly for π in a neighborhood of its compact range, with $m = O(\sigma^2)$ and $n = -\sigma/4 + O(\sigma^3)$. On this plane the matrix is $B_s = \left(\begin{smallmatrix} 0 & 1 \\ -\pi/2 + \pi m & \pi n \end{smallmatrix}\right)$. The remaining root is ν/σ, where $\nu = -2 + \sigma^2 g$ and $\nu^2 g + \pi(\nu + 1) = 0$. Its eigenvector, normalized in the R row, is $(-2\sigma^2 g/\nu, -2\sigma g, 1)^T$. Together with $(1, 0, m)^T, (0, 1, n)^T$ these columns form an analytic invertible matrix at $\sigma = 0$. The exact solution thus consists of the analytic slow block $e^{B_s s}$ and a fast term proportional to $e^{\nu s/\sigma}$, with $\nu < -1$. On every late interval $s \geq s_* > 0$, that fast term and its derivatives of any fixed order extend flatly to zero. The slip row is integrated without dividing by π: $U' = h_0 Y/2 - h_0 R$. Its fast contribution contains $(\sigma/\nu)(e^{\nu s/\sigma} - 1)$, which has the same smooth late extension. No diagonalization of the slow block is required at $\pi = 0$.

At zero scale and $y_0 = z_0 = 0$, the arrest has a common positive time and negative slip slope, by (B.7). Its slip stays positive before a fixed arrest bracket. Uniform convergence of the integrated solution on the whole preceding interval preserves this first-event property for the small independent perturbations. The implicit function theorem on the late bracket gives a C^2 arrest and inherited state. Lemma B.1 and the linear formula $e^{A_D \rho} N_0(0)$ give the same conclusion for its first release, with the positive-force segment and transverse bracket preserved. At $\sigma = y_0 = z_0 = 0$ its velocity is exactly the first term of (B.36). The bounded second derivatives and the mean value formula give both the value and the stated first-derivative bounds. □

For subsequent prepared touchdowns set

$$\pi = -a, \quad Q = r_0 W, \quad S = d_\lambda U^r, \quad r_0 = c_\lambda/3, \qquad W = 1 + \delta^{1/3}\eta_W, \quad U^r = 1 + \delta^{1/3}\eta_U. \tag{B.37}$$

The prepared cone retains its complete dependent acceleration coordinate, $|\zeta| \leq Z_*$ and $r_0 W - \zeta \geq \gamma_0$; its inverse is taken at $u_0 = d_\lambda \delta U^r + r$ with $0 \leq r \leq A_{\mathrm{off}} \delta^{4/3}$. These

hypotheses give (B.2). The reference is exactly (B.5) with its finite h_{in} and complete initial memory, so (B.13) applies to this same input and offset. For the tuned limiting reference define

$$q_\lambda = \nu_\# \sqrt{r_0 d_\lambda / h + a d_\lambda^2/(2h^2)} > 0. \tag{B.38}$$

The analytic inverse of $p(\phi_\delta) = -a\delta$ gives $h_\delta = h + O(\delta)$ and $\delta h'_\delta = O(\delta)$. Lemma B.3, restricted to $\sigma = \delta^{1/2}$, $h_0 = h_\delta$, $y_0 = z_0 = \sigma\zeta$ and (B.37), gives

$$\mathcal{Q} = q_\lambda + O_{C^1_*}(\delta^{1/3}), \qquad |D_\xi \mathcal{Q}| \le C\delta^{1/3}, \qquad \xi = (\zeta, \eta_U, \eta_W). \tag{B.39}$$

Here C^1_* uses $\delta\partial_\delta$ and D_ξ on any fixed bounded auxiliary fibre set. The chain includes both initial memory coordinates: its radial part is $(\sigma/2)\partial_\sigma + \delta h'_\delta \partial_{h_0} + (y_0/2)\partial_{y_0} + (z_0/2)\partial_{z_0}$, together with $\delta\partial_\delta Q$ and $\delta\partial_\delta S$ from (B.37). Thus the reference is defined on the open neighbourhood needed for fibre differentiation, even where physical closing is not imposed. The free-flight construction below will establish the complete weighted return factorization.

The first physical touchdown uses the wider offset range $\delta = \sqrt{\varepsilon}$, $r = k\delta$. Its complete input and the strict signed arrest margin are verified in Proposition C.2; later prepared inputs are verified in Section D.

C The two preparatory flights

The first release has normal velocity of order ε, and its flight ends on the scale $\sqrt{\varepsilon}$. The ensuing contact amplifies the velocity to order $\varepsilon^{3/4}$, so the next flight is longer, of order $\varepsilon^{3/8}$. Both flights inherit a nonzero normal memory. We retain it in the initial opening layer and identify each actual first touchdown before normalizing its complete incoming state.

Throughout $r = k\sqrt{\varepsilon} = -\eta$ is the same physical belt offset and $u_r = u_0 - r$ is the physical slip. The belt coefficient k, the remaining parameters and the initial preparation stay fixed.

The scales established below distinguish the first signed contact from the recurring returns:

touchdown	incoming scale	offset ratio
TD_1	$\sqrt{\varepsilon}$	$r/\sqrt{\varepsilon} = k$
TD_2	$\delta_2 \asymp \varepsilon^{3/8}$	$r/\delta_2^{4/3} \asymp 1$
TD_3	$\delta_3 \asymp \varepsilon^{9/32}$	$r/\delta_3^{4/3} = O(\varepsilon^{1/8})$

The second touchdown has a bounded complete cone. One further actual return makes the offset ratio small and places the trajectory in the domain used for iteration.

In a flight,

$$\dot y = v, \quad \dot v = b(X), \quad \dot z = -z/\varepsilon, \quad \dot u_r = a_T(X), \qquad g = y - z. \tag{C.1}$$

Lemma C.1 (The gap of a local free extension)**.** *Suppose* $|X_L - G| \le C\rho$, $\rho \to 0$, *and* $y_L = z_L$. *On a common fixed local interval, the smooth free extension satisfies*

$$\begin{aligned} p(t) &= p_L - at + O(\rho t + t^2), \qquad b(t) = b_L - ct + O(\rho t + t^2), \\ u_r(t) &= u_{r,L} + d_\lambda t + O(\rho t + t^2), \\ v(t) &= v_L + b_L t - ct^2/2 + O(\rho t^2 + t^3), \qquad \text{(C.2)} \\ g(t) &= z_L(1 - e^{-t/\varepsilon}) + v_L t + b_L t^2/2 - ct^3/6 + \mathcal{R}(t), \\ |\mathcal{R}(t)| &\le C(\rho t^3 + t^4), \qquad |\mathcal{R}'(t)| \le C(\rho t^2 + t^3). \qquad \text{(C.3)} \end{aligned}$$

If $\min\{v_L, v_L + z_L/\varepsilon\} \ge v_* > 0$, *then*

$$\dot g(t) \ge v_* - C\rho t - Ct^2. \qquad \text{(C.4)}$$

The extension is the actual unilateral solution on every initial interval on which $g > 0$. *At any positive root* T *of this free gap within the local interval, put* $D = -p(T)/a$ *and* $H_b = \dot b + c$. *Then*

$$-v(T) - \frac{c}{3}D^2 = \frac{y_L - z(T)}{T} - \frac{1}{2}b_L T + \frac{c}{3}(T^2 - D^2) - \frac{1}{2T}\int_0^T (T^2 - t^2)H_b(t)\,dt. \qquad \text{(C.5)}$$

Proof. Stop at a common chart boundary. The bounded free field gives $|X(t) - G| \le C(\rho + t)$, which improves the stop on a smaller fixed time interval. Its critical rates give $\dot p = -a + O(\rho + t)$, $\dot b = -c + O(\rho + t)$ and $a_T = d_\lambda + O(\rho + t)$. Successive integrations give all the mechanical estimates and the gap remainder. The memory is exactly $z_L e^{-t/\varepsilon}$. Moreover $v_L + (z_L/\varepsilon)e^{-t/\varepsilon} \ge \min(v_L, v_L + z_L/\varepsilon)$, which proves (C.4). While $g > 0$ both contact forces vanish; uniqueness identifies the extension with the actual free leg. For the last identity integrate $\dot b = -c + H_b$ twice:

$$\begin{aligned} v(T) &= v_L + b_L T - cT^2/2 + \int_0^T (T - t)H_b(t)\,dt, \\ z(T) &= y_L + v_L T + b_L T^2/2 - cT^3/6 + \tfrac{1}{2}\int_0^T (T - t)^2 H_b(t)\,dt. \end{aligned}$$

Eliminate v_L with the second equality, which is the actual root condition $y(T) = z(T)$. No asymptotic estimate is used in this identity. □

C.1 The first touchdown and signed compression

Use the critical-release data of Theorem A.4 and put $s = \sqrt{\varepsilon}$. Define

$$H = ak/\gamma, \qquad q = q_{\mathrm{in}}, \qquad T_f = \frac{\sqrt{9\ell^2 H^2 + 24cq} - 3\ell H}{2c}, \qquad \pi = H - aT_f. \qquad \text{(C.6)}$$

Thus $cT_f^2 + 3\ell H T_f = 6q$. These quantities have common finite bounds and T_f, q, H have common positive lower bounds on the fixed compact label classes.

Proposition C.2 (First flight and its amplified release)**.** *The first free leg after the captured passage has a transverse first touchdown, at elapsed time t_1, with*

$$t_1/s \to T_f, \qquad p_1/s \to \pi, \qquad b_1/s \to -\ell H - cT_f, \qquad u_{r,1}/s \to S := d_\lambda T_f, \tag{C.7}$$

$$-v_1/\varepsilon \to Q := T_f(\ell H/2 + cT_f/3) > 0, \qquad y_1 = z_1 = o(\varepsilon^2), \qquad |X_1 - G| = O(s). \tag{C.8}$$

The subsequent contact has its first release with

$$|X_L - G| = O(s), \qquad p_L, b_L = O(s), \qquad u_{r,L} = O(\varepsilon), \tag{C.9}$$

$$v_L = \varepsilon^{3/4}(R + o(1)), \qquad y_L = z_L = \varepsilon^{7/4}(Z + o(1)) < 0, \tag{C.10}$$

$$R = A_s\nu_\#, \quad Z = -A_s\gamma_\#, \qquad A_s^2 = QS/h - \pi S^2/(2h^2) > 0. \tag{C.11}$$

In particular R and $R+Z$ have common positive lower bounds. The first touchdown parameter π may be positive, zero or negative.

Proof. The release theorem gives $p_L/s \to H$, $b_L/s \to -\ell H$, $v_L/\varepsilon \to q$ and a common positive opening margin $(v_L + z_L/\varepsilon)/\varepsilon$. Apply the free-gap lemma with $\rho = s$. For $t = s\tau$, uniformly on bounded intervals,

$$s^{-3}g(s\tau) \longrightarrow P_f(\tau) := q\tau - \ell H\tau^2/2 - c\tau^3/6.$$

The normalized memory value is $O(s)$; away from $\tau = 0$ its normalized derivative tends to zero exponentially. The opening estimate gives $\dot g > 0$ on $0 \le t \le s_0 s$ for a common small $s_0 > 0$. The polynomial is positive until its unique positive zero T_f, where $P_f'(T_f) = -T_f(\ell H/2 + cT_f/3) < 0$. Choose a fixed small bracket with a common negative derivative margin. Uniform value convergence preserves positivity between the opening interval and the bracket and the opposite endpoint signs; uniform derivative convergence in the bracket gives exactly one zero. It is the actual first transverse touchdown. The mechanical expansions give (C.8); exact memory decay gives, more precisely, $|z_1| \le C\varepsilon^2 e^{-c/\sqrt{\varepsilon}}$ and makes $\dot g(t_1)$ have the same negative leading term as v_1.

For the ensuing contact set $\delta = s$. Its exact normalized incoming data are $\pi_\varepsilon = p_1/\delta$, $Q_\varepsilon = -v_1/\delta^2$, $S_\varepsilon = u_{r,1}/\delta$ and $\zeta_\varepsilon = z_1/(\varepsilon\delta^2)$. They converge to $(\pi, Q, S, 0)$, and

$$2hQ - \pi S = T_f\{(h\ell - d_\lambda)H + (2hc/3 + ad_\lambda)T_f\} > 0,$$

with a common strict margin since $h\ell - d_\lambda = \gamma > 0$. Thus they lie in a fixed compact subset of the signed contact domain of Theorem B.2; the incoming gap-closing margin also holds. Here $\varepsilon/\delta^2 = 1$ and $r = k\delta$. That theorem, applied to this actual state, proves the first release, the complete normal memory and (C.10). Its normal reference has $\nu_\# > \gamma_\# > 0$, giving the opening margin $R + Z > 0$. The strict amplitude inequality and compactness bound A_s away from zero. No new state has been prepared. □

At this first touchdown $r = k\delta$, so the signed contact theorem supplies the amplified release. The next flight brings the offset into the range $r \le A\delta^{4/3}$ required for recurring returns.

C.2 The second touchdown in complete coordinates

The normal displacement at the amplified release is too large to use the first-flight normalization. Its exact decay still ensures an initial free interval; on the longer flight the cubic acceleration term dominates the initial $O(\sqrt{\varepsilon})$ acceleration. At touchdown, the smaller fibre scale requires cancellation of the actual release velocity before division.

Lemma C.3 (Complete preparation at the second touchdown)**.** *Suppose an actual release has $|X_L - G| = O(\sqrt{\varepsilon})$, $u_{r,L} = O(\varepsilon)$ and satisfies* (C.10)*, with bounded R, Z and common positive lower bounds for $R, R+Z$. Its next touchdown is the transverse first positive zero of the free gap. If T is its elapsed time, $D = -p(T)/a$ and $C_R = \sqrt{6R/c}$, then*

$$T = \varepsilon^{3/8} C_R(1 + o(1)), \qquad D \sim T, \qquad D - T = O(\sqrt{\varepsilon}), \qquad v(T)/\varepsilon^{3/4} \to -2R, \tag{C.12}$$

$$\varepsilon/D^2 = O(\varepsilon^{1/4}) \to 0. \tag{C.13}$$

Put $r_0 = c/3$ and define the complete actual incoming coordinates

$$U_2^r = \frac{u_r(T)}{d_\lambda D}, \quad B_2 = \frac{-b(T)}{cD}, \quad W_2 = \frac{-v(T)}{r_0 D^2}, \quad \zeta_2 = \frac{z(T)}{\varepsilon D^2}, \tag{C.14}$$

$$\eta_{U,2}^r = \frac{U_2^r - 1}{D^{1/3}}, \quad \eta_{B,2} = \frac{B_2 - 1}{D^{1/3}}, \quad \eta_{W,2} = \frac{W_2 - 1}{D^{1/3}}, \qquad \xi_2^r = (\zeta_2, \eta_{U,2}^r, \eta_{W,2}). \tag{C.15}$$

There exist fixed constants $M_2 < \infty$ and $\gamma_2 > 0$ such that

$$|\xi_2^r| + |\eta_{B,2}| \le M_2, \qquad r_0 W_2 - \zeta_2 \ge \gamma_2, \qquad (B_2, U_2^r, W_2) \to (1,1,1), \tag{C.16}$$

$$-Ce^{-c_1 \varepsilon^{-5/8}} \le \zeta_2 < 0, \qquad \frac{r}{D^{4/3}} = \frac{k}{C_R^{4/3}}(1 + o(1)). \tag{C.17}$$

The offset ratio has positive finite common upper and lower bounds.

Proof. Put $P = \sqrt{\varepsilon}$, $L = \varepsilon^{3/8}$, so the release velocity scale is L^2 and its memory scale is εL^2. For $t = L\tau$, the free-gap lemma gives

$$L^{-3} g(L\tau) \longrightarrow R\tau - c\tau^3/6.$$

The initial quadratic term is $O(P/L)$, memory value is $O(\varepsilon/L)$ and the remainder is $O(P + L)$, all tending to zero. Away from zero the normalized memory derivative decays exponentially. On $0 \le t \le s_0 L$, (C.4) gives $\dot{g} \ge L^2\{m - Cs_0 P/L - Cs_0^2\} > 0$ after choosing s_0 small and then ε small. The cubic has its unique positive zero C_R with derivative $-2R$. Its positive interval, opposite root bracket signs and negative derivative margin transfer to the extension. Thus the bracket contains precisely its first zero and is an actual touchdown, with $T/L \to C_R$ and $v(T)/L^2 \to -2R$.

All finer estimates are on free prefixes. Stop at a chart boundary or HL, where $H > \sup C_R + 1$ is fixed. The bounded free field gives $|X(t) - G| \le C(P + t)$, improving the chart stop; hence

$$A_p := \dot{p} + a = O(P + t), \quad H_b := \dot{b} + c = O(P + t), \quad \dot{u}_r - d_\lambda = O(P + t), \qquad 0 \le t \le HL. \tag{C.18}$$

In particular $D - T = -p_L/a - a^{-1}\int_0^T A_p\,dt = O(P)$, since $T \asymp L$ and $L^2 = o(P)$. This proves $D \sim T > 0$. Retain the critical combinations

$$J_U = u_r + (d_\lambda/a)p, \qquad J_B = -b + (c/a)p, \qquad J_W = -v - (c/3)(p/a)^2. \tag{C.19}$$

The free-field derivatives of J_U and J_B vanish at G, so integration with (C.18) gives $J_U(T), J_B(T) = O(P + PT + T^2) = O(P)$.

For $J_W = -v(T) - (c/3)D^2$, use (C.5), which eliminates the release velocity through the actual touchdown identity $y(T) = z(T)$. The four terms have sizes $O(\varepsilon^{11/8})$, $O(\varepsilon^{7/8})$, $O(\varepsilon^{7/8})$ and $O(\varepsilon^{9/8})$. For the last use the bound on every prefix: $C(PT^2 + T^3) = O(\varepsilon^{9/8})$. Therefore $J_W(T) = O(PL)$. No rate for the $o(1)$ in the release velocity is required. Only now divide, using the exact identities

$$\eta^r_{U,2} = \frac{J_U(T)}{d_\lambda D^{4/3}}, \quad \eta_{B,2} = \frac{J_B(T)}{cD^{4/3}}, \quad \eta_{W,2} = \frac{J_W(T)}{r_0 D^{7/3}}.$$

Their denominators are respectively of sizes P, P, PL. The three fibres are bounded and the three unscaled ratios tend to one. Finally $z(T) = z_L e^{-T/\varepsilon} < 0$ and $|z_L|/(\varepsilon D^2) \le C$; since $T/\varepsilon \ge c_1\varepsilon^{-5/8}$, this proves the memory bound and the strict closing margin. The last offset identity follows from $D \sim C_R\varepsilon^{3/8}$ and $r = k\sqrt{\varepsilon}$. □

Corollary C.4 (The physical second touchdown)**.** *The original physical trajectory has this complete second-touchdown state, with $\delta_2 = D \asymp \varepsilon^{3/8}$ and $p_2 < 0$. All bounds are uniform on its fixed compact label class, including labels for which the leading first-touchdown parameter π vanishes.*

Proof. Proposition C.2 supplies every release hypothesis of Lemma C.3, including $u_{r,L} = O(\varepsilon)$, negative memory, and the common $R, R + Z$ margins. The lemma applies to that same trajectory. □

Corollary C.5 (Entry to the recurring returns)**.** *Write $\eta_{U,*,\lambda}$ for the slip component of $\xi_{*,\lambda}$ and put $\eta_{B,*,\lambda} = (g_b/c)\eta_{U,*,\lambda}$, with $g_b = -d_\lambda b_u(G)$ as in Lemma D.1. The next actual contact–flight return reaches* TD_3 *with*

$$\delta_3 \asymp \varepsilon^{9/32}, \qquad \xi^r_3 \in \operatorname{int}\mathcal{K}, \qquad r/\delta_3^{4/3} = O(\varepsilon^{1/8}), \tag{C.20}$$

$$\left|\log\frac{\delta_3}{K_\lambda\delta_2^{3/4}}\right| + |\xi^r_3 - \xi_{*,\lambda}| + |\eta_{B,3} - \eta_{B,*,\lambda}| \le C(\varepsilon^{3/32} + \varepsilon^{7/16} + \varepsilon^{1/8}). \tag{C.21}$$

It retains the complete negative normal memory and every strict physical margin of Theorem D.5, and is an admissible input to a further return. In particular $\log(1/\delta_3) = (9/32)\log(1/\varepsilon) + O(1)$ *uniformly.*

Proof. Choose a fixed finite cone containing the full second-touchdown bounds, and fixed A covering $r/\delta_2^{4/3}$. Since $\varepsilon/\delta_2^2 = O(\varepsilon^{1/4}) \to 0$ and $\delta_2 \to 0$, Theorem D.5 applies to this actual input for all sufficiently small ε. Its three error terms are $\delta_2^{1/4} = O(\varepsilon^{3/32})$, $\varepsilon\delta_2^{-3/2} = O(\varepsilon^{7/16})$,

and $r/\delta_2 = O(\varepsilon^{1/8})$. It gives $\delta_3 \asymp \delta_2^{3/4}$ before output admission. Lemma D.1 recovers the dependent acceleration from the actual output. Its additional terms are $O(r/\delta_3^{4/3}) = O(\varepsilon^{1/8})$ and $O(\delta_3^{2/3}) = O(\varepsilon^{3/16})$, within the displayed bound. The strict output cone and memory are supplied by that theorem. Since $\delta_3 \to 0$, the output eventually also satisfies its input cutoff, permitting the next actual return. The uniform positive upper and lower scale bounds prove the logarithmic statement, with the original labels and belt parameter on the same physical trajectory. □

D The return in normalized touchdown coordinates

The contact theorem gives a release with normal velocity of order $\delta^{3/2}$. The subsequent free gap is the integral of a normal acceleration whose leading rate is $-c$, so its next zero has time and scale of order $\delta^{3/4}$. To control the finer output fibres, we cancel the leading free-flight terms before normalizing. We then compare the compliant and reference returns at identical values of the independent coordinates, and establish weighted state derivatives for the exact reference return. These two estimates will control repeated returns.

D.1 Independent touchdown data and the physical inverse

Use the independent tuple (4.1), physical inverse (4.2) and input bounds (4.3), writing $c = c_\lambda$ and $\gamma = \gamma_{\mathrm{cl}}$. The body coordinates are recovered by

$$\psi = y - 1 + \cos\phi_\delta, \qquad \dot{\psi} = v - \sin\phi_\delta\,\omega.$$

For the inverse and comparison estimates below, the closing bound may be any fixed $\gamma > 0$. Constants depend only on the fixed input bounds and compact model classes, with no positive lower bound on e or r.

Lemma D.1 (The dependent acceleration)**.** *On* (4.3)*, the physical inverse is* $G + O(\delta)$ *and*

$$\begin{aligned} b &= -c\delta + d_\lambda b_u(G)\delta^{4/3}\eta_U^r + b_u(G)r + O(\delta^2), \\ \eta_B &= \frac{g_b}{c}\left(\eta_U^r + \frac{r}{d_\lambda\delta^{4/3}}\right) + O(\delta^{2/3}), \qquad g_b = -d_\lambda b_u(G). \end{aligned} \tag{D.1}$$

In particular $|\eta_B| \le M_B$ *for a common finite* M_B *depending on* M_{in}, A *and the compact model classes, after reducing the common scale cutoff. This bound is fixed before applying the contact theorem.*

Proof. In the analytic chart (p, y, u_0, v) the free rates at G are $(-a, 0, d_\lambda, 0)$. Therefore $F_{\mathrm{free}} b(G) = -c$ gives

$$-ab_p(G) + d_\lambda b_u(G) = -c. \tag{D.2}$$

Substitute (4.1) with $u_0 = u_r + r$ into the Taylor expansion of b. Here $p, u_0 = O(\delta)$, $v = O(\delta^2)$ and $y = O(\delta^4)$; the remaining linear terms and quadratic remainder are $O(\delta^2)$,

uniformly on the fixed domain. Equation (D.2) yields the first formula, and division by $-c\delta^{4/3}$ yields the second. The explicit inverse gives the asserted bound on the mechanical state distance from G, together with common bounds on all derivatives used. □

The exact reference $T_{0,\lambda} = (f_{0,\lambda}, h_{0,\lambda})$ is defined by (4.5)–(4.6) and the subsequent release reconstruction and free flight in Section 4. Write $Q = V(R_0)$ for its normal release amplitude. The actual return constructed below is $T_{\varepsilon,r} = (f_{\varepsilon,r}, h^r_{\varepsilon,r})$.

D.2 The first free event and normalization

The normal fibre has denominator $D^{7/3} \asymp \delta^{7/4}$, whereas its uncancelled velocity terms have size $\delta^{3/2}$. We eliminate the release velocity with the touchdown equation before dividing. For a free leg on its own elapsed release clock define

$$\begin{aligned} D &= -p(T)/a, \qquad J_U = u_0(T) + (d_\lambda/a)p(T), \\ J_W &= -v(T) - (c/3)D^2, \qquad A_p = \dot p + a, \quad B_b = \dot b + c. \end{aligned} \tag{D.3}$$

Integrating the free equations and using $y(T) = z(T)$ gives the exact identities

$$D - T = -p_L/a - a^{-1}\int_0^T A_p(t)\,\mathrm{d}t, \tag{D.4}$$

$$J_U = u_{0,L} + (d_\lambda/a)p_L + \int_0^T \{a_T + (d_\lambda/a)\dot p\}\,\mathrm{d}t, \tag{D.5}$$

For the normal numerator use the exact identity (C.5), with $H_b = B_b$ and $J_W = -v(T) - (c/3)D^2$. Thus the actual release velocity is eliminated before either normalized output is formed. The tuned output fibres are

$$\eta^+_U = \frac{J_U}{d_\lambda D^{4/3}}, \qquad \eta^+_W = \frac{J_W}{r_0 D^{7/3}}, \qquad \zeta^+ = \frac{z_L e^{-T/\varepsilon}}{\varepsilon D^2}. \tag{D.6}$$

For zero reference memory the last expression means $\zeta^+ = 0$.

Lemma D.2 (Paired free flights)**.** *Consider either the actual and reference releases of Theorem B.2 specialized by* (B.37) *at the same native input and offset, or two exact releases with the same $p_L = -a\delta$, $y_L = z_L = 0$, $v_L = \delta^{3/2}Q$ and slips $u_{0,L} = r, 0$, where Q lies in a fixed positive compact interval. Put $L = \delta^{3/4}$ and set $\vartheta = e$ in the first case and $\vartheta = r/\delta$ in the second. Both legs have first subsequent touchdowns, positive gap between events, and*

$$\begin{aligned} T_i \asymp D_i \asymp L, &\qquad g_i'(T_i), v_i(T_i) \le -\kappa_f\delta^{3/2}, \\ |\Delta T| + |\Delta D| \le C\vartheta L, &\qquad |\log(D_1/D_0)| + |\xi_1^{0,+} - \xi_0^{0,+}| \le C\vartheta. \end{aligned} \tag{D.7}$$

On each full free prefix, $|X_i(\ell) - G| \le C(\delta + \ell)$ and $T_i \le HL$. In the actual leg

$$z(\ell) = z_L e^{-\ell/\varepsilon} < 0, \qquad |z(\ell)|/\varepsilon \le C\delta^{3/2}, \qquad |\zeta^+| \le C\exp\{-\kappa_f/(eL)\} \le CeL. \tag{D.8}$$

Here $\kappa_f > 0$ is a common lower-bound constant. All output coordinates in (D.7) *use u_0.*

Proof. The native contact estimate and the smooth inverse, respectively, give

$$\begin{aligned} |X_{L,i} - G| \le C\delta, \quad b_{L,i} = O(\delta), \quad |\Delta p_L| + |\Delta u_0| + |\Delta b_L| \le C\vartheta\delta, \\ |\Delta v_L| \le C\vartheta\delta^{3/2}, \quad y_{L,0} = z_{L,0} = 0, \quad y_{L,1} = z_{L,1}, \quad |z_{L,1}| \le C\vartheta\delta^3. \end{aligned} \tag{D.9}$$

The actual release also has $|z_L| \le C\varepsilon\delta^{3/2}$ and $v_L + z_L/\varepsilon \ge \kappa_{\mathrm{op}}\delta^{3/2}$. In the second pair, $\Delta u_0 = r$ and $\Delta b_L = O(r)$. Indeed, at fixed angle the body velocity differences are exactly

$$\Delta\omega = r/\cos\phi_\delta, \qquad \Delta\dot\psi = -r\tan\phi_\delta, \quad \Delta b_L = b_u(G)r + O(r\delta + r^2). \tag{D.10}$$

The two references have the same normal release coordinates, but different mechanical velocities.

Stop the analytic free extensions at HL or the chart boundary. Their common smooth field gives $|X_i(t) - G| \le C(\delta + t)$, excluding that boundary, and $|\Delta X(t)| \le C\vartheta\delta$ by Gronwall's inequality. Each residual $P \in \{A_p, B_b, a_T + (d_\lambda/a)\dot p\}$ vanishes at G, so $|P_i(t)| \le C(\delta + t)$ and $|\Delta P(t)| \le C\vartheta\delta$. The exact gap is

$$g_i(t) = v_{L,i}t + z_{L,i}(1 - e^{-t/\varepsilon}) + \int_0^t (t - s)b_i(s)\,\mathrm{d}s. \tag{D.11}$$

The memory term is absent when $z_L = 0$. Since $z_L < 0$, its contribution is at least $(z_L/\varepsilon)t$. The opening margin therefore gives $g_i(t) \ge t(\kappa_f\delta^{3/2} - C\delta t - Ct^2) > 0$ for $0 < t \le \eta L$, with fixed small $\eta > 0$. On $\eta \le x \le H$,

$$\delta^{-9/4} g_i(Lx) = Qx - cx^3/6 + O_{C^1_x}(\delta^{1/4} + \vartheta).$$

Here the differentiated memory is bounded using $e^{-\eta/(eL)} \le Ce$. The positive interior portion and a common transverse bracket about $\sqrt{6Q/c}$ identify the first zero. The release and field differences give

$$|\Delta g(t)| \le C\vartheta(\delta^3 + \delta^{3/2}t + \delta t^2) \le C\vartheta\delta^{9/4}.$$

Dividing at the transverse bracket and then moving the event in p proves the time, scale and logarithmic bounds. It also gives $u_0(T_i) = d_\lambda T_i + O(\delta + T_i^2) > 0$; subtracting $r = O(\delta^{4/3})$ preserves positive native slip.

For the normalized rows, substitute $t = T_i q$ in each residual integral. The residuals there have size $O(L)$ and paired difference $O(\vartheta L)$, including the changed argument. Equation (D.4) gives $H_i = D_i - T_i = O(\delta)$ and $\Delta H = O(\vartheta\delta)$. Thus $J_{U,i} = O(\delta)$ and $\Delta J_U = O(\vartheta\delta)$. In (C.5) keep

$$T_i^2 - D_i^2 = -H_i(2T_i + H_i)$$

before dividing. The $b_L T$ and squared-scale terms have size $O(\delta L)$ and difference $O(\vartheta\delta L)$. The last integral has paired difference $O(\vartheta L^3)$, and the memory term has size $O(\vartheta\delta^3/L)$. Since $L^2 \le \delta$, these give $J_{W,i} = O(\delta L)$ and $\Delta J_W = O(\vartheta\delta L)$. For either exponent $\alpha = 4/3, 7/3$ use

$$\frac{J_1}{D_1^\alpha} - \frac{J_0}{D_0^\alpha} = \frac{\Delta J}{D_1^\alpha} + \frac{J_0}{D_0^\alpha}\{(D_1/D_0)^{-\alpha} - 1\}. \tag{D.12}$$

Because $D_i^{4/3} \asymp \delta$ and $D_i^{7/3} \asymp \delta L$, this proves the fibre bound with their respective denominators. The exact memory formula and $D_i^2 \asymp \delta^{3/2}$ prove (D.8). □

D.3 Weighted regularity of the exact limiting return

Use the constants (4.7) and the weighted norm of Section 4. Put $B_\lambda = -ab_p(G) = -c + g_b$ and $\mathscr{D}_\delta = \delta\partial_\delta$. The reference derivatives are needed on a bounded open fibre domain, including inputs outside the physical closing condition.

Proposition D.3 (Weighted limiting factorization)**.** *Let $\Omega \Subset \Omega_1$ be fixed bounded convex open sets containing the independent input ball $|\xi| \le M_{\mathrm{in}}$ and the ball $\mathcal{K}$ specified in Proposition 4.2. On a common $(0, \delta_w] \times \overline{\Omega}$, the reference has the C^1 extension, weighted bounds* (4.13) *and joint λ-continuity stated there. The memory component of h_0 vanishes. Moreover, $f_0 > \delta$ and $\mathscr{D}_\delta \log f_0 = 3/4 + O(\delta^{1/4})$.*

On the auxiliary domain this is the algebraic extension of the signed linear contact reference: compression is continued even if its force is initially negative. No physical contact assertion is made there. The actual return is used only under the closing condition in (4.3).

Proof. Lemma B.3 treats the filter scale, frozen contact coefficients, initial normal displacement and filter datum, slip and incoming normal velocity as independent parameters. Restrict it to the native coefficient $\pi = -a$ and write the remaining parameters as $(\sigma, \widehat{\delta}, \alpha_U, \alpha_W, \beta_m)$, with $h_0 = h_{\widehat{\delta}}$, initial displacement and filter datum both β_m, initial slip $d_\lambda(1+\alpha_U)$ and incoming speed $r_0(1+\alpha_W)$. Its first-arrest and first-release maps are C^1 on a common neighborhood of zero in these independent parameters. In particular the normal release amplitude is C^1 there and has positive limit q_λ. These derivative bounds include the independent initial displacement and filter datum at $\sigma = 0$; they do not use physical force positivity for arbitrary auxiliary data. Use the regularizing coordinate

$$\rho = \delta^{1/12}, \quad \sigma = \rho^6, \quad \widehat{\delta} = \rho^{12}, \quad \alpha_i = \rho^4\eta_i, \quad \beta_m = \rho^6\zeta. \tag{D.13}$$

The chain rule, including the term $(\beta_m/2)\partial_{\beta_m}$ in $\mathscr{D}_\delta$, gives $\|Q - q_\lambda\|_{1,*} \le C\rho^4$ and $|D_\xi Q| \le C\rho^4$. Only the cutoff is reduced to cover the fixed Ω_1.

For an independent positive Q_* near q_λ, reconstruct $X_L = (-a\rho^{12}, 0, 0, \rho^{18}Q_*)$ in the chart (p, y, u_0, v), centered at $G = 0$. With $t = \rho^9 s$ and $X = \rho^9\widehat{X}$, its free equation becomes $\widehat{X}_s = F_{\mathrm{free}}(\rho^9\widehat{X})$ with initial state $(-a\rho^3, 0, 0, \rho^9 Q_*)$. This equation extends smoothly to $\rho = 0$. For every residual P vanishing at G,

$$P(\rho^9\widehat{X}) = \rho^9 \int_0^1 DP(t\rho^9\widehat{X})\widehat{X}\,\mathrm{d}t.$$

This divided form controls its remainders and derivatives. In particular $b_L = \rho^{12}(B_\lambda + \rho^6 B_1)$ and

$$\rho^{-27} y(\rho^9 s) = Q_* s + \tfrac{1}{2}\rho^3(B_\lambda + \rho^6 B_1)s^2 - cs^3/6 + \rho^9 R_f, \tag{D.14}$$

with smooth bounded remainders on fixed s intervals. The positive cubic root $\sqrt{6Q_*/c}$ has derivative denominator $-2Q_*$. The opening interval and positive intervening gap in the

proof of Lemma D.2 apply throughout this primitive domain; thus the implicit root is the first positive root. It has the form $S = \sqrt{6Q_*/c} + \rho^3 S_1$. Integrating p gives

$$T_0 = \rho^9 S, \quad D_0 = \rho^9 \mathcal{D}, \quad \mathcal{D} = S + \rho^3 + \rho^9 H_1, \quad D_0 - T_0 = \rho^{12}(1 + \rho^6 H_1). \tag{D.15}$$

After substituting the C^1 contact amplitude $Q(\rho, \xi)$, $\mathcal{D}$ is bounded below and $\|\mathcal{D} - K_\lambda\|_{1,*} \leq C\rho^3$, $|D_\xi \mathcal{D}| \leq C\rho^4$.

The two normal cancellations remain visible in (C.5): $D_0 - T_0 = \delta + O(\delta^{3/2})$ and $b_L = B_\lambda \delta + O(\delta^{3/2})$ give

$$J_W = (-B_\lambda/2 - 2c/3)\delta D_0 + (B_\lambda/2 + c/3)\delta^2 + O(\delta^{9/4}).$$

Using the divided remainders above makes this an identity with bounded C^1 remainder, rather than differentiation of an order estimate. Consequently there are bounded C^1 functions U_1, W_1 such that

$$\begin{aligned} \eta^+_{U,0} &= (-1 + \rho^6 U_1/d_\lambda)\mathcal{D}^{-4/3}, \\ \eta^+_{W,0} &= -\varkappa_W \mathcal{D}^{-4/3} + A_1 \rho^3 \mathcal{D}^{-7/3} + \frac{\rho^6 W_1}{r_0 \mathcal{D}^{7/3}}, \quad A_1 = 1 + \frac{B_\lambda}{2r_0}, \qquad \zeta^+_0 = 0. \end{aligned} \tag{D.16}$$

Their input-fibre dependence is only through Q. Product and chain rules prove (4.13); the radial derivative of $\log \rho^9$ is $3/4$. □

D.4 A direct comparison at identical coordinate values

The actual inverse (4.2) uses $u_0 = u_r + r$, whereas $T_0(\delta, \xi^r)$ uses the tuned inverse at identical values of the independent coordinates. These are generally different physical touchdowns. We pass between their returns through a reference with the actual offset r, and recode the actual output after its scale is known.

Theorem D.4 (Value comparison at matched normalized coordinates)**.** *For sufficiently small inputs in* (4.3)*, the actual first contact–flight return exists, has $D = f_{\varepsilon,r} \asymp \delta^{3/4}$, and satisfies* (4.14)*. The actual memory is the exact negative memory of* (D.8)*. The first events, their transverse margins, and all compact labels are retained.*

Proof. Lemma D.1 places the actual input in a fixed finite physical cone, so the native specialization (B.37) of Theorem B.2 applies. Reconstruct its reference release at $u_{0,L} = r$, with the same finite-σ compression and inherited release, and call its exact free return R^r. The first pair in Lemma D.2 gives

$$|\log(f_{\varepsilon,r}/f_{R^r})| + |h^0_{\varepsilon,r} - h^0_{R^r}| \leq Ce.$$

For fixed δ the frozen angle and h_δ do not depend on r. Hence R^r and $T_0(\delta, \xi^r)$ share exactly the entire compression solution, inherited release, first release root and amplitude Q, not

only their leading amplitudes. Their reconstructed slips are r and 0. The second free pair, including the $O(r)$ body and acceleration changes in (D.10), gives

$$|\log(f_{R^r}/f_0)| + |h^0_{R^r} - h_0| \le Cr/\delta.$$

In particular $\Delta b_L T = O(rL)$ is retained in the normal numerator before division by $D^{7/3} \asymp \delta L$. Finally recode only the actual output:

$$h^r_{\varepsilon,r} = h^0_{\varepsilon,r} - \frac{r}{d_\lambda D^{4/3}}(0,1,0). \tag{D.17}$$

Its cost is Cr/δ because the paired free flights have already established $D^{4/3} \asymp \delta$. Summing these three comparisons proves the theorem. □

For complete physical-state comparisons, recover the acceleration from the same analytic inverse chart. Define

$$\mathcal{B}_{\varepsilon,r}(s,\xi) = -\frac{b(-as, \varepsilon s^2\zeta, d_\lambda s + d_\lambda s^{4/3}\eta_U + r, -r_0 s^2 - r_0 s^{7/3}\eta_W) + cs}{cs^{4/3}}.$$

Taylor's formula and (D.2) give the first four bounds below at $r = 0$, together with the offset comparison in the last bound:

$$|\partial_{\eta_U}\mathcal{B}| \le C, \quad |\partial_{\eta_W}\mathcal{B}| \le Cs, \quad |\partial_\zeta \mathcal{B}| \le C\varepsilon s^{2/3}, \quad |s\partial_s \mathcal{B}| \le Cs^{2/3}, \quad |\mathcal{B}_{\varepsilon,r} - \mathcal{B}_{\varepsilon,0}| \le Cr/s^{4/3}. \tag{D.18}$$

The radial derivative keeps ε, r fixed; its leading cancellation is exactly (D.2). Integrating these estimates on the logarithmic scale interval between the two output scales and on the bounded fibre segment gives $|\eta^+_B - \eta^+_{B,0}| \le C(e + r/\delta)$. The memory component of h_0 vanishes, so inserting the same ε into its inverse is exact. This recovery uses both output scales; the independent state remains $(\log\delta, \zeta, \eta^r_U, \eta_W)$.

D.5 Admissibility of the returned state

For the already fixed closed ball $\mathcal{K}$, put $d_{\mathcal{K}} = \inf_\lambda \operatorname{dist}(\xi_{*,\lambda}, \partial\mathcal{K}) > 0$, and retain $\gamma = \gamma_{\mathrm{cl}}$ and M_{in} from (4.3). The ball and independent bound are fixed once and for all before the return estimates; the latter contains the second-touchdown tuples of (C.17) and $\mathcal{K}$ in its interior. The dependent acceleration is recovered from the physical inverse.

Theorem D.5 (Admissibility of the returned state)**.** *With these choices, the common radii, output bounds* (4.15)*, signs and inverse branch in Proposition 4.2 hold, including its inner and outer domain qualifications and order of radius choices. The closing margin is* $r_0 W^+ - \zeta^+ > \gamma$*, and the dependent acceleration is uniformly bounded as in Lemma D.1.*

Proof. The compact classes, independent bound M_{in}, $\mathcal{K}, \Omega, \Omega_1, A, C_0, \gamma$ and inverse neighborhood are fixed. Lemma D.1 gives a finite M_B depending on M_{in}, A and the compact model classes. Choose a physical cone containing these independent and dependent bounds,

and then obtain the contact, free and reference constants and a common δ_{call}. Choose any sufficiently small $\delta_{\mathrm{g}} < \delta_{\mathrm{call}}$ within the limiting domain, with $\delta_{\mathrm{g}}^{1/3} \sup_{\mathcal{K}}(|\eta_U| + |\eta_W|) \le 1/4$. Theorem D.4 and Proposition D.3 prove the estimate in (4.15). Its error tends to zero uniformly because $e \le C_0\delta^{1/2}$ and $r/\delta \le A\delta^{1/3}$. Choose δ_{reset} last, making this error less than $d_{\mathcal{K}}/2$ and ensuring, for the common scale bounds $c_D\delta^{3/4} \le D \le C_D\delta^{3/4}$, $c_D\delta_{\mathrm{reset}}^{-1/4} > 1$ and $C_D\delta_{\mathrm{reset}}^{3/4} < \delta_{\mathrm{g}}$. Then $D > \delta$, so the compliance and offset conditions propagate. The ratio bounds give $U^{r,+}, W^+ \ge 3/4$ and hence the velocity signs. Since $\zeta^+ < 0$, $r_0 W^+ - \zeta^+ \ge (3/4)\min r_0 > \gamma$. The same inverse applies, and Lemma D.1 recovers its dependent coordinate on the enlarged output scale. The label and physical parameters were fixed throughout. □

The two smallness conditions in (4.3) have different roles: $\varepsilon \le C_0\delta^2$ controls compliance, while $r \le A\delta^{4/3}$ bounds the dependent acceleration in the physical inverse. The first contact uses the broader range $\delta = \sqrt{\varepsilon}$, $r = k\delta$ of the contact theorem; the two preparatory flights bring the same trajectory into (4.3).

E Normal memory along the release endpoint path

Corollary E.1 refines the release-endpoint path limit by describing its scaled normal memory and effective-gap velocities.

Let R_s be the first release time of the exact inherited normal reference at scale s, and write its normalized coordinates as $(Y_s, V_s^{\mathrm{n}}, Z_s)$. Define

$$V_s = s^{3/2} V_s^{\mathrm{n}}(R_s), \qquad M_s = s^{3/2} Z_s(R_s). \tag{E.1}$$

These constants retain the exact finite-$\sqrt{s}$ reference, and satisfy $M_s < 0 < V_s + M_s < V_s$.

Corollary E.1 (Memory above one physical path)**.** *For every fixed complete label and every $w \in [V_s + M_s, V_s]$, there is a sequence $\varepsilon_i \downarrow 0$ such that t_s lies strictly between release from the last contact, at t_L, and the ensuing touchdown,*

$$d_{\mathrm{F}}(f_{\varepsilon_i}, \mathcal{W}_{\lambda,\nu,s}(s,2)) \to 0, \qquad v(t_s) + z(t_s)/\varepsilon_i \to w.$$

For each fixed $0 < \tau < \infty$, the value $w = V_s + M_s e^{-\tau}$ can be realized with the exact delay $t_s - t_L = \varepsilon_i \tau$, and then

$$(v(t_s), z(t_s)/\varepsilon_i, \dot{g}(t_s)) \longrightarrow (V_s, M_s e^{-\tau}, V_s + M_s e^{-\tau}). \tag{E.2}$$

The memory coordinate vanishes in the physical path but its normalized value survives at release and decays on the fast time. Thus the same release endpoint path supports an interval of limiting effective-gap velocities.

Proof of Corollary E.1. Fix the complete label and $\tau \in (0, \infty)$. On the actual sweep interval J_j consider

$$t_{j,\tau}(\varepsilon) = t_{L,j}(\varepsilon) + \varepsilon\tau, \qquad G_{j,\tau}(\varepsilon) = p(\mathfrak{X}_\varepsilon(t_{j,\tau}(\varepsilon))) + as.$$

The comparison on $[s/2, 2s]$ gives a common positive lower bound $T_{\min}(s)$ for actual flight duration. Thus, for large j, these times lie strictly after the actual first release and before the next first touchdown, throughout J_j. The function is continuous, and bounded $\dot{p}$ gives

$$G_{j,\tau}(a_j) = as/2 + O_{s,\tau}(a_j) > 0, \qquad G_{j,\tau}(c_j) = -as + O_{s,\tau}(c_j) < 0.$$

At an actual root, strict decrease of p and the predecessor and successor bounds give $N_s = j + 1$ and

$$d_\varepsilon \to s, \qquad t_s = t_L + \varepsilon\tau, \qquad \alpha_\varepsilon = 2 + \varepsilon\tau/T_\varepsilon \longrightarrow 2. \tag{E.3}$$

All these sequences have the same complete path limit $\mathcal{W}_{\lambda,\nu,s}(s, 2)$.

At their moving first-release roots the complete normalized comparison (6.4) gives

$$v_L \longrightarrow V_s = s^{3/2} V_s^{\mathrm{n}}(R_s), \qquad z_L/\varepsilon \longrightarrow M_s = s^{3/2} Z_s(R_s). \tag{E.4}$$

These are the exact finite-$\sqrt{s}$ inherited release data. The release margins imply $M_s < 0 < V_s + M_s < V_s$. In the actual free leg $z(t_s)/\varepsilon = (z_L/\varepsilon)e^{-\tau}$; the bounded free field and $\varepsilon\tau \to 0$ give $v(t_s) \to V_s$. Since $g = y - z$ and $\dot{g} = v + z/\varepsilon$ in flight,

$$(v(t_s), z(t_s)/\varepsilon, \dot{g}(t_s)) \longrightarrow (V_s, M_s e^{-\tau}, V_s + M_s e^{-\tau}). \tag{E.5}$$

The scaled-memory assertion follows from (E.4), not merely from physical $z_L \to 0$.

For the lower endpoint take $\tau_i = 1/i > 0$ and choose a sufficiently late root from each fixed-τ_i sequence, with all path and normalized-state errors below $1/i$ and decreasing compliance. Every section remains strictly inside the free leg, and its fast delay tends to zero. For the upper endpoint take $\tau_i \to \infty$, first fixed in each row, and choose roots so late that the same errors are below $1/i$ and

$$\varepsilon_i\tau_i < 1/i, \qquad \varepsilon_i\tau_i < T_{\min}(s)/2, \qquad \varepsilon_i < \varepsilon_{i-1}/2.$$

Exact exponential decay gives $z(t_s)/\varepsilon_i \to 0$, while the bounded mechanical field gives $v(t_s) \to V_s$ and $\alpha_{\varepsilon_i} \to 2$. No comparison on an unbounded normal-time window is required. Together with the finite positive delays, these diagonals realize the whole closed interval $[V_s + M_s, V_s]$ above the same whole physical path, with the section strictly in free flight at every chosen compliance. □

F Separation and the geometry of the accumulation families

We first establish the single nontrivial identification for each of the terminal-state, ordered-path and compact-image parameterizations, then use continuity and compactness to identify their images with the quotient circle.

Lemma F.1 (Separation of terminal states and path images)**.** *Fix λ, ν and $0 < s \le s_0$ as in Theorem A, and write $d_* = F_\lambda^{-1}(s)$. The terminal-state map*

$$E_s : \mathcal{A}_s \longrightarrow \mathbb{R}^5, \qquad E_s(d, \alpha) = \mathcal{C}_{\lambda,d}(\alpha),$$

is one-to-one except for

$$E_s(d_*, 3) = E_s(s, 0).$$

The ordered-path and compact-image maps in (3.6)–(3.7) *have exactly the same single non-trivial identification.*

Proof. The contact injectivity established in (5.15) applies for every $0 < s \le s_0$. It separates admission, the interior contact points and release.

For $d_* < d < s$, the stopped endpoint lies strictly inside its first free flight and has $y > 0$. Contact endpoints have $y = 0$. Two free endpoints cannot coincide. Indeed, suppose

$$\Phi_{\mathrm{f}}(t_1; \mathcal{R}_{d_1}) = \Phi_{\mathrm{f}}(t_2; \mathcal{R}_{d_2}), \qquad 0 < t_i < T_{d_i},$$

and take $t_1 \le t_2$. Uniqueness for the autonomous free mechanical flow gives $\mathcal{R}_{d_1} = \Phi_{\mathrm{f}}(t_2 - t_1; \mathcal{R}_{d_2})$. If $t_1 < t_2$, the right side has $y > 0$, whereas the left side has $y = 0$. Hence $t_1 = t_2$ and the release states agree. Their p values give $d_1 = d_2$, and strict decrease of p on the free flight gives the same stopping point. At the free endpoints, graph invariance gives $\mathcal{C}_{\lambda,d_*}(3) = \mathcal{C}_{\lambda,s}(0)$, while $\mathcal{C}_{\lambda,s}(2)$ is already the common point of the graph and vertical branches of $\mathcal{A}_s$. Injectivity of the contact curve now proves the assertion for E_s.

The two descriptions $(d_*, 3)$ and $(s, 0)$ give the same ordered path, since the final return at d_* is the last full return in L_s. Conversely, the ordered distance bounds the distance between terminal states. The terminal-state assertion therefore gives the assertion for ordered paths.

For images, endpoint separation alone would not suffice. Put $\Sigma_s = \{\mathfrak{X} : p(\mathfrak{X}) = -as\}$. The incoming arc has $p \ge 0$. When $d < s$, all completed earlier returns and the final contact have $p > -as$, and the last free prefix has strictly decreasing p. Consequently

$$W_{\lambda,\nu,s}(d, j_s(d)) \cap \Sigma_s = \{E_s(d, j_s(d))\}, \qquad d_* \le d < s.$$

For a contact prefix, the only earlier point on Σ_s is its admission point: the last full return in L_s ends there, and all other earlier points have $p > -as$. Hence

$$W_{\lambda,\nu,s}(s, \alpha) \cap \Sigma_s = \mathcal{C}_{\lambda,s}([0, \alpha]), \qquad 0 \le \alpha \le 2.$$

These sets are strictly nested as α increases, by contact injectivity. A contact prefix of positive length cannot have a singleton intersection with Σ_s. The case $\alpha = 0$ has the same singleton as $(d_*, 3)$; every interior free endpoint has $y > 0$. Finally two free singletons coincide only in the already identified terminal-state cases. This proves separation of the compact images as well. □

Corollary F.2 (The accumulation families are topological circles)**.** *For fixed λ, ν, s as in Lemma F.1, the terminal-state set, the ordered-path accumulation family with the distance* (3.1)*, and the compact-image accumulation family with its Hausdorff distance are each homeomorphic to S^1. More precisely, their parameter space is*

$$\mathcal{A}_s / \big((d_*, 3) \sim (s, 0)\big).$$

These statements concern the full physical state (ϕ, y, ω, v, z).

Proof. The graph branch of $\mathcal{A}_s$ is an arc from $(d_*, 3)$ to $(s, 2)$, and its vertical branch is an arc from $(s, 2)$ to $(s, 0)$. Their only common point is $(s, 2)$, so $\mathcal{A}_s$ is homeomorphic to a closed interval with endpoints $(d_*, 3)$ and $(s, 0)$. Identifying these endpoints gives S^1.

The terminal-state map is continuous by the construction of the reference curves. The ordered-path map is continuous by the finite-final-part and small-tail argument following (6.18); image continuity follows from $d_{\mathrm{H}} \leq d_{\mathrm{F}}$. Lemma F.1 says that all three maps identify exactly the two interval endpoints. Each therefore induces a continuous bijection from the compact quotient circle to its image in a metric space, and is a homeomorphism. □

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Data availability Data sharing is not applicable to this article as no datasets were generated or analysed during the current study.

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